HABITAT CONSERVATION TRUST FOUNDATION

SOCIETY FOR ECOSYSTEM RESTORATION IN NORTHERN BRITISH COLUMBIA

Restoring Fish Passage in the Skeena Region - 2025

Prepared for
Habitat Conservation Trust Foundation - CAT26-0-636
Ministry of Transportation and Infrastructure

Prepared by
Al Irvine, B.Sc., R.P.Bio. and Lucy Schick, B.Sc.
New Graph Environment Ltd.
on behalf of
Society for Ecosystem Restoration in Northern British Columbia

Version 0.6.0 DRAFT 2026-08-06

Claude Opus 5 (Anthropic) assisted with literature synthesis, drafting, and technical writing. All scientific interpretation, data analysis, and conclusions are the responsibility of the authors.

new graph environment

Table of Contents

List of Figures

Acknowledgement

At New Graph Environment, we understand our well-being as inseparable from the health of the land and waters we work within. When we care for ecosystems, we care for ourselves and for the communities connected to them.

Modern civilization has a long journey ahead to acknowledge and address the historic and ongoing impacts of colonialism that have resulted in harm to the cultures and livelihoods living interconnected with our ecosystems for many thousands of years.

Executive Summary

Executive Summary (PDF)

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from fish_passage_skeena_2025_reporting.pdf.

Since 2020, the Society for Ecosystem Restoration Northern British Columbia (SERNbc) has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Bulkley River and Morice River watershed groups, which are sub-basins of the Skeena River watershed. In 2022, the study area was expanded to include the Zymoetz River watershed group and the Kispiox River watershed group, followed by an extension in 2023 to encompass sections of the Kitsumkalum River watershed group, particularly where Highway 16 intersects the watershed.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document 2025 field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time.

- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning (2020)
- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021
- Bulkley River Watershed Fish Passage Restoration Planning 2022
- Skeena Watershed Fish Passage Restoration Planning 2022
- Skeena Watershed Fish Passage Restoration Planning 2023
- Skeena Watershed Fish Passage Restoration Planning 2024

Fish passage assessment procedures conducted through SERNbc in the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups since 2020 are amalgamated online within the Assessment Data Summary appendix of the report found here which includes links to project reporting for each site.

Environmental DNA (eDNA) sampling was the substantive field programme this season. A total of 19 samples were collected across 10 streams, alongside 1 field and office blanks run as protocol controls.

Of the 5 species assayed, rainbow trout (9 of 13 sites), bull trout (7 of 18 sites), and coho salmon (3 of 19 sites) were detected. chinook salmon and sockeye salmon were not detected in any sample. Detections use the four-droplet call threshold applied by the laboratory. A single season of sampling establishes presence, not absence — a species may go undetected because it is absent, or because it was not shedding detectable DNA at the place and moment sampled — which is why the value of this layer grows with repetition.

Monitoring was conducted at two sites, including post-remediation evaluations at Waterfall Creek (PSCIS crossing 124421) and Peacock Creek (PSCIS crossing 197962). Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

A climate departure analysis was completed for the study area, comparing the recent decade against a pre-warming reference period of 1951 to 1980. Mean annual air temperature is 2.1 °C above that reference, and the warming is strikingly even across the region — the 6 ecoregions spanning the study area differ by only 0.28 °C in warming accumulated since 1951. The clearest signal is in the snowpack, and it is one of timing rather than of quantity: the snowmelt midpoint now falls 15 days earlier in the year and summer snow water equivalent has fallen 60 per cent. The freshet that crossing structures were sized against is arriving substantially earlier than the record they were designed from, and the late-summer cold water that snowpack once supplied is declining. Because the warming is so uniform, the thermal outlook for a given barrier is a question of elevation and local reach conditions rather than of which watershed group it sits in.

Functional floodplain extent was mapped for the Bulkley River watershed group, the first of the study area's watershed groups to be delineated. Modelled floodplain covers 48,344 ha, or 6.2 per cent of the 7,762 km² group. Passage and floodplain condition are the same problem viewed from two angles — reconnecting a stream longitudinally achieves less where the floodplain is severed laterally — so the layer indicates where remediation should account for more than the structure itself. The delineation is published to a shared spatial catalogue and read into this report rather than regenerated for it, so one modelled product serves every report that needs it; Morice and Kispiox are already published and can follow without re-running the model.

These are not separate studies. Each answers part of one question: what condition are these watersheds in, and where will restoration effort do the most good. Passage assessments and the provincial model establish where fish can reach; habitat confirmations, fish sampling and environmental DNA establish where they are; floodplain delineation and aerial imagery establish how infrastructure has altered the valley bottom; climate departure and water temperature establish the trajectory; effectiveness monitoring tests whether the work delivered. Assembling these layers makes explicit what is known, what is not, and what should be measured next. Making that evidence usable is part of the work — data collected on structured digital forms, methods documented alongside the code that implements them, layers published to open catalogues, and reporting written so partners, First Nations, regulators and tenure holders can question it and build on it — so understanding accumulates across the program rather than being rebuilt each season.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, because the criteria that determine whether a crossing is worth remediating are not measured in common units, and the deciding constraint sits outside the project in the capacity and willingness of infrastructure owners and tenure holders to support implementation across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Looking ahead, priorities for the program are:

- Continue and extend effectiveness monitoring across the region's remediated crossings, spanning a range of structure types, treatments and stream sizes, and continue developing a cost-effective framework that ties baseline and post-remediation data to measurable productivity gains.
- Continue environmental DNA sampling as an annual, region-wide layer so single-year detections become distribution and trend information, and resolve species use through targeted fish sampling where habitat ranks high but detections are absent or unresolved.

- Extend functional floodplain mapping from the Bulkley to the remaining watershed groups, and carry climate departure and water temperature into prioritization, so watershed condition and trajectory sit alongside barrier status wherever ranking is done.
- Resume Phase 1 assessments and continue calibrating the provincial habitat model against observed conditions, so the candidate set that prioritization draws on keeps growing and improving.
- Commission engineering designs at the Ministry of Transportation and Infrastructure crossings identified in previous reporting, and advance the highest ranked sites assessed this year.
- Continue developing the shared data and reporting infrastructure — open spatial catalogues, a public web map, and common repositories for water temperature and environmental DNA — so results are findable and reusable by everyone working in these watersheds.
- Continue working with First Nations, stewardship groups, educational institutions and tenure holders, feeding monitoring results back into prioritization and design as adaptive management.

1 Introduction

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest PDF from fish_passage_skeena_2025_reporting.pdf.

Since 2020, the Society for Ecosystem Restoration Northern British Columbia (SERNbc) has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Bulkley River and Morice River watershed groups, which are sub-basins of the Skeena River watershed. In 2022, the study area was expanded to include the Zymoetz River watershed group and the Kispiox River watershed group, followed by an extension in 2023 to encompass sections of the Kitsumkalum River watershed group, particularly where Highway 16 intersects the watershed.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

This project builds on Society for Ecosystem Restoration Northern BC (SERNbc) work in:

- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning (2020) (Irvine 2021)
- Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021 (Irvine [2021] 2022)
- Bulkley River Watershed Fish Passage Restoration Planning 2022 (Irvine 2023)
- Skeena Watershed Fish Passage Restoration Planning 2022 (Irvine and Wintersheidt 2023)
- Skeena Watershed Fish Passage Restoration Planning 2023 (Irvine and Schick 2023)
- Skeena Watershed Fish Passage Restoration Planning 2024 (Irvine and Schick 2025)

The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al. 2002; Swales and Levings 1989; Diebel et al. 2015). Culverts can present barriers to fish migration due to low water depth, increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney and Zaldokas 1997; Cote et al. 2005). As road crossing structures are commonly upgraded or removed there are numerous opportunities to restore connectivity by ensuring that fish passage considerations are incorporated into repair, replacement, relocation and deactivation designs.

Although remediation and replacement of stream crossing structures can have benefits to local fish populations, the costs of remedial works can be significant and the impacts of the work often complex to evaluate and quantify. Additionally, allocation of ecosystem restoration funding towards infrastructure upgrades on transportation right of ways are not always considered ethical under all circumstances from all perspectives. When funds are finite and invested groups are engaged in fund raising, cost benefits and the ethics of crossing replacements should be explored collaboratively alongside the cost benefits and ethics of alternative investment activities including transportation corridor relocation/deactivation, land procurement/covenant, cattle exclusion, riparian/floodplain restoration, habitat complexing, water conservation, commercial/recreational fishing management, salt water interventions and research.

Please note that at the time of reporting this document can be considered a living document. See the Open Source Reporting Framework (page 36) section for details on version tracking and project documentation.

2 Background

2.1 Approach

Fish passage assessments in British Columbia follow the provincial protocol (MoE 2011), which scores barrier risk at road-stream crossings from structural parameters — culvert embedment, slope, outlet drop, and culvert diameter relative to channel width. The “Barrier” and “Potential Barrier” classifications represent passability for juvenile salmon or small resident rainbow trout under any flow conditions that may occur throughout the year (Clarkin et al. 2005; Bell 1991; Thompson 2013). The thresholds are deliberately conservative, so the classifications are risk scores rather than determinations of whether fish actually move through. Many structures classified as “Barrier” still pass some species and life stages at some flows, and natural features in adjacent reaches can be as limiting as the structure itself. Passability can be quantified in other ways — fish length, swim speed, and species physiology all affect connectivity outcomes — and finer-grained scoring methods exist for structures already classified as barriers (Bourne et al. 2011; Kemp and O’Hanley 2010; Washington Department of Fish & Wildlife 2009). Whether a remediation changes outcomes for fish therefore depends on the habitat upstream, the species and life stages in question, and the broader condition of the reach the structure connects to — not the classification alone.

The provincial inventory contains thousands of structures on fish-bearing streams. Replacement at a typical forest road culvert is in the six-figure range; structural replacements on highway and rail corridors run into seven and eight figures. Funding for restoration is finite and split across multiple programs and partners. Prioritization across the inventory therefore carries more weight than any individual assessment — the assessment establishes what could be fixed, while prioritization decides what should be.

Authority and funding for these structures are distributed across rail authorities, highway ministries, forestry licensees, and private landowners. Each operates under its own mandate, performance metrics, and budget cycle. Advancing restoration involves working inside organizations whose primary mandate is moving goods, maintaining roads, or harvesting timber — not aquatic ecological outcomes. The work proceeds through shared data, sustained relationships, and incremental progress across multiple budget years and field seasons.

Many road and rail corridors were built without consideration of fish passage or floodplain function. Damage frequently extends beyond the structure at the crossing — channelized streams, dykes that sever lateral connection, drained wetlands, cleared riparian buffers. Replacing a culvert in-kind can restore longitudinal passage while leaving the broader corridor-scale disconnection unaddressed. Understanding where infrastructure constrains watershed function helps distinguish cases where structural replacement is sufficient from cases where remediation needs to address more than the structure itself.

This report compiles the layers used for prioritization in the watershed groups where the work is active: structural assessment data, habitat modelling, climate trajectory, and floodplain extent. Each layer contributes a piece of the picture — what is present, what it could support, how it is changing, and the surrounding watershed context. The work happens over many years across many partners; this report documents one year's contribution to that effort.

2.2 Project Location

The study area includes the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups (Figure 2.1) and is within the traditional territories of the Wet'suwet'en, Gitxsan and Tsimshian.

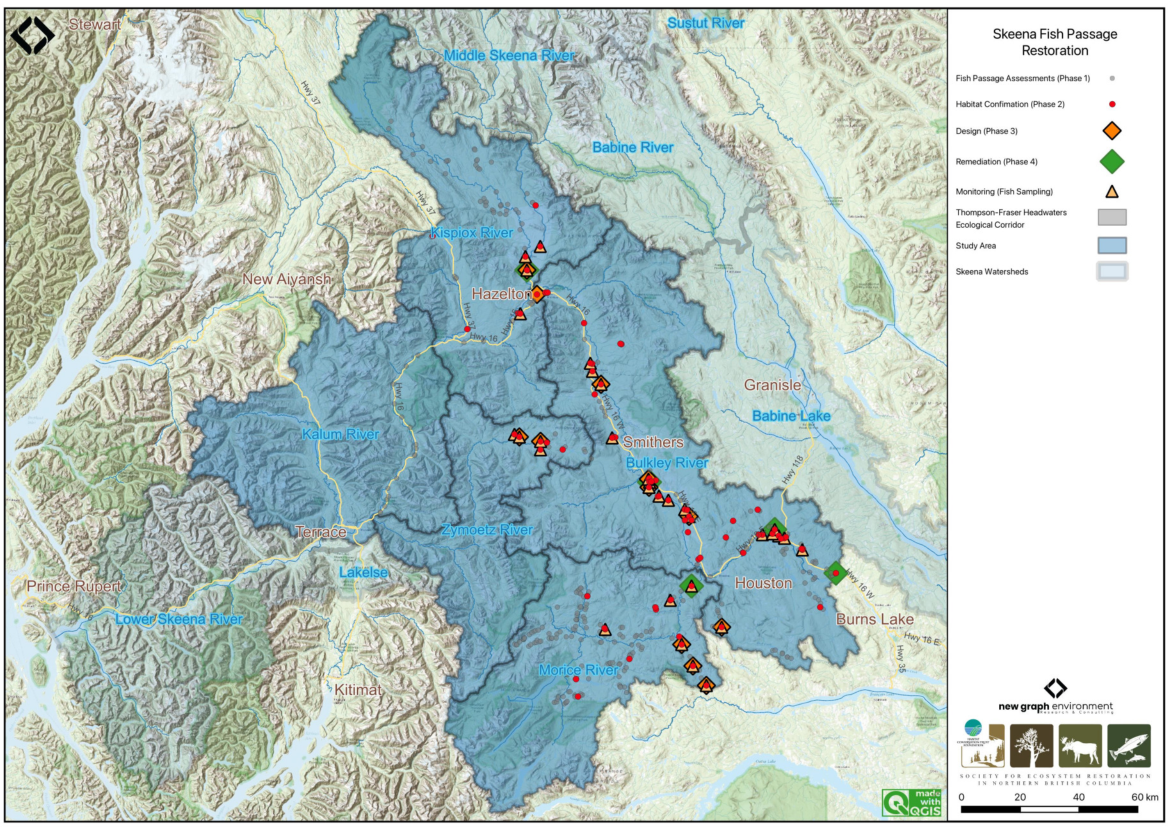

Figure 2.1: Overview map of the Skeena Region study areas

2.3 Wet'suwet'en

Wet'suwet'en hereditary territory covers an area of 22,000km^2 including the Bulkley River and Morice River watersheds and portions of the Nechako River watershed. The Wet'suwet'en people are a matrilineal society organized into the Gilseyhu (Big Frog), Laksilyu (Small Frog), Tsayu (Beaver clan), Gitdumden (Wolf/Bear) and Laksamshu (Fireweed) clans. Within each of the clans there are are a number of kin-based groups known as Yikhs or House groups. The Yikh is a partnership between the people and the territory. Thirteen Yikhs with Hereditary Chiefs manage a total of 38 distinct territories upon which they have jurisdiction. Within a clan, the head Chief is entrusted with the stewardship of the House territory to ensure the Land is managed in a sustainable manner. Inuk Nu'at'en (Wet'suwet'en law) governing the harvesting of fish within their lands are based on values founded on thousands of years of social, subsistence and environmental dynamics. The Yintahk (Land) is the centre of life as well as culture and it's management is intended to provide security for sustaining salmon, wildlife, and natural foods to ensure the health and well-being of the Wet'suwet'en (Office of the Wet'suwet'en 2013; "Office of the Wet'suwet'en" 2021; FLNRORD 2017).

2.4 Gitxsan

Gitxsan means "People of the River Mist". The Gitxsan Laxyip (traditional territories) covers an area of 33,000km^2 within the Skeena River and Nass River watersheds. The Laxyip is governed by 60 Simgiigyet (Hereditary Chiefs), within the traditional hereditary system made up of Wilps (House groups). Anaat are fisheries tenures found throughout the Laxyip. Traditional governance within a matrilineal society operates under the principles of Ayookw (Gitxsan law) ("Gitxsan Huwilp Government," n.d.). Many band members live in Hazelton, Kispiox and Glen Vowell (the Eastern Gitxsan) as well as within Kitwanga, Kitwankool and Kitsegukla (the Western Gitxsan) (Powell et al. 2018).

2.5 Tsimshian

The Kitsumkalum community, part of the Tsimshian Nation, maintains a rich cultural heritage rooted in ancient traditions and values. Their society, governed by Tsimshian Law (ayaawx), emphasizes strong connections through marriages, adoptions, and resource sharing with other Tsimshian tribes. The community upholds its cultural and spiritual practices, including fishing, harvesting, and land stewardship, despite the impacts of colonization (Kitsumkalum Band, n.d.).

Kitsumkalum's social structure is based on matrilineal kinship, with significant emphasis on family ties through the mother's lineage. Their cultural identity is expressed through crest groups (pteex), lineage houses (waap), and the importance of landed property (laxyuup), which ties them to their ancestral territories. The community combines traditional governance with modern administrative functions, reflecting their resilience and commitment to preserving their heritage (Kitsumkalum Band, n.d.).

The Kitsumkalum River salmon populations have been an important part of their culture and economy (Gottesfeld and Rabnett 2007).

2.6 Bulkley River

The colonial naming conventions for the Bulkley and Morice rivers can be misleading. The Neexdzii Kwah (Upper Bulkley River) is a tributary to Widzin Kwah (Morice River), contributing on average less than one-third of the total flow at their confluence in Houston, BC. However, historical naming during early mapping and settlement resulted in “Bulkley River” being applied to the mainstem flowing from Houston to its confluence with the Skeena River at Hazelton. From a hydrological perspective, it may be more accurate to consider the Morice River as the mainstem and the Upper Bulkley River as one of its tributaries. These naming differences can obscure the true flow dynamics of the watershed. By using Wet’suwet’en place names, we not only recognize the deep connection the Wet’suwet’en have to their clan territories but also convey a more accurate understanding of the geography and hydrology of the region.

The Bulkley River (Neexdzii Kwah), as defined by colonial naming, is an 8th order stream that drains an area of 7,762km^2 in a generally northerly direction from Bulkley Lake on the Nechako Plateau to its confluence with the Skeena River at Hazelton. It has a mean annual discharge of 139.2m^3/s at station 08EE004 near Quick (~27km south of Telkwa), and 18m^3/s at station 08EE003 upstream near Houston. Flow at Quick is heavily influenced by the Morice River (Widzin Kwah), which joins the Bulkley just downstream of Houston. As a result, discharge patterns at Quick are characteristic of high-elevation, snow-dominated watersheds, with peak flows typically occurring from May to July (Figure 2.2). Upstream of Houston, the hydrograph peaks earlier (May–June), consistent with lower-elevation terrain and reduced snowpack influence (Figure 2.3).

Changes to the climate systems are causing impacts to natural and human systems on all continents with alterations to hydrological systems caused by changing precipitation or melting snow and ice increasing the frequency and magnitude of extreme events such as floods and droughts (Calvin et al. 2023; ECCC 2016). These changes are resulting in modifications to the quantity and quality of water resources throughout British Columbia and are likely to compound issues related to drought and flooding in the Bulkley River watershed where numerous water licenses are held with a potential over-allocation of flows identified during low flow periods (ILMB 2007).

The valley bottom has seen extensive settlement over the past hundred years with major population centers including the Village of Hazelton, the Town of Smithers, the Village of Telkwa and the District Municipality of Houston. As a major access corridor to northwestern British Columbia, Highway 16 and the Canadian National Railway are major linear developments that run along the Bulkley River within and adjacent to the floodplain with numerous crossing structures impeding fish access into and potentially out from important fish habitats. Additionally, as the valley bottom contains some of the most productive land in the area, there has been extensive conversion of riparian ecosystems to hayfields and pastures leading to alterations in flow regimes, increases in water temperatures, reduced streambank stability, loss of overstream cover and channelization (ILMB 2007; Wilson and Rabnett 2007).

2.6 Bulkley River

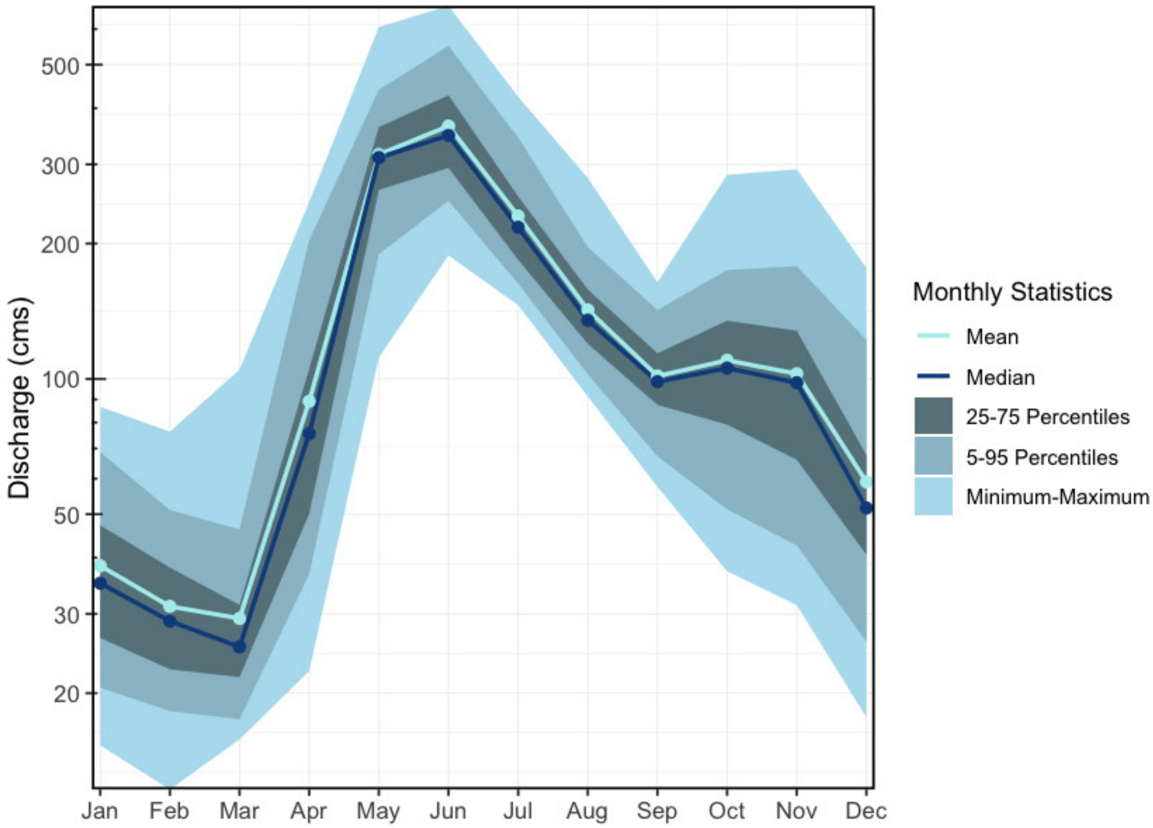

Figure 2.2: Hydrograph for Bulkley River at Quick (Station #08EE004). Available mean daily discharge data from 1930 to 2023.

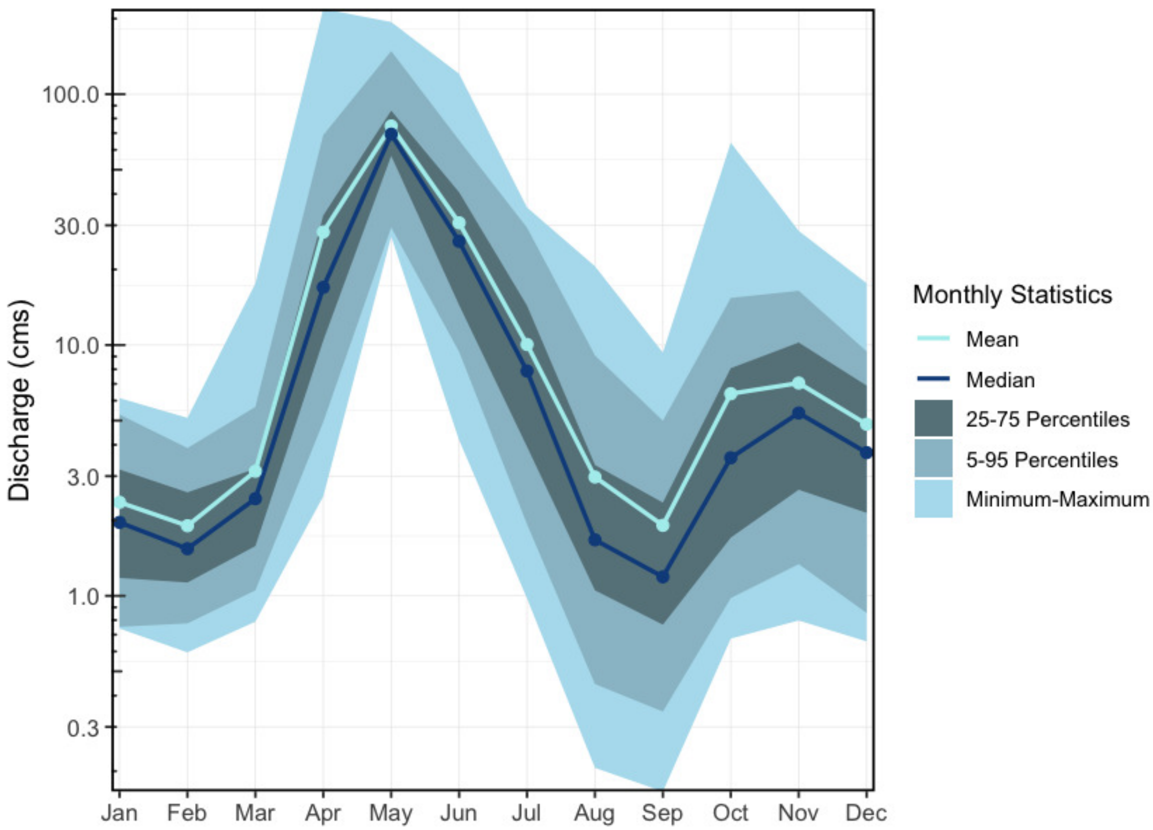

Figure 2.3: Hydrograph for Upper Bulkley River near Houston, upstream of the Morice River confluence (Station #08EE003). Available mean daily discharge data from 1930 to 2023.

2.7 Morice River

The Morice River, known to the Wet'suwet'en as Widzin Kwah and meaning river of clear, blue-green waters, flows from its headwaters at Morice Lake (Widzin Bin) northeast to its confluence with the Upper Bulkley River (Neexdzii Kwah) just north of Houston, BC. The river is an 8th order stream that drains 4,379km^2 of the Coast Mountains and Interior Plateau. Major tributaries include the Nanika, Atna, and Thautil rivers, as well as Gosnell Creek. Numerous large lakes are located in the southern portion of the watershed, including Morice Lake, McBride Lake, Stepp Lake, Nanika Lake, Kid Price Lake, and Owen Lake. One active hydrometric station (08ED002), located near the outlet of Morice Lake, records a mean annual discharge of 75m^3/s (Canada 2024). Flow patterns are characteristic of high-elevation coastal-influenced watersheds, with peak discharge from snowmelt occurring between May and July, and additional high flows from fall precipitation and rain-on-snow events (Figure 2.4).

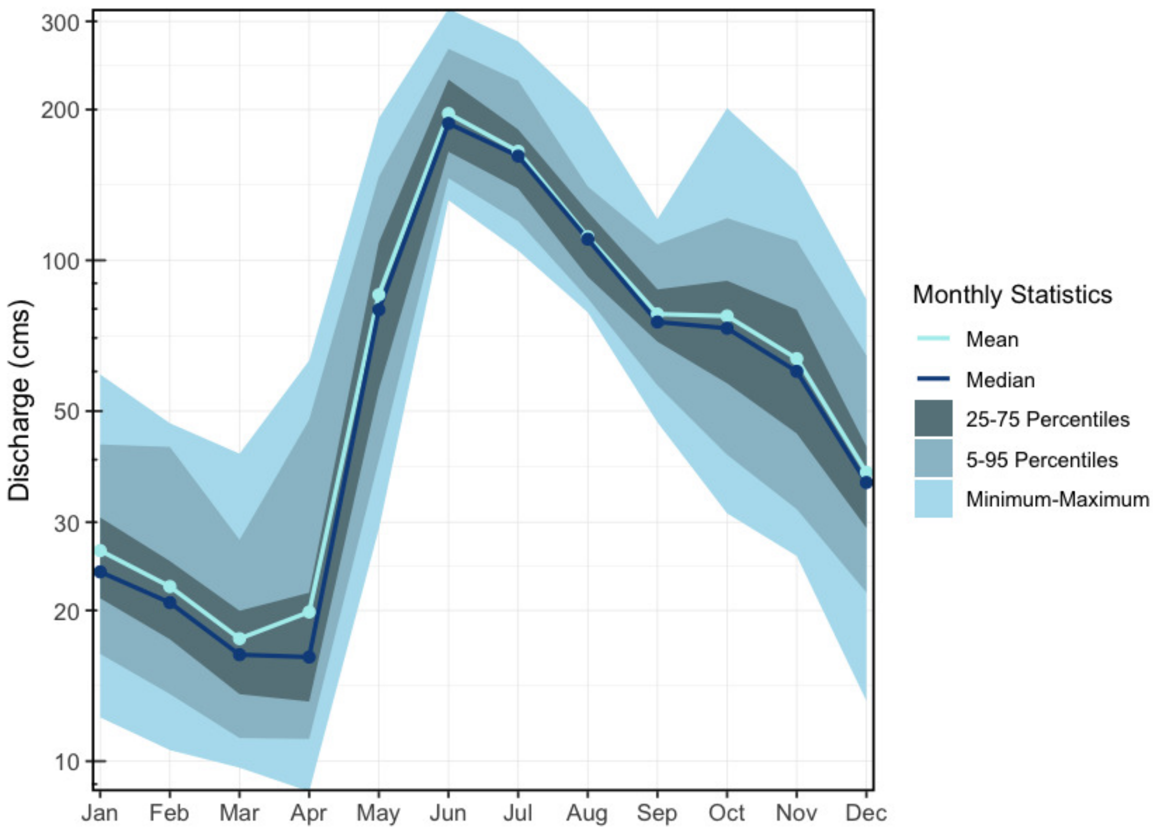

Figure 2.4: Hydrograph for the Morice River near outlet of Morice Lake (Station #08ED002). Available mean daily discharge data from 1961 to 2023.

2.8 Zymoetz River

The Zymoetz River (known locally as the Copper River) is an 8th order stream that drains an area of 3,026km^2 in a generally westerly direction and is a major tributary to the Skeena River, contributing approximately 10% of its flow. Flowing roughly 120km from its headwaters near Hudson Bay Mountain west of Smithers, to its confluence with the Skeena River about 8km northeast of Terrace, with headwater lakes including Aldrich, Dennis, and McDonell lakes. Elevations in the watershed range from 120m at the confluence, to 2740m in the Howson Range. The watershed is accessed via logging roads from Highway 16 near Smithers and Terrace, though access to the mid-watershed is limited due to road washouts. The Duthie Mine, active during the 1930s and 1950s on the southwest slope of Hudson Bay Mountain, has been linked to contamination in surrounding streams and lakes (Gottesfeld et al. 2002), while the lower watershed has experienced riparian habitat loss due to wildfires, forestry activity, and pipeline and road construction (Gottesfeld et al. 2002). Snowmelt dominates the hydrology, with peak flows from May to early June and a mean annual discharge of 105m^3/s at station 08EF005 near Smithers (Figure 2.5).

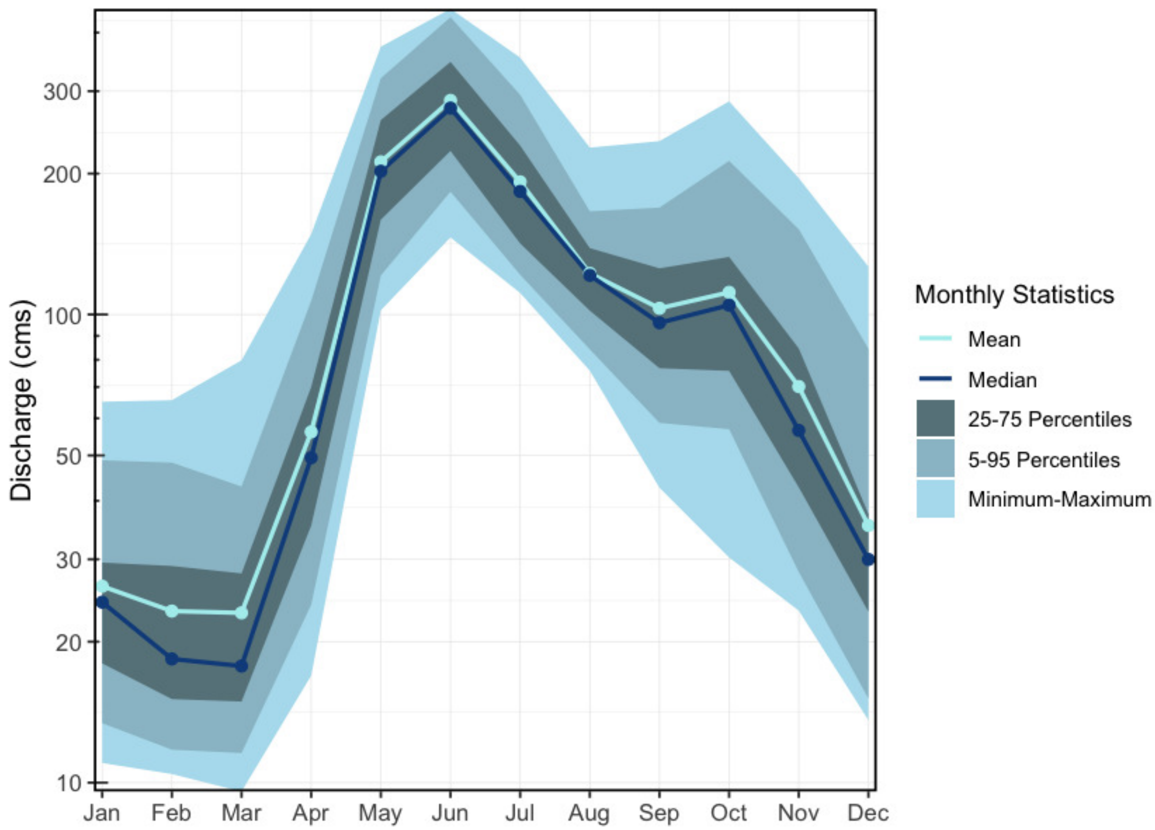

Figure 2.5: Hydrograph for the Zymoetz River above O.K. Creek (Station #08EF005 - Lat 54.49363 Lon -128.32466). Available mean daily discharge data from 1963 to 2022.

2.9 Kispiox River

The Kispiox River is a 7th order stream that flows approximately 140km southeast to its confluence with the Skeena River near Kispiox Village, draining a watershed area of 2,100km^2 and contributing an estimated 9% of the Skeena River's flow. Elevations in the watershed range from 200m at the mouth to 2,090m on Kispiox Mountain. The river is primarily fed by glacier and high-elevation snowmelt, with peak discharge occurring in May and June (Figure 2.6), and a mean annual discharge of 45m^3/s recorded at station 08EB004 near Hazelton. Important sockeye systems in the upper watershed include Swan and Stephens lakes; Swan Lake drains via Club Lake into Stephens Lake, which flows via Stephens Creek into the Kispiox mainstem. The upper third of the watershed, upstream of the Nangeese River, is relatively undisturbed due to limited road access and protection within Swan Lake Kispiox River Provincial Park (Gottesfeld et al. 2002). Reported threats to aquatic ecosystems in the Kispiox Valley include erosion, obstructions, sedimentation, and altered water yield.

2.9 Kispiox River

The Kispiox watershed is also in danger due to the proposed Prince Rupert Gas Transmission (PRGT) pipeline which would run approximately 900km from Hudson's Hope in northeast BC to the proposed Ksi Lisims LNG facility near the Nass River estuary. The proposed pipeline would not only cut through Kispiox River, but also the Babine, Suskwa, Salmon/Shegunia, Skeena, and Nass rivers, all of which have salmon and steelhead habitat ("LNG Pipelines I Skeena Watershed Conservation Coalition," n.d.). The project has raised significant concern due to risks to water quality, fish habitat, and indigenous rights. The Kispiox Band, Skeena Watershed Conservation Coalition, Kispiox Valley Community Centre Association, and other local groups have publicly opposed the project and launched legal action challenging the BC Energy Regulator's permitting decisions. As of March 2025, the pipeline had not yet been constructed, and legal proceedings remain ongoing ("Groups Challenge Rush Start of Construction on Pipeline" 2025).

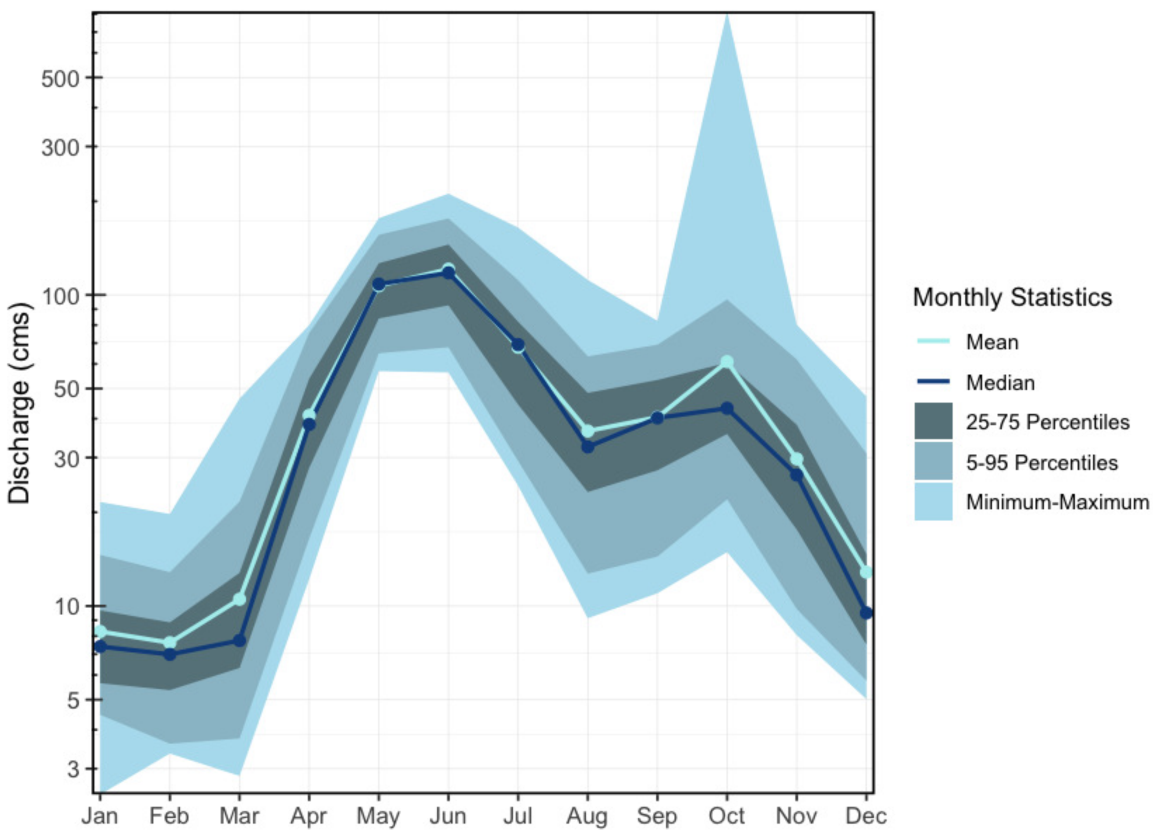

Figure 2.6: Hydrograph for the Kispiox River near Hazelton (Station #08EB004 - Lat 55.43385 Lon -127.71616). Available mean daily discharge data from 1963 to 2024.

2.10 Kitsumkalum River

The Kitsumkalum River is a 6th order stream that flows north-east from the Coast Mountains into Kitsumkalum Lake, then south to its confluence with the Skeena River at Terrace, draining a watershed area of 2,289km^2. Major tributaries include the Cedar River, Nelson River, Mayo Creek, Goat Creek, Lean-To Creek, and Deep Creek (McElhanney 2022). Peak flows occur in May–June from snowmelt, with additional peaks in the fall due to rainfall events (McElhanney 2022). A hydrometric station located below Kitsumkalum Lake has been active since 2018 and records a mean annual discharge of 123m^3/s (Figure 2.7).

The Kitsumkalum River watershed has been highly impacted by logging. Many of the tributaries to the Kitsumkalum River have altered channel morphology, increased bedload movement, bank failures, sediment loading, and debris accumulation (Gottesfeld and Rabnett 2007).

There has been a significant amount of work done to enhance salmon populations within the watershed. The SkeenaWild Conservation Trust is conducting riparian restoration surveys on several tributaries to the Kitsumkalum River, including Willow Creek, Spring Creek, Lean-To Creek, and Deep Creek (Healthy Watersheds Initiative 2021). The Deep Creek Hatchery, operated by the Terrace Salmonid Enhancement Society, has been supporting Kitsumkalum River chinook populations since 1984 (Gottesfeld and Rabnett 2007). Additionally, there is a small groundwater facility for the incubation and rearing coho and chum, run by the Kitsumkalum First Nation (Gottesfeld and Rabnett 2007). In 2000, the Clear Creek Eastern Side Channel was constructed to enhance juvenile rearing habitat and adult spawning habitat for coho salmon on Clear Creek, a tributary to the Kitsumkalum River, however the site has not been maintained and beaver activity has obstructed fish accessibility to much of the channel (Elmer 2021).

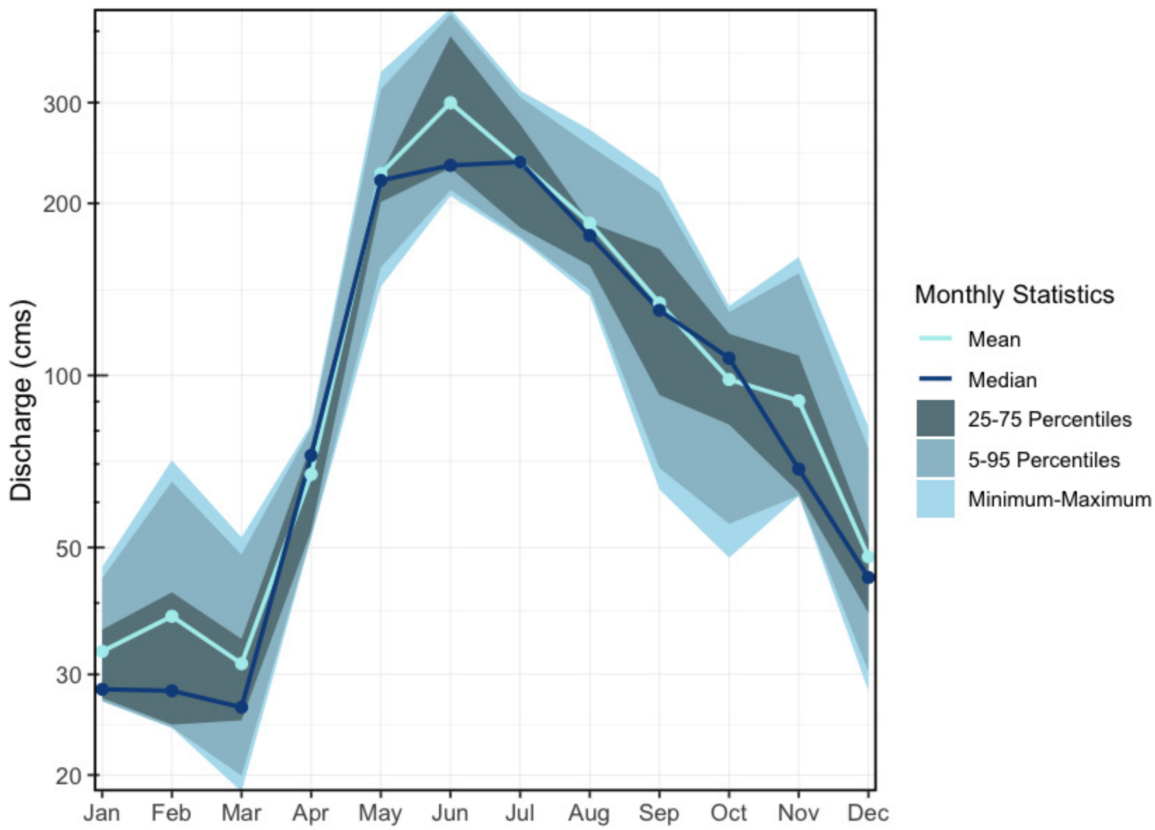

Figure 2.7: Hydrograph for the Kitsumkalum River Below Alice Creek (Station #08EG019 - Lat 54.6793 Lon -128.74396). Available mean daily discharge data from 2018 to 2023.

2.11 Fisheries

In 2004, IBM Business Consulting Services (2006) estimated the value of Skeena Fisheries at an annual average of $110 million dollars. The Bulkley-Morice watershed is an integral part of the salmon production in the Skeena drainage and supports an internationally renown steelhead, chinook and coho sport fishery (Tamblyn 2005).

2.11.1 Bulkley River

Traditionally, the salmon stocks passing through and spawning in the greater Bulkley River were the principal food source for the Gitxsan and Wet'suwet'en people living there (Wilson and Rabnett 2007). Anadromous lamprey passing through and spawning in the upper Bulkley River were traditionally also an important food source for the Wet'suwet'en (Gottesfeld and Rabnett (2007); pers comm. Mike Ridsdale, Environmental Assessment Coordinator, Office of the Wet'suwet'en). Gottesfeld and Rabnett (2007) report sourceing information from Department of Fisheries and Oceans (1991) that principal spawning areas for chinook in the Neexdzii Kwah include the mainstem above and below Buck and McQuarrie Creeks, between Cesford and Watson Creeks, and the reaches upstream and downstream of Bulkley Falls.

Renowned as a world class recreational steelhead and coho fishery, the greater Bulkley River receives some of the heaviest angling pressure in the province. In response to longstanding angler concerns with respect to overcrowding, quality of experience and conflict amongst anglers, an Angling Management Plan was drafted for the river following the initiation of the Skeena Quality Waters Strategy process in 2006 and an extensive multi-year consultation process. The plan introduces a number of regulatory measures with the intent to provide Canadian resident anglers with quality steelhead fishing opportunities. Regulatory measures introduced with the Angling Management Plan include prohibited angling for non-guided non-resident aliens on Saturdays and Sundays, Sept 1 - Oct 31 within the Bulkley River, angling prohibited for non-guided non-resident aliens on Saturdays and Sundays, all year within the Suskwa River and angling prohibited for non-guided non-resident aliens Sept 1 - Oct 31 in the Telkwa River. The Neexdzii Kwah is considered Class II water and there is no fshing permitted upstream of the Morice/Bulkley River Confluence (FLNRO 2013a, 2013b; FLNRORD 2019).

2.11.1.1 Upper Bulkley Falls

A detailed field assessment and write up regarding the upper Bulkley falls was conducted as part of fish passage restoration in the watershed - is presented in Irvine (2021) with a condensed summary here. The site was assessed on October 28, 2021 by Nallas Nikal, B.i.T, and Chad Lewis, Environmental Technician. The top of the falls is located at 11U.678269.6038266 at an elevation of 697m approximatley 11.3km downstream of Bulkley Lake and upstream of Ailport Creek. Within the Bulkley River immediately below the 12 - 15m high bedrock falls, the channel width was 17.4m and the wetted width was 15.6m. Two channels comprised the falls. The primary channel was 20m long, had a channel/wetted width of 8.5m, a 16% grade and water depths ranging from 35 - 63cm. The secondary channel was 25m long, with channel/wetted widths of 7.5m, a grade of 12% and water depths ranging from 3 - 13cm (Irvine 2021).

Dyson (1949) and Stokes (1956) report substantial use of habitat above Bulkley Falls by steelhead, chinook, coho and sockeye utilization in the past (pre-1950) based on spawning reports. Both authors concluded that the Bulkley Falls pose a partial obstruction to migrating fish based on flow levels. Chinook, which migrate early in the summer when water levels are high, have been noted as able to ascend the falls in normal to high water years and in high water years it was thought that coho and steelhead could ascend. Gottesfeld and Rabnett (2007) report that the falls are almost completely impassable to all salmon during low water flows. Stokes (1956) reports that there was high value spawning habitat located within the first 3km of the Neexdzii Kwah from the outlet of Bulkley Lake.

Wilson and Rabnett (2007) reported that approximately 11.3 km downstream of the Bulkley Lake outlet and just upstream of Watson Creek, the upper Bulkley falls is an approximately 4m high narrow rock sill that crosses the Neexdzii Kwah, producing a steep cascade section. This obstacle to fish passage is recorded as an almost complete barrier to fish passage for salmon during low water flows. Wilson and Rabnett (2007) also reported that coho have not been observed beyond the falls since 1972.

2.11.2 Morice River

Detailed reviews of Morice River watershed fisheries can be found in Bustard and Schell (2002), Gottesfeld et al. (2002), Schell (2003), Gottesfeld and Rabnett (2007), and ILMB (2007) with a comprehensive review of water quality by Oliver (2018) Overall, the Morice watershed contains high fisheries values as a major producer of chinook, pink, sockeye, coho and steelhead.

2.11.3 Zymoetz River

Within the Zymoetz Watershed, there are many areas with high fishery values. Steelhead are the most extensively documented fish species in the Zymoetz River watershed. Adults enter the river from July to November and then go on to spawn the following year in late spring to early summer. The Zymoetz River is a relatively steep system. Two canyons are located 6.4 and 19.6 kilometers upstream of the Skeena River confluence. These canyons make access to the Zymoetz difficult for pink and chum salmon (Gottesfeld et al. 2002). The Zymoetz River is renowned for its aggressive steelhead that have been known to take flies or lures. There is a 50km stretch upstream of Limonite Creek that's very remote and offers high quality fishing opportunities for anglers (FLNRORD 2013).

Traditional First Nations use of the upper Zymoetz River watershed by the Gitxsan and Wet'suwet'en people differed between community sites, residences, and fish houses, and was large and diverse. From the upper to lower Zymoetz River and to the Skeena River, a significant ancient grease trail connected, with a branch track forking through Limonite Creek and flowing down the Telkwa River. The fishery used a weir at the mouth of McDonell Lake and spears at Six Mile Flats, near Dennis Lake. There is no information on native fisheries on the lower Zymoetz River. The Zymoetz is considered to be one of the top ten steelhead rivers in BC (Gottesfeld et al. 2002).

2.11.4 Kispiox River

Kispiox River salmon are a important food source and cultural symbol for the Gitxsan people with sockeye and coho historically the two most significant species. Gitangwalk and Lax Didax, two significant villages that were both abandoned in the early 1900s, were situated on the Kispiox in such a way as to block the sockeye and coho salmon's upstream migration to the Upper Kispiox River spawning grounds providing opportunities to gather and preserve a significant amount of high-quality food over relatively short time periods (Gottesfeld et al. 2002). The 100 km of mainstem and 300km of tributary streams in the Kispiox River Watershed are considering high value fish habitat supporting migration, spawning and rearing for many fish species. The Kispiox fisheries supports both recreational and commercial fishing while also enhancing the ecology, nutrient regime, and structural diversity of the drainage. Since 1992, sockeye and coho escapements from the Kispiox Watershed have been documented by the Gitxsan Watershed Authorities as they creates strong cultural, economic, and symbolic ties for the local communities (Gottesfeld et al. 2002).

2.11.5 Kitsumkalum River

The Kitsumkalum River is an important waterway for all species of salmon. It is one of the three main chinook producing rivers in the Skeena watershed and supports all five species of pacific salmon, steelhead, and other resident trout and char species (McElhanney 2022). Most notably, the Kitsumkalum River has consistently produced the largest-bodied chinook in the Skeena Watershed, as well as on most of the Pacific coast (Gottesfeld and Rabnett 2007). The watershed supports strong recreational coho, steelhead, and chinook fishing. Kitsumkalum salmon also play an important role in the culture and economy of the Kitsumkalum Band (Gottesfeld and Rabnett 2007).

2.11.6 Salmon Stock Assessment Data

Fisheries and Oceans Canada stock assessment data for the study area was accessed via the NuSEDS-New Salmon Escapement Database System through the Open Government Portal (Fisheries and Oceans Canada, n.d.). Records are presented in Appendix - Salmon Stock Assessment Data (page 51).

2.11.7 Fish Species

Fish species recorded in the study area watershed groups are detailed in Appendix - Fish Species (page 53) (MoE 2024a). Coastal cutthrout trout and bull trout are considered of special concern (blue-listed) provincially. Summaries of some of the Skeena fish species life history, biology, stock status, and traditional use are documented in Schell (2003), Wilson and Rabnett (2007), Gottesfeld et al. (2002) and Office of the Wet'suwet'en (2013). Wilson and Rabnett (2007) discuss chinook, pink, sockeye, coho, steelhead and indigenous freshwater Bulkley River fish stocks within the context of key lower and upper Bulkley River habitats such as the Suskwa River, Station Creek, Harold Price Creek, Telkwa River and Buck Creek. Key areas within the upper Bulkley River watershed with high fishery values, documented in Schell (2003), are the upper Bulkley mainstem, Buck Creek, Dungate Creek, Barren Creek, McQuarrie Creek, Byman Creek, Richfield Creek, Johnny David Creek, Aitken Creek and Emerson Creek.

Some key areas of high fisheries values for chinook, sockeye and coho are noted in Bustard and Schell (2002) as McBride Lake, Nanika Lake, and Morice Lake watersheds. A draft gantt chart for select species in the Morice River watershed were derived from reviews of the aforementioned references and is included as Figure 2.8. The data is considered in draft form and will be refined over the spring and summer of 2021 with local fisheries technicians and knowledge holders during the collaboratory assessment planning and fieldwork activities planned.

In the 1990's the Morice River watershed, Gottesfeld and Rabnett (2007) estimated that chinook comprised 30% of the total Skeena system chinook escapements. It is estimated that Morice River coho comprise approximatley 4% of the Skeena escapement with a declining trend noted since the 1950 in Gottesfeld and Rabnett (2007). Coho spawn in major tributaries and small streams ideally at locations where downstream dispersal can result in seeding of prime off channel habitats including warm productive sloughs and side channels. Of all the salmon species, coho rely on small tributaries the most (Bustard and Schell 2002). Bustard and Schell (2002) report that much of the distribution of coho into non-natal tributaries occurs during high flow periods of May - early July with road culverts blocking migration into these habitats.

Summaries of historical fish observations in the Bulkley River and Morice River watershed groups (n=4033), graphed by remotely sensed average gradient as well as measured or modelled channel width categories for their associated stream segments where calculated with `bcfishpass` and `bcfishobs` and are provided in Figures 2.9 - 2.10.

Fish species recorded in the study area watershed groups are presented in Appendix - Fish Species (page 53).

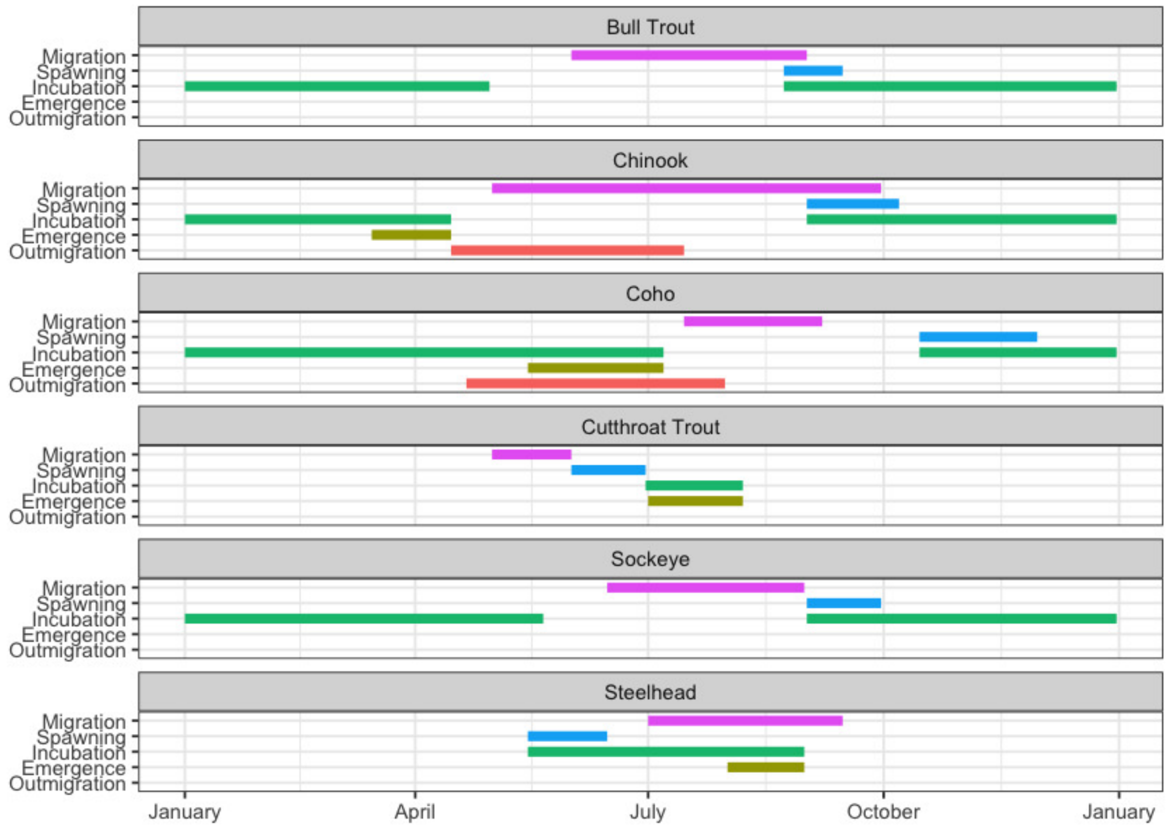

Figure 2.8: Gantt chart for select species in the Morice River watershed. To be updated in consultation with local fisheries techicians and knowledge holders.

2.11 Fisheries

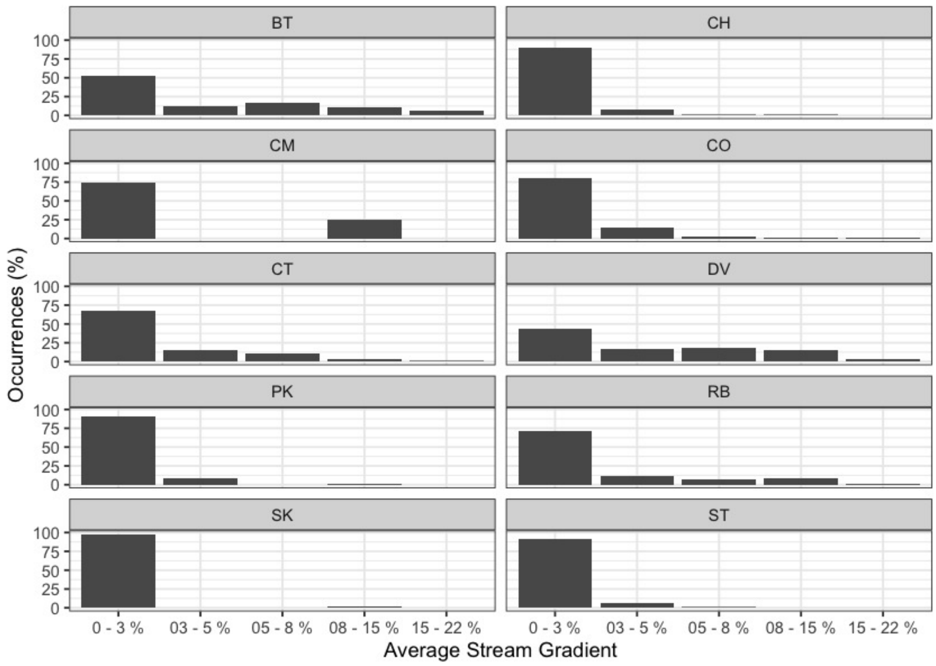

Figure 2.9: Summary of historic salmonid observations vs. stream gradient category for the Bulkley River watershed group.

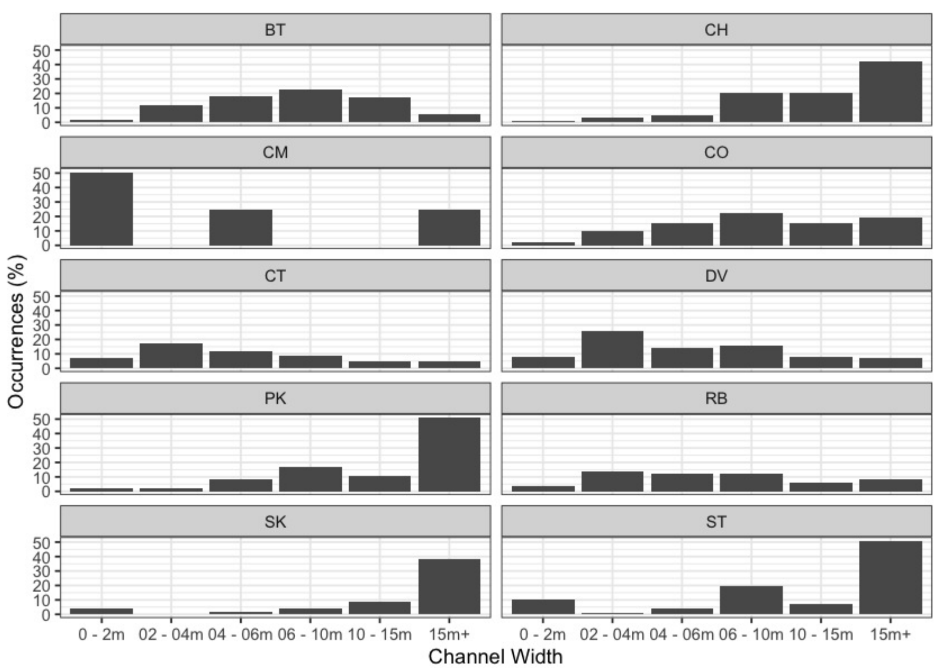

Figure 2.10: Summary of historic salmonid observations vs. channel width category for the Bulkley River watershed group.

3 Methods

3.1 Climate Departure

Climate departure measures how far recent climate has shifted from a historical baseline — in this case, how much warmer, drier, or snowpack-altered the study area has become relative to the pre-warming period (1951–1980). Where traditional fish-passage assessments focus on the physical structures in and around streams, climate departure adds the atmospheric and hydrologic context those structures sit inside: warming air temperatures that drive warmer water, shifting snowpack that reshapes the freshet, and atmospheric drying that reduces late-summer flows. Understanding these departures at the watershed scale lets us ask whether the habitat above a barrier is gaining or losing value as climate changes — and weight barrier prioritisation accordingly.

We examined 15 climate variables spanning temperature (annual mean, daytime maximum, overnight minimum), moisture (precipitation, vapour pressure deficit, soil moisture, relative humidity), and snowpack (snow water equivalent, snowfall, snowmelt, snow cover, snowfall fraction, peak snow water equivalent, snowmelt timing, and peak melt rate). Each variable was assessed at annual and seasonal scales, across the full study area and within each of its ecoregions.

Climate departures were computed with the cd R package, which reads ERA5-Land hourly reanalysis (Muñoz Sabater 2019) (1950–present, ~9 km native grid) aggregated to monthly, seasonal, and annual periods on a single grid covering British Columbia, served as Cloud-Optimised GeoTIFFs from a public S3 catalog. For each variable the package crops the published rasters to the area of interest, takes a yearly spatial mean, and computes anomalies against the 1951–1980 pre-warming reference period — the same base period Hansen et al. (2012) used for global temperature emergence. Long-term trends are tested with Mann-Kendall significance and quantified with Theil-Sen slopes; the recent-decade (2015–2025) versus pre-warming window comparison uses a Welch two-sample t-test on the annual values. The same pipeline runs at any polygon scale — region, ecoregion, or watershed group. The regional and ecoregion-level analyses are summarised in the results; the watershed-group breakdown is carried through to Appendix - Climate Departure (page 57), which holds the full set of plots and tables.

3.2 Planning — Habitat, Connectivity and Floodplain Modelling

3.2.1 Habitat Modelling

Habitat modelling used to help guide planning for field assessments is generated by `bcfishpass` (Norris [2020] 2024) which has been designed to prioritize potential fish passage barriers for assessment or remediation by generating a simple model of aquatic habitat connectivity. We utilize the `bcfishpass` access model, linear spawning/rearing habitat model and lateral habitat connectivity for planning purposes. These models provide a valuable starting point, but their results are not definitive and should always be considered with professional judgment. Detailed information regarding model methodology, select parameters and known model limitations are detailed in Norris ([2020] 2024) with key documentation linked below:

- Access model
- Linear spawning/rearing habitat models
- Lateral habitat model

Table 3.1 documents the custom species-specific thresholds for stream gradient and channel width applied to the linear spawning and rearing habitat model for this year's project planning.

Table 3.1: Stream gradient and channel width thresholds used to model potentially highest value fish habitat.

Variable	Chinook Salmon	Chum Salmon	Coho Salmon	Pink Salmon	Sockeye Salmon	Steelhead
Spawning Gradient Max (%)	4.5	6.5	5.5	6.5	2.5	4.5
Spawning Width Min (m)	4.0	2.1	2.0	2.1	2.0	4.0
Rearing Width Min (m)	1.5	–	1.5	–	1.5	1.5
Rearing Gradient Max (%)	5.5	–	5.5	–	–	8.5

*

3.2.2 Floodplain Delineation

To complement the linear barrier inventory, we mapped functional floodplain extent at watershed scale for the Bulkley River Watershed Group. Overlaying floodplain footprint on the crossing inventory lets us ask not just whether fish can pass a structure, but how much off-channel habitat sits within the affected corridor and whether infrastructure has compromised the lateral connectivity that sustains it.

The delineation uses the `flooded` R package (Nagel et al. 2014), which models the area a river actively inundates during regular high-flow events — the **functional floodplain**. This is distinct from the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain is the scale where off-channel habitat is created and maintained: where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Bankfull depth is estimated from the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams. Inputs to the published delineation were the MRDEM-30 30 m DEM, a coho accessible stream network of stream order ≥ 3 built with `link` over the BC Freshwater Atlas, and Freshwater Atlas lakes and wetlands on that network.

The modelled result is not regenerated for this report. Delineations are published once to the `stac-floodplains-bc` collection — a SpatioTemporal Asset Catalog (STAC) served at https://images.a11s.one through `stac-fastapi-pgstac` and queryable with `rstac` (Radiantearth 2024; stac-utils 2025; Simoes et al. 2021), the same catalogue that serves the project's aerial imagery (Section 3.5.4 (page 36)). The publishing pipeline, the current coverage, and instructions for querying the collection from `rstac` or QGIS are documented at https://github.com/NewGraphEnvironment/stac_floodplains_bc. This report reads the published item and caches it, rather than re-running the model or committing its own copy of the rasters. Doing it this way means one modelled product serves every report that draws on it, and the reports cannot drift from each other or from the catalogue.

Items are identified as `<watershed group>_<species>_ff<nn>`, recording the watershed group, the species-accessible network the delineation was built on, and the flood factor. This report uses `bulk_co_ff04`. Full methodology and watershed-scale mapping are in Appendix - Floodplain Delineation (page 81).

3.3 Statistical Support for Habitat Modelling

The statistical-modelling work described in this section underpins our intrinsic-habitat-potential and connectivity-prioritization outputs. It **extends** — rather than supersedes — the canonical BC freshwater modelling stack (`fwapg` for the Freshwater Atlas in PostgreSQL, `bcfishpass` for habitat and barrier modelling, and `bcfishobs` for fish-observation records). Our role is downstream: project-specific parameterization of habitat thresholds, integration of water-temperature and water-quality covariates, and calibration of model predictions against observed fish-presence evidence.

3.3.1 Channel Width

Channel width is a primary determinant of habitat suitability in `bcfishpass`. In early 2021, Bayesian statistical methods were developed to predict channel width across BC Freshwater Atlas stream segments where field measurements were unavailable. The original model related channel width to upstream watershed area and watershed-area-weighted mean annual precipitation (Thorley and Irvine 2021). In December 2021 the methodology was updated using the power-law form derived by Finnegan et al. (2005), relating stream discharge to watershed area and precipitation; the updated model (Thorley et al. 2021) is the channel-width estimator used in `bcfishpass` at the time of reporting. Detailed documentation of the data-collection and statistical-analysis methodology is given in Irvine ([2021] 2022) and Thorley et al. (2021).

3.3.2 Stream Temperature and Growing-Season Degree Days

Water temperature drives productivity for cold-water salmonids — too cold and growth stalls; too warm and thermal stress reduces survival. We work with Poisson Consulting on Bayesian Spatial Stream Network (SSN) models that predict stream temperature at fine spatial resolution across the FWA network, using a hybrid `air2stream` formulation that bridges fully mechanistic models (data-intensive, hard to parameterize) and purely statistical approaches (no physical justification, extrapolate poorly) (Toffolon and Piccolroaz 2015). The 2024 Nechako modelling effort settled on a four-parameter `air2stream` with site-level random effects (Hill et al. 2024).

The 2025 modelling effort was carried out in this study area. It extended the approach to a larger spatial network and added Growing-Season Degree Days (GSDD) — accumulated thermal units during the season when stream temperature remains at or above 5 °C — as the productivity-relevant metric for age-0 salmonid growth (Hill et al. 2025). Both studies are linked in Attachment - Water Temperature Modelling (page 125).

Stream-temperature and discharge observations are now organized centrally in `water-temp-bc` (Irvine [2025] 2025b) — a monthly-automated ingestion pipeline that publishes Environment Canada hydrometric and water-quality records as a partitioned-parquet layer on Amazon S3. Stations metadata, observations by parameter (water temperature, discharge, water level), and per-snapshot dedupe (canonical value retained at the latest `harvested_at` timestamp) are queryable via the package's `query_canonical()` helper without database setup. This replaces the per-project Zenodo and ECCC downloads used in earlier modelling work — one accessible source for any subsequent stream network or temperature-suitability analysis in BC.

3.4 Fish Passage Assessments

3.4.1 Natural Barriers to Fish Passage

Our assessments may include natural features such as waterfalls that could limit fish passage. This informs whether upstream culvert upgrades would restore access for anadromous species (e.g., salmon) or primarily benefit resident fish already upstream. We document these features by measuring height, gradient, and pool depth, recording field observations, capturing site photographs, and reviewing background sources for context.

3.4.2 Road Stream Crossings

In the field, crossings prioritized for follow-up were first assessed for fish passage following the procedures outlined in “Field Assessment for Determining Fish Passage Status of Closed Bottomed Structures” (MoE 2011). The reader is referred to (MoE 2011) for detailed methodology. Crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered “other” (i.e. fords). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any other relevant features. The following information was recorded for all surveyed crossings: date of inspection, crossing reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for OBS. A more detailed “full assessment” was completed for all closed bottom structures and included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, road fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.2).

Table 3.2: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	Fish Habitat Criteria
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

Fish passage potential was determined for each stream crossing identified as a closed bottom structure as per MoE (2011). The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier to some fish species and life stages (Table 3.3, Table 3.4). These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al. 2005; Bell 1991; Thompson 2013). For crossings determined to be potential barriers or barriers based on the data, a culvert fix and recommended diameter/span was proposed.

Table 3.3: Fish Barrier Risk Assessment (MoE 2011).

Risk	LOW	MOD	HIGH
Embedded	>30cm or >20% of diameter and continuous	<30cm or 20% of diameter but continuous	No embedment or discontinuous
Value	0	5	10
Outlet Drop (cm)	<15	15-30	>30
Value	0	5	10
SWR	<1.0	1.0-1.3	>1.3
Value	0	3	6
Slope (%)	<1	1-3	>3
Value	0	5	10
Length (m)	<15	15-30	>30
Value	0	3	6

Table 3.4: Fish Barrier Scoring Results (MoE 2011).

Cumlative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project, a gradient threshold between accessible and non-accessible habitat was set at 20% (for a minimimum length of 100m) intended to represent the maximum gradient of which the strongest swimmers of anadromous species (steelhead) are likely to be able to migrate upstream.

For reporting of Phase 1 - fish passage assessments within the body of this report (Table 3.3), a “total” value of habitat <20% output from `bcfishpass` was used to estimate the amount of habitat upstream of each crossing less than 20% gradient before a falls of height >5m - as recorded in MoE (2024b) or documented in other `bcfishpass` online documentation. For Phase 2 - habitat confirmation sites, conservative estimates of the linear quantity of habitat to be potentially gained by fish passage restoration, steelhead rearing maximum gradient threshold (8.5%) was used. To generate estimates for area of habitat upstream (m^2), the estimated linear length was multiplied by half the downstream channel width measured (overall triangular channel shape) as part of the fish passage assessment protocol. Although these estimates are not generally conservative, have low accuracy and do not account for upstream stream crossing structures they allow a rough idea of the best candidates for follow up.

Potential options to remediate fish passage were selected from MoE (2011) and included:

- Removal (RM) - Complete removal of the structure and deactivation of the road.
- Open Bottom Structure (OBS) - Replacement of the culvert with a bridge or other open bottom structure. Based on consultation with FLNR road crossing engineering experts, for this project we considered bridges as the only viable option for OBS type .
- Streambed Simulation (SS) - Replacement of the structure with a streambed simulation design culvert. Often achieved by embedding the culvert by 40% or more. Based on consultation with FLNR engineering experts, we considered crossings on streams with a channel width of <2m and a stream gradient of <8% as candidates for replacement with streambed simulations.
- Additional Substrate Material (EM) - Add additional substrate to the culvert and/or downstream weir to embed culvert and reduce overall velocity/turbulence. This option was considered only when outlet drop = 0, culvert slope <1.0% and stream width ratio < 1.0.
- Backwater (BW) - Backwatering of the structure to reduce velocity and turbulence. This option was considered only when outlet drop < 0.3m, culvert slope <2.0%, stream width ratio < 1.2 and stream profiling indicates it would be effective..

3.4.3 Cost Estimates

Cost estimates for structure replacement with bridges and embedded culverts were generated based on the channel width, slope of the culvert, depth of fill, road class and road surface type. Road details were sourced from FLNRORD (2020b) and FLNRORD (2020a) through `bcfishpass`. Interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay, Steve Page, Area Engineer - FLNR - Northern Engineering Group and Matt Hawkins - MoTi - Design Supervisor for Highway Design and Survey - Nelson were utilized to help refine estimates which have since been adjusted for inflation in 2020 and based on past experience.

Base costs for installation of bridges on forest service roads and permit roads with surfaces specified in provincial GIS road layers as rough and loose was estimated at $30000/linear m and assumed that the road could be closed during construction and a minimum bridge span of 15m. For streams with channel widths <2m, embedded culverts were reported as an effective solution with total installation costs estimated at $100k/crossing (pers. comm. Phil MacDonald, Steve Page then adjusted for inflation in 2020). For larger streams (>6m), estimated span width increased proportionally to the size of the stream. For crossings with large amounts of fill (>3m), the replacement bridge span was increased by an additional 3m for each 1m of fill >3m to account for cutslopes to the stream at a 1.5:1 ratio. To account for road type, a multiplier table was generated to estimate incremental cost increases with costs estimated for structure replacement on paved surfaces, railways and arterial/highways costing up to 15 times more than forest service roads due to expenses associate with design/engineering requirements, traffic control and paving. The cost multiplier table (Table 3.5) should be considered very approximate with refinement recommended for future projects.

Table 3.5: Cost multiplier table based on road class and surface type.

Class	Surface	Class Multiplier	Surface Multiplier	Bridge $/15m	Streambed Simulation $
FSR	Rough	1	1	450,000	100,000
FSR	Loose	1	1	450,000	100,000
Resource	Loose	1	1	450,000	100,000
Collector	Loose	1	1	450,000	100,000
Recreation	Loose	1	1	450,000	100,000
Resource	Rough	1	1	450,000	100,000
Permit	Unknown	1	1	450,000	100,000
Permit	Loose	1	1	450,000	100,000
Permit	Rough	1	1	450,000	100,000
Unclassified	Loose	1	1	450,000	100,000
Unclassified	Rough	1	1	450,000	100,000
Unclassified	Paved	1	2	750,000	150,000
Unclassified	Unknown	1	2	750,000	150,000
Local	Loose	4	1	1,500,000	200,000
Local	Paved	4	2	3,000,000	400,000
Collector	Paved	4	2	3,000,000	400,000
Arterial	Paved	15	2	11,250,000	1,500,000
Highway	Paved	15	2	11,250,000	1,500,000
Rail	Rail	15	2	11,250,000	1,500,000

3.4.4 Climate Change Risk Assessment

In collaboration with the Ministry of Transportation and Infrastructure (MoTi), a new climate change replacement program aims to prioritize vulnerable culverts for replacement (pers. comm Sean Wong, 2022) based on data collected and ranked related to three categories - culvert condition, vulnerability and priority. Within the “condition” risk category - data was collected and crossings were ranked based on erosion, embankment and blockage issues. The “climate” risk category included ranked assessments of the likelihood of both a flood event affecting the culvert as well as the consequence of a flood event affecting the culvert. Within the “priority” category the following factors were ranked - traffic volume, community access, cost, constructability, fish bearing status and environmental impacts (Table 3.6). This project is still in its early stages with methodology changes going forward.

Table 3.6: Climate change data collected at MoTi culvert sites

Parameter	Description
erosion_issues	Erosion (scale 1 low - 5 high)
embankment_fill_issues	Embankment fill issues 1 (low) 2 (medium) 3 (high)
blockage_issues	Blockage Issues 1 (0-30%) 2 (>30-75%) 3 (>75%)
condition_rank	Condition Rank = embankment + blockage + erosion
condition_notes	Describe details and rational for condition rankings
likelihood_flood_event_affecting_culvert	Likelihood Flood Event Affecting Culvert (scale 1 low - 5 high)
consequence_flood_event_affecting_culvert	Consequence Flood Event Affecting Culvert (scale 1 low - 5 high)
climate_change_flood_risk	Climate Change Flood Risk (likelihood x consequence) 1-6 (low) 6-12 (medium) 10-25 (high)
vulnerability_rank	Vulnerability Rank = Condition Rank + Climate Rank
climate_notes	Describe details and rational for climate risk rankings
traffic_volume	Traffic Volume 1 (low) 5 (medium) 10 (high)
community_access	Community Access - Scale - 1 (high - multiple road access) 5 (medium - some road access) 10 (low - one road access)
cost	Cost (scale: 1 high - 10 low)
constructability	Constructibility (scale: 1 difficult -10 easy)
fish_bearing	Fish Bearing 10 (Yes) 0 (No) - see maps for fish points
environmental_impacts	Environmental Impacts (scale: 1 high -10 low)
priority_rank	Priority Rank = traffic volume + community access + cost + constructability + fish bearing + environmental impacts
overall_rank	Overall Rank = Vulnerability Rank + Priority Rank
priority_notes	Describe details and rational for priority rankings

3.5 Habitat Confirmation Assessments

Following fish passage assessments, habitat confirmations were completed in accordance with procedures outlined in the document “A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation of a Stream Crossing” (Fish Passage Technical Working Group 2011). The main objective of the field surveys was to document upstream habitat quantity and quality and to determine if any other obstructions exist above or below the crossing. Habitat value was assessed based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation. Criteria used to rank habitat value was based on guidelines in Fish Passage Technical Working Group (2011) (Table 3.2).

During habitat confirmations, to standardize data collected and facilitate submission of the data to provincial databases, information was collected on digital field forms adapted from provincial “Site Cards”. Habitat characteristics recorded included channel widths, wetted widths, residual pool depths, gradients, bankfull depths, stage, temperature, conductivity, pH, cover by type, substrate and channel morphology (among others). When possible, the crew surveyed downstream of the crossing to a minimum distance 300m and upstream to a minimum distance of 500 - 600m. Any potential obstacles to fish passage were inventoried with photos, physical descriptions and locations recorded on site cards. Surveyed routes were recorded with time-signatures on handheld GPS units.

3.5.1 Fish Sampling

3.5.1.1 Electrofishing

Fish sampling was conducted on a subset of sites when biological data was considered to add significant value to the physical habitat assessment information. Electrofishing was utilized for fish sampling according to stream inventory standards and procedures found in the Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Manual (Resource Inventory Standards Committee 2001). A Haltech 2000 backpack electrofisher was used within discrete site units both upstream and downstream of the subject crossing with electrofisher settings and seconds, water quality parameters (i.e. conductivity, temperature and ph), start and end locations, length of site and wetted widths (average of a minimum of three) recorded.

3.5.1.2 Fish Handling and Processing

Captured fish were held in buckets with sufficient water to minimize stress until processing, and multiple buckets were used when catch numbers were high. For each fish captured, fork length, weight and species was recorded with results documented in the fish data submission spreadsheet.

3.5.1.3 Pit Tagging

Fish with a fork length greater than 60 mm and belonging to species approved under the scientific fish collection permit SM25-984576 were tagged with Passive Integrated Transponders (PIT tags) using the Abdominal Cavity method outlined by Biomark. To anesthetize fish prior to pit tagging, we used a solution of approximately 0.1 mL of clove oil per 1 L of water (1:10,000). This concentration was selected for its efficiency in providing effective sedation with minimal residual effects, making it ideal for studies in which fish are released back into their natural habitats (Fernandes et al. 2017). The clove oil solution was prepared in advance by dissolving pure clove oil in ethyl alcohol in a 1:9 ratio (clove oil: ethyl alcohol) to enhance solubility, then mixed into the water bucket (Fernandes et al. 2017). Fish were immersed in this solution until they reached an appropriate level of anesthesia for handling and then were tagged. To maintain needle sharpness and minimize injury risk, needles were replaced approximately every 10 fish. Each tagged fish was scanned with the PIT reader, and both the PIT tag ID and row ID were recorded. Once tagged, fish were placed into a bucket of fresh water and allowed to recover before being released back into the stream. Fish information and habitat data will be submitted to the province under scientific fish collection permit SM25-984576.

3.5.2 Environmental DNA (eDNA) sampling

3.5.2.1 eDNA Sampling and Filtration

Environmental DNA (eDNA) samples were collected using two approaches: in-stream filtration with a Haltech EMOS eDNA sampler and a grab-and-go bottle collection method following Hobbs et al. (2025). For both approaches, all water collected at a site (typically 1–3L) was treated as a single sample and filtered onto one MCE filter, resulting in one eDNA replicate per site. Typically, one sample was collected upstream and one downstream of the road–stream crossing, although in some cases only a downstream sample was collected. Field blanks were collected at the start of each sampling day using distilled water.

In-stream filtration was completed using Haltech reusable aluminum housings pre-loaded with mixed cellulose ester (MCE) filters (0.45μm, 1μm, or 1.5μm pore sizes). When turbidity was high, a Haltech pre-filter equipped with a 100μm nitex screen was installed upstream of the MCE filter to remove coarse material and maintain filtration efficiency. After filtration, aluminum housings were returned to their labelled bags and stored in a cooler with ice until the end of the day. Each evening, filters were removed from housings in a clean environment, transferred into coin envelopes placed inside labelled ziplock bags with silica desiccant, and stored in a freezer until shipment. Aluminum housings were sanitized with a bleach solution, reloaded with new filters, and stored in clean ziplock bags for future use.

At sites where the grab-and-go method was used, stream water was collected in sanitized 1L HDPE Nalgene bottles. For detailed procedures, see Hobbs et al. (2025). Bottles were stored in a cooler with ice until the end of the day and then filtered through MCE filters identical to those used for the EMOS sampler. Post-filtration handling followed the same procedures as the in-stream method, with filters transferred to labelled bags with silica desiccant and frozen until shipment. Nalgene bottles were sanitized with a bleach solution after use.

All eDNA samples were shipped frozen to the University of Northern British Columbia (UNBC) laboratory for analysis.

3.5.2.2 eDNA Analysis

Samples were analysed at the UNBC Genetics Facility. Full laboratory methods, per-sample results and the laboratory's interpretation are in their technical report, available here. The points below are the ones needed to read the results in this report; the technical report is the authority on everything else.

Species-specific assays were run on a droplet digital PCR (ddPCR) platform, paired as duplex reactions with FAM and VIC reporter dyes. Each filter was extracted in two elutions and both were analysed. **A sample was called positive for a species where four or more positive droplets were detected.** Samples returning two or three droplets were retested against an additional portion of the elution volume; where the additional droplets did not bring the total to four or more, the sample was called negative. Results below the four-droplet threshold are reported here as not detected.

Exogenous lambda phage DNA and an ePlant assay were used to test for PCR inhibition and eDNA degradation in samples returning no detections. Where those tests indicated inhibition or degradation, elutions were combined and cleaned before retesting.

The bull trout assay does not distinguish Bull Trout from Dolly Varden, and results for that assay should be read as applying to either species.

3.5.3 Open Source Reporting Framework

This report is produced using open-source tools (R, bookdown) with full version control via git, enabling iterative updates. All code, data, and revision history are publicly accessible in the project repository, with ongoing tasks tracked via GitHub issues and version history maintained in the project changelog.

3.5.4 Aerial Imagery

Scripted processing and serving of UAV imagery collected during the project is available at https://github.com/NewGraphEnvironment/stac_uav_bc/ (Irvine [2025] 2025a). `OpenDroneMap` was utilized to produce orthomosaics, digital surface models (DSMs), and digital terrain models (DTMs) (OpenDroneMap Authors [2014] 2025). To support efficient web-based access - imagery products were converted to cloud-optimized GeoTIFFs (COGs) using `rio-cogeo`, then collated accordiong to the [SpatioTemporal Asset Catalog (STAC)] specification with `pystac` and uploaded to S3 storage Amazon Web Services (2025). A `titiler` tile server was set up to facilitate interactive viewing of the orthoimagery and an Application Program Interface (API) leveraging `stac-fastapi-pgstac` is served at https://images.a11s.one to enable linking of collection images through QGIS as well as remote spatial and temporal querying using open source software such as `rstac`(Development Seed [2019] 2025; stac-utils 2025; Simoes et al. 2021).

3.6 Engineering Design

Engineering designs were signed and sealed by professional engineers. When possible - completed designs are loaded to the PSCIS data portal.

3.7 Remediations

Structure replacement was conducted by project specific contractors. If not already completed, as-built drawings will be loaded to the PSCIS data portal.

3.8 Monitoring

Monitoring of fish passage restoration sites — both proposed and completed - is essential to ensure restoration investments lead to meaningful ecological outcomes. Monitoring enables evaluation of whether remediation actions improve connectivity for fish and provides critical feedback to refine future prioritization and restoration strategies.

Baseline data collection, including fish sampling and aerial surveys via drone, are core components of this monitoring. While detailed methods for these activities are included in the previous habitat confirmation section, they are also fundamental to effectiveness monitoring, as they provide context to not only facilitate prioritization and communications but also for detecting change following restoration.

To support consistent and targeted assessment, a custom field form was developed for routine effectiveness monitoring based loosely on Forest Investment Account (2003) but tailored specifically for fish passage projects. Table 3.7) outlines the monitoring metrics used.

Table 3.7: Description of monitoring metrics used for effectiveness monitoring.

Parameter	Description
Dewatering	Have the remediation works led to dewatering of the channel due to substrate aggradation or other factors?
Velocity	Are flow velocities similar to those within the natural channel? Are they expected to exceed swim speeds of particular fish species/life stages of interest?
Constriction	Have the remediation works led to constriction of the channel. Compare channel width underneath structure and within construction footprint to average channel widths upstream and downstream?

Parameter	Description
Substrate	Is the substrate within/under and adjacent to the remediated structure generally equivalent to that found upstream and downstream where natural channel conditions exist?
Riparian	What is the condition of the riparian area within the construction footprint?
UAV Flight	Was a flight conducted with unmmanned aerial vehicle to document conditions at time of monitoring?
Flow_depth	What are the flow depths at the time of assessment within project footprint. Are depths expected to be sufficient to facilitat upstream passage for specific species/life stages of interest?
Stability	Does the structure appear to be stable or is there evidence of erosion/shifting?
Revegetation	How were riparian areas rehabillitated and are they improving fish habitat value?
Cover	Is cover available for fish within the construction footprint in the form of overhanging vegetation, large/small woody debris, boulders, undercut banks, etc?
Maintenance	If required, provide maintenance recommendations.
Recommendations	General recommendations for follow up. Could include revegetation, addition of substrate, fish sampling, etc.

3.9 Collaborative GIS Environment

Geographical Information Systems are essential for developing and communicating restoration plans as well as the reasons they are required and how they are developed. Without the ability to visualize the landscape and the data that is used to make decisions, it is difficult to conduct and communicate the need for restoration, the details of past and future plans as well as and the potential results of physical works.

To facilitate the planning and implementation of restoration activities, a collaborative GIS environment has been established using QGIS and is served on the cloud using source code stored here. This environment is intended to be a space where project team members can access, view, and contribute to the amalgamation of background spatial data and the development of restoration as well as monitoring for the project. The collaborative GIS environment allows users to view, edit, and analyze shared, up to date spatial data on personal computers in an office setting as well as on phones and tablets in the field. At the time of reporting, the environment was being used to develop and share maps, conduct spatial analyses, communicate restoration plans to stakeholders as well as to provide a central place to store methodologies and tools for conducting field assessments on standardized pre-developed digital forms. The platform can also be used to track the progress of restoration activities and monitor changes in the landscape over time, helping encourage the record keeping of past and future restoration activities in a coordinated manner.

The shared QGIS project was created using scripts currently kept in `dff-2022` with the precise calls to project creation scripts tracked in the `project_creation_and_permissions.txt` document kept in the main QGIS project directory. Information about the scripts used for GIS project creation and updates can be viewed here with outcomes of their use summarized below:

watershed groups and load to a geopackage called `background_layers.gpkg` stored in the main directory of the project.

- A project directory is created to hold the spatial data and `QGIS` project information (ie. layer symbology and naming conventions, metadata, etc.).
- Metadata for individual project spatial layers is kept in the `rfp_tracking` table within the `background_layers.gpkg` along with tables related to user supplied stream width/gradient inputs to `bcfishpass` to model potentially high value habitat that is accessible to fish species of interest.

3.9.1 Mapping

The workflows to produce the georeferenced `pdf` maps include using a QGIS layer file defining and symbolizing all layers required and are continuously evolving. At the time of reporting - mapping scripts and associated layer file were kept under version control within `bcfishpass` here. Loading the QGIS layer file within a QGIS project, allows load and representation of all map component layers provided the user points to a postgresql database populated via `bcfishpass` outputs.

4 Results and Discussion

4.1 Site Assessment Data

A summary of fish passage assessment procedures conducted through SERNbc since 2020 — including assessments, habitat confirmations, design, remediation, monitoring, fish sampling, eDNA sampling, and drone imagery — is provided in Appendix - Assessment Data Summary (page 87).

Orthoimagery and elevation model rasters generated since 2020 are catalogued in Appendix - UAV Imagery (page 89).

4.2 Climate Departure

Four findings from the regional climate-departure analysis carry to fish-passage prioritisation.

Warming is broad and significant region-wide. All six ecoregions of the study area warmed about 2.1 °C over the recent decade relative to the pre-warming reference — annual mean (+2.1 °C), daytime maximum (+1.9 °C), and overnight minimum (+2.2 °C) all tell the same story (Mann-Kendall $p < 0.001$ in every ecoregion). The rate has been steady rather than accelerating — the 1981–present slope (0.026 °C/yr) is shallower than the 1951–present slope (0.035 °C/yr). What is notable here is how *little* the signal varies across the study area: cumulative warming since 1951 spans only 0.28 °C between the most and least affected ecoregion, so the warming applies to every watershed group alike and the thermal question falls to elevation and reach scale — see Appendix - Climate Departure (page 57) Figure 5.11.

The atmosphere is drying despite flat precipitation. Vapour pressure deficit rose 0.34 hPa region-wide, significant in every ecoregion. Annual precipitation rose +2 % regionally but the long-term trend is not statistically significant in any ecoregion; soil moisture is essentially flat because the warmer atmosphere is drinking the modest precipitation increase back through evaporation and transpiration.

The snowpack signal is about timing, and about volume. Total annual snowfall is essentially unchanged (-8 %, not significant), but annual snow water equivalent is down 20 % and that decline *is* significant ($p = 0.007$). The sharpest signals are on the summer side and at the midpoint: summer SWE collapsed 60 % and the snowmelt midpoint moved 15 days earlier ($p < 0.001$ regionally and in every ecoregion). The counterbalancing rise is concentrated in spring, which gained 52 % and is highly significant; winter snowmelt rose a comparable 46 % but does not clear significance on its own. The freshet is arriving more than two weeks earlier and the high-elevation snowpack that historically lingered into summer no longer does.

Day-night asymmetry is present. Overnight minimums warmed 2.2 °C while daytime maximums warmed 1.9 °C — overnight outpacing daytime is the textbook fingerprint of greenhouse warming (Karl et al. 1993). Mean diurnal range has narrowed by 0.3 °C across the record.

For barrier prioritisation these signals cut both ways. Where upstream reaches are cold-limited, added growing-degree-days can raise the intrinsic potential of habitat above the barrier and *increase* the value of restoring passage; where upstream reaches sit closer to the upper edge of the cold-water thermal niche, the same warming reduces usable habitat and lowers the payoff. Because the departure is so even across this study area, which of those applies is an elevation and reach-scale question rather than a watershed-group one. The six-ecoregion signal is mapped onto the 5 watershed groups via a percentage crosswalk in Appendix - Climate Departure (page 57).

4.3 Planning — Habitat, Connectivity and Floodplain Modelling

4.3.1 Habitat Modelling

Habitat modelling from `bcfishpass` including access model, linear spawning/rearing habitat model and lateral habitat connectivity models for watershed groups within our study area were updated for the spring of 2025 and are included spatially in the collaborative GIS project. A snapshot of these outputs related to each modeled and PSCIS stream crossing structure are also included within an `sqlite` database within this year's project reporting/code repository here.

4.3.2 Statistical Support for Habitat Modelling

The statistical modelling underpinning these outputs — channel-width prediction, and Bayesian Spatial Stream Network modelling of stream temperature with Poisson Consulting (Hill et al. 2024; Hill et al. 2025) — is described in the Methods chapter. The 2025 effort was carried out in this study area, extending the approach to a larger network and adding Growing-Season Degree Days as the productivity-relevant thermal metric, with observations drawn from `water-temp-bc` (Irvine [2025] 2025b). Both studies are linked in Attachment - Water Temperature Modelling (page 125).

These predictions are the route by which thermal condition enters prioritization: reaches too cold to support production can be discounted from intrinsic habitat models, and reaches becoming warm enough to support it recognised as gaining value.

4.3.3 Floodplain Delineation

Modelled functional floodplain in the Bulkley River Watershed Group covers 48,344 ha — 6.2 % of the 7,762 km² watershed group — attached to 1,587 km of coho accessible order-3+ streams. Within that footprint sit 180 lakes and 345 mapped wetland polygons — the off-channel inventory that crossing remediations in these reaches stand to reconnect or leave severed, depending on whether design accounts for the floodplain context around the structure. The full metric table, watershed-scale map, and a detail map of the valley south of Smithers are in Appendix - Floodplain Delineation (page 81).

4.4 Fish Passage Assessments

Field assessments were conducted between September 29, 2025 and October 04, 2025 by Allan Irvine, R.P.Bio. The 2025 field season was directed at effectiveness monitoring and environmental DNA sampling rather than at new fish passage assessments. A total of 1 crossings were assessed, all of them reassessments of structures already inventoried in PSCIS. No Phase 1 assessments were completed.

Reassessments update the provincial record where a structure has been replaced or remediated, or where the existing PSCIS entry no longer reflects conditions at the crossing. They do not add new crossings to the inventory, so they do not carry the cost estimates and priority rankings that accompany Phase 1 results. The amalgamated assessment record for the study area, spanning 2020 to 2025, is presented in Appendix - Assessment Data Summary (page 87).

4.5 Habitat Confirmation Assessments

No Phase 2 habitat confirmation assessments were conducted in 2025. Habitat confirmations characterise the quantity and quality of habitat upstream and downstream of a crossing prioritised during Phase 1 assessment; with no Phase 1 work completed this season, none were scheduled.

Habitat confirmation results from previous years in the Bulkley River, Kispiox River, Kalum River, Morice River, and Zymoetz River watershed groups are carried in the amalgamated record in Appendix - Assessment Data Summary (page 87), and in the reporting for those years.

Environmental DNA (eDNA) Sampling

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 20 collection events were undertaken across 11 streams, comprising 19 real environmental samples, 1 field blanks, and 0 office blanks collected for quality assurance.

Per-species detection results across the 19 real environmental samples are summarized in Table 4.1. Detections use the four-droplet call threshold applied by the laboratory; results below that threshold, including two- and three-droplet results that were retested without reaching the cutoff, are reported as not detected. Laboratory methods and per-sample results are in the UNBC technical report, available here. Site-level detail — the per-site detection table, field blanks, and retests — is provided in Appendix - Environmental DNA Results (page 93), and the interactive eDNA map shows toggleable per-species layers.

Table 4.1: Per-species summary of eDNA results across real environmental samples in the project area (excludes field and office blanks).

Species	Code	Sites tested	Detected (≥4 droplets)	Not detected
Rainbow Trout	RAIN	13	9	4

Species	Code	Sites tested	Detected (≥4 droplets)	Not detected
Bull Trout	BULT	18	7	11
Coho Salmon	COHO	19	3	16
Chinook Salmon	CHIN	17	0	17
Sockeye Salmon	SOCK	9	0	9

4.6 Engineering Design

No new designs were commissioned in 2025 due to uncertainty related to forest harvesting activities and lack of funding for the 50% costs of replacing structures. All sites with a design for remediation of fish passage can be found online within the Assessment Data Summary appendix of the report found here.

4.7 Monitoring

In 2025, baseline or follow up monitoring data was gathered through completion of an effectivness montoring form (sites where remediation has been completed) and/or through eDNA sampling and aquistion of aerial imagery at the following sites (Table 4.2:

- Effectiveness monitoring was conducted on Peacock Creek (PSCIS crossing 197962) on the Morice FSR, where the oval culvert was replaced with a bridge in 2021. Environmental DNA sampling upstream and downstream of the crossing confirmed coho, rainbow trout and bull trout in the reach. Results are presented in the Peacock Creek - 197962 - Monitoring Appendix (page 103).
- Effectiveness monitoring was conducted on Waterfall Creek (PSCIS crossing 124421) at 11th Avenue, where the culvert was removed by the Gitksan Watershed Authorities. Riprap at the site has been buried and seeded, with good regrowth of grasses established and shrub planting planned. Results are presented in the Waterfall Creek - 124421 - Monitoring Appendix (page 97).

Table 4.2: Summary of monitoring sites.

PSCIS ID	Stream	Road	Crossing type	Crossing subtype	Diameter or span (m)	Length or width (m)	Assessment comments
197962	Peacock Creek	Morice FSR	Open Bottom Structure	BRIDGE	20	10	Bridge installed in 2020 or 2021. Bridge span estimated from photos.
124421	Waterfall Creek	11th Avenue	Open Bottom Structure	FORD	–	–	Culvert was removed by Gitksan Watershed Authorities. Site riprap was buried with dirt and seated with good regrowth of grasses. Shrub planting is planned for this fall.

4.8 Collaborative GIS Environment

Fieldwork activities and background spatial layers are maintained in a collaborative QGIS and Mergin Maps environment (sern_skeena_2023), which field crews and partners use to view, edit and add data. The layers loaded at the time of writing are listed in Appendix - Collaborative GIS Layers (page 91).

5 Recommendations

This program is organized around a single question: what condition are these watersheds in, and where will restoration effort do the most good. No one line of evidence answers it. Barrier assessments and the provincial habitat model establish where fish can reach and what lies above each crossing. Environmental DNA, fish sampling and habitat confirmations establish where fish are and what the habitat supports. Floodplain delineation and aerial imagery establish how infrastructure has altered the valley bottom. Climate departure and water temperature establish the trajectory, and how the flows that formed these channels are changing. Effectiveness monitoring tests whether the decisions taken were the right ones. The recommendations below are directed at strengthening each of those lines of evidence, and at making what they produce usable by the people whose decisions shape these watersheds.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, and the reason is worth stating plainly: the criteria that determine whether a crossing is worth remediating — species present and species possible, habitat quantity and quality, barrier severity, floodplain condition, constructability, and cost — are not measured in common units, so any ordering depends on which criterion is allowed to lead. The deciding constraint sits outside the project entirely, in the capacity and willingness of infrastructure owners and tenure holders to support implementation, both financially and across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Government, community groups, landowners, non-profits, industry and other stakeholders should work collaboratively to address the high and moderate priority barriers identified online within the report found here. Progress on any front is meaningful, and aiming to remediate at least one high-priority site per year per watershed group—regardless of its overall rank—is a practical and effective approach.

To advance fish passage restoration in the Skeena region:

Extending the lines of evidence

- Resume Phase 1 fish passage assessments. 2025 added two reassessments and no new inventory, and the candidate set that prioritization draws on only grows through Phase 1 work.

- Continue environmental DNA sampling as an annual, region-wide layer so single-year detections become distribution and trend information, and carry the results into prioritization rather than reporting them alongside it. The 2025 detections establish species presence at sites with no habitat confirmation assessment, which belongs in the ranking. Follow up the multi-species detections at Peacock Creek, Corya Creek and Sandstone Creek with targeted fish sampling to resolve what they mean.

- Extend functional floodplain mapping from the Bulkley to Morice, Zymoetz and Kispiox. Passage and floodplain health are the same problem viewed from two angles: reconnecting a stream longitudinally achieves less where the floodplain is severed laterally. The delivery route is now proven — the Bulkley delineation is published to a spatial catalogue that this report reads from rather than carrying a copy of its own — and Morice and Kispiox are already published, so extending to them is a matter of reading them in. Zymoetz still needs the model run.

- Continue integrating climate departure and water temperature into prioritization, in both directions. Opening access to cold, drought-resistant habitat addresses reaches at the warm edge of their thermal range; the same data identifies reaches currently too cold to support production, where warming may be raising rather than lowering the restoration value of a barrier below them.

- Calibrate intrinsic habitat models against areas of known high-value spawning and rearing identified through fish presence observations, quantifying model uncertainty and refining prioritization.

Testing whether the work delivers

- Continue and extend effectiveness monitoring across the region's remediated crossings, and continue developing a cost-effective framework that ties baseline and post-remediation data to measurable productivity gains. Outcomes are only visible over multiple years, and carrying enough sites to span a range of structure types, treatments and stream sizes is what allows them to be compared rather than read one at a time.

- Feed monitoring results back into prioritization and design. The purpose of monitoring is adaptive management, and what is learned at monitored sites should change which treatments are proposed elsewhere.

Making the knowledge usable

- Develop a public web map to communicate fish passage priorities, baseline conditions, monitoring results and partner activities to collaborators, First Nations, industry, regulators and the public. Analysis that cannot be found or interrogated does not inform decisions.

- Continue publishing project data to shared, open catalogues rather than to individual project repositories, extending the pattern already used for aerial imagery and floodplain extent to environmental DNA and to regional water temperature. Each layer placed where others can find it compounds the value of the work that produced it.

- Keep methods documented alongside the code and field forms that implement them, so results stay reproducible and each season's work is legible to the next.

- Share monitoring outcomes openly with tenure holders, regulators, First Nations and other practitioners, so decisions elsewhere are informed by what has and has not worked here.

Partnerships and capacity

- Continue working with First Nations across the study area — Wet'suwet'en, Gitxsan and Tsimshian — on site selection, monitoring, and interpretation of what the data show.

- Continue partnerships with stewardship groups and educational institutions, including the University of Northern British Columbia, so field programs double as training and research opportunities, and support a community of practice around regional water temperature and fish presence data rather than every program reconstructing the same picture from fragmented sources.

- Build capacity in partner organizations to interpret and use the floodplain, climate and habitat layers, so the opportunities and constraints they surface are visible to the people whose decisions shape watershed-scale outcomes.

- Maintain partnerships with rail, highway and forestry tenure holders, and with the Habitat Conservation Trust Foundation and the Ministry of Transportation and Infrastructure, to support funding, site selection, remediation and monitoring.

- Commission engineering designs at the Ministry of Transportation and Infrastructure crossings identified in previous reporting, and advance the highest ranked sites assessed this year.

Appendix - Salmon Stock Assessment Data

Fisheries and Oceans Canada stock assessment data was accessed via the NuSEDS-New Salmon Escapement Database System through the Open Government Portal with results presented online in this appendix at app-stock-assessment.html (Fisheries and Oceans Canada, n.d.). A brief memo on the data extraction process is available here.

Appendix - Fish Species by Watershed Group

Historical fish observations recorded in the watershed groups covered by this project (MoE 2024a). The print version collapses presence to a single column listing the watershed groups each species has been recorded in; the online version gives one column per watershed group.

Table 5.1: Fish species recorded in the Bulkley, Kispiox, Kalum, Morice, and Zymoetz watershed groups.

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
Catostomus catostomus	Longnose Sucker	Yellow	–	Bulkley, Kispiox, Morice, Zymoetz
Catostomus commersonii	White Sucker	Yellow	–	Bulkley, Kalum, Kispiox, Morice
Catostomus macrocheilus	Largescale Sucker	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Chrosomus eos	Northern Redbelly Dace	Yellow	–	Bulkley
Coregonus clupeaformis	Lake Whitefish	Yellow	–	Bulkley, Kispiox, Morice
Coregonus sardinella	Least Cisco	Blue	–	Kalum
Cottus aleuticus	Coastrange Sculpin (formerly Aleutian Sculpin)	Yellow	–	Bulkley, Kalum, Kispiox, Morice
Cottus asper	Prickly Sculpin	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Cottus cognatus	Slimy Sculpin	Yellow	–	Kalum, Kispiox
Couesius plumbeus	Lake Chub	Yellow	DD	Bulkley, Kalum, Kispiox, Morice
Entosphenus tridentatus	Pacific Lamprey	Yellow	–	Bulkley, Kalum, Morice
Gasterosteus aculeatus	Threespine Stickleback	Yellow	–	Kalum, Kispiox
Hybognathus hankinsoni	Brassy Minnow	No Status	–	Bulkley
Lampetra ayresii	River Lamprey	Yellow	–	Kalum
Lota lota	Burbot	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Mylocheilus caurinus	Peamouth Chub	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus clarkii	Cutthroat Trout	No Status	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus clarkii	Cutthroat Trout (Anadromous)	No Status	–	Bulkley, Kispiox, Zymoetz
Oncorhynchus clarkii clarkii	Coastal Cutthroat Trout	Blue	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus clarkii lewisi	Westslope (Yellowstone) Cutthroat Trout	Blue	SC (Nov 2016)	Kalum, Kispiox

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
Oncorhynchus gorbuscha	Pink Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus keta	Chum Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus kisutch	Coho Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus mykiss	Rainbow Trout	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus mykiss	Steelhead	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus mykiss	Steelhead (Summer-run)	Yellow	–	Bulkley, Morice
Oncorhynchus mykiss	Steelhead (Winter-run)	Yellow	–	Kispiox, Zymoetz
Oncorhynchus nerka	Kokanee	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus nerka	Sockeye Salmon	Not Reviewed	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Oncorhynchus tshawytscha	Chinook Salmon	Not Reviewed	E/T/SC/DD/NAR (Nov 2020)	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Prosopium coulterii	Pygmy Whitefish	Yellow	NAR (Nov 2016)	Bulkley, Kispiox, Morice
Prosopium coulterii pop. 3	Giant Pygmy Whitefish	Yellow	NAR (Nov 2016)	Bulkley
Prosopium cylindraceum	Round Whitefish	Yellow	–	Kalum
Prosopium williamsoni	Mountain Whitefish	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Ptychocheilus oregonensis	Northern Pikeminnow	Yellow	–	Bulkley, Kispiox, Morice, Zymoetz
Pungitius pungitius	Ninespine Stickleback	Unknown	–	Bulkley
Rhinichthys cataractae	Longnose Dace	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Rhinichthys falcatus	Leopard Dace	Yellow	NAR (May 1990)	Morice
Richardsonius balteatus	Redside Shiner	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Salvelinus confluentus	Bull Trout	Blue	SC (Nov 2012)	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Salvelinus fontinalis	Brook Trout	Exotic	–	Bulkley, Morice
Salvelinus malma	Dolly Varden	Yellow	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
Salvelinus namaycush	Lake Trout	Yellow	–	Bulkley, Kispiox, Morice
–	All Salmon	–	–	Kalum, Kispiox
–	Arctic Char	–	–	Morice
–	Chub (General)	–	–	Kispiox

Scientific Name	Species Name	BC List	COSEWIC	Present in WSGs
–	Cutthroat/Rainbow cross	–	–	Bulkley, Kalum, Kispiox
–	Dace (General)	–	–	Morice
–	Lamprey (General)	–	–	Bulkley, Kalum, Kispiox, Morice
–	Minnow (General)	–	–	Bulkley, Kispiox, Morice
–	Mottled Sculpin	–	–	Bulkley
–	Salmon (General)	–	–	Bulkley, Kispiox, Morice, Zymoetz
–	Sculpin (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
–	Squanga	–	–	Kispiox
–	Stickleback (General)	–	–	Kalum, Kispiox
–	Sucker (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz
–	Verified DV BT hybrid	–	–	Kalum
–	Whitefish (General)	–	–	Bulkley, Kalum, Kispiox, Morice, Zymoetz

* COSEWIC abbreviations :
SC - Special concern
DD - Data deficient
NAR - Not at risk
E - Endangered
T - Threatened

BC List definitions :
Yellow - Species that is apparently secure
Blue - Species that is of special concern
Exotic - Species that have been moved beyond their natural range as a result of human activity

Appendix - Climate Departure

Climate departure and fish passage

Stream crossings are the visible structures in a fish-passage inventory: culverts, bridges, weirs, fords. The streams they cross sit inside a climate that is changing — warmer air, shifting snowpack, longer shoulder seasons — and those climate changes set the hydrologic and thermal context the structures pass fish *through*. A culvert installed in the 1980s is now passing a different hydrograph: earlier freshet, lower late-summer flows, warmer water. The barrier inventory does not see that on its own.

Warmer air drives warmer streams. For the cold-water salmonids the study area supports — steelhead, coho, chinook, bull trout, Dolly Varden and rainbow trout — that can move habitat in either direction: cold-limited headwaters gain growing-degree-days (larger fish, better overwinter survival, stronger out-migration), while reaches near the upper edge of the thermal niche lose usable habitat. The recovery value of removing a downstream barrier scales with which side of that line each upstream reach sits on. Climate departure tells you which side that is, watershed by watershed.

Beyond temperature, hydrologic timing matters. Earlier and faster snowmelt shifts the freshet weeks earlier in the calendar and compresses the high-flow pulse. That changes design assumptions for crossing structures, and it changes the timing of migration, gravel mobilisation, and off-channel rearing — the in-stream processes barrier removal is meant to restore.

Quantifying departure at the watershed scale lets us carry climate context into barrier prioritisation alongside the structural and biological inputs.

Area of interest

The study area for this climate-departure analysis is the union of five Freshwater Atlas watershed groups in the Skeena drainage: Bulkley River (BULK), Kispiox River (KISP), Kalum River (KLUM), Morice River (MORR), and Zymoetz River (ZYMO). Together they cover 25,332 km² and span 6 ecoregions (Figure 5.1) — broad climate-physiography zones based on latitude, elevation, and dominant vegetation. We carry that ecoregion partition through the analysis because climate departure can vary across them in ways the regional average hides, then map the ecoregion signal back onto the five watershed groups at the end of the appendix.

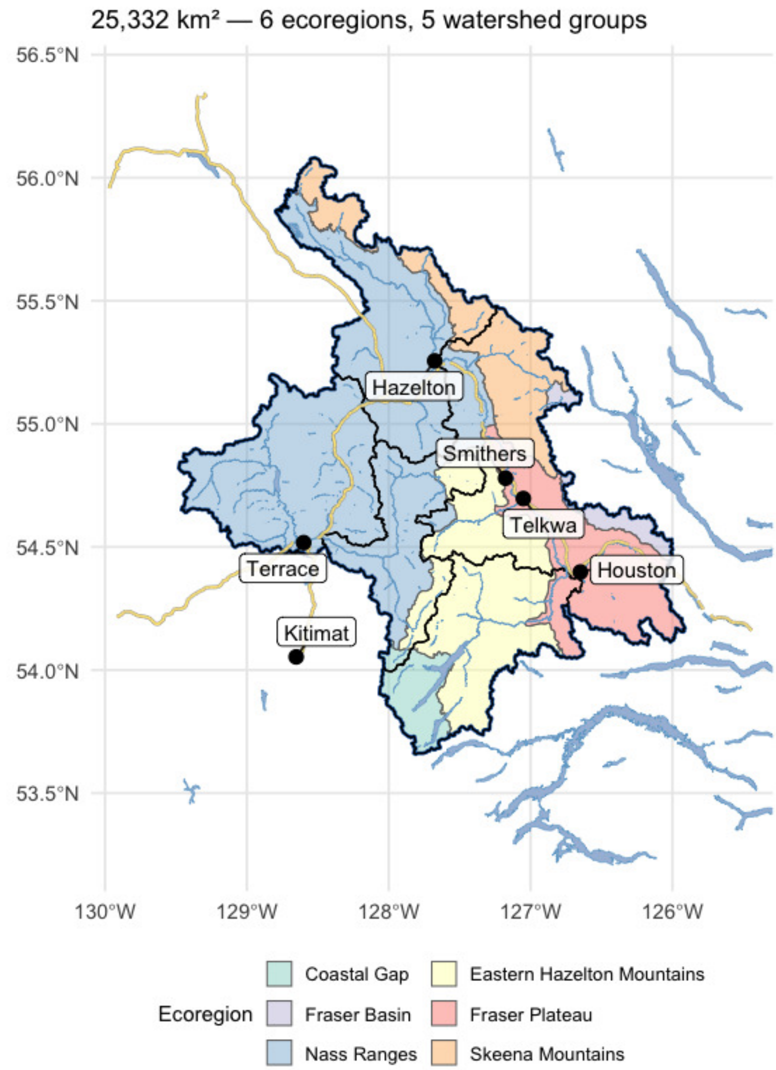

Figure 5.1: Study area in northwest British Columbia, coloured by ecoregion, with the five watershed groups (heavy black outlines) that define the analysis extent. The Skeena and its major tributaries drain west to the coast; the study area spans the Nass Ranges and Skeena Mountains in the north and west, the Eastern Hazelton Mountains and Coastal Gap in the south, and reaches the interior plateau along its eastern edge.

Recent decade versus pre-warming reference

The table below compares two windows directly — the recent decade (2015–2025) against the pre-warming reference (1951–1980). The difference is reported in the variable's native units, alongside percent change where it is meaningful (precipitation, soil moisture, vapour pressure deficit, relative humidity). Two p-values appear, answering different questions. **Δ p (windows)** is the Welch t-test on the annual values within each window — this tells us if the recent decade differs from the pre-warming reference. **Trend p (75-yr)** is the Mann-Kendall test on the full 1951–present series — this tells us if there is a steady year-on-year ramp. Step changes show up as significant Δ p with non-significant trend p; gradual ramps show up as both significant.

Table 5.2: Recent decade (2015-2025) versus pre-warming reference (1951-1980) for all 15 climate variables and all available periods. Two p-values answer different questions — see text.

Variable	Period	Recent (2015–2025)	Pre-warming (1951–1980)	Δ absolute	Δ %	Δ p (windows)	Trend p (75-yr)
prcp	annual	1459.58	1433.77	25.81	1.8	0.685	0.841
prcp	fall	483.58	497.28	-13.70	-2.8	0.706	0.395
prcp	spring	270.70	261.51	9.19	3.5	0.677	0.405
prcp	summer	287.84	247.19	40.65	16.4	0.150	0.023
prcp	winter	417.47	427.79	-10.32	-2.4	0.747	0.534
rh	annual	77.93	78.49	-0.56	-0.7	0.228	0.213
rh	fall	82.87	84.18	-1.31	-1.6	0.058	0.024
rh	spring	73.02	73.60	-0.58	-0.8	0.507	0.891
rh	summer	72.07	74.33	-2.26	-3.0	0.097	0.018

The recent decade ran 2.1 °C warmer than the pre-warming reference for annual mean temperature, with both window and trend p-values below 0.001. Vapour pressure deficit rose 0.34 hPa, significant on both tests. Annual precipitation rose about +2 % regionally but neither test confirms a real shift (window p ≈ 0.5, trend p ≈ 0.8). Soil moisture and relative humidity show no significant change. The snowpack rows (swe summer, snowmelt spring, snowmelt winter, snowmelt_doy_50) carry the seasonal-redistribution and freshet-timing signals described below.

Trends over the long record

Anomalies are computed against the 1951–1980 pre-warming reference. Saying a year is “+1.5 °C” means it was 1.5 °C warmer than the average year between 1951 and 1980. The trend table below has two rows per variable — trends computed from two different start years:

- **1951–present (75 years)** — the long view. Captures the full magnitude of warming since the pre-warming reference.
- **1981–present (45 years)** — starts at the beginning of the current 30-year climate normal (1981–2010), the reference used in most published climate products.

Comparing the two slopes is informative. If the 45-year slope is steeper than the 75-year slope, warming has accelerated; if shallower, warming has slowed (though “slower” almost never means “stopped”). Similar slopes mean a roughly steady rate of change across the full record. Total change is slope multiplied by the number of years — the cumulative shift over the trend window.

Table 5.3: Trend statistics for all variables and periods — Theil-Sen slope, Mann-Kendall p-value, and cumulative change over each trend window.

Parameter	Period	Slope	Years	Total Change	Unit	p-value
Precipitation	Annual	0.010	75	0.8	%	0.8405
Relative humidity	Annual	-0.010	75	-0.7	%	0.2134
Snow cover	Annual	-0.101	75	-7.6	%	0.0000
Snowfall	Annual	-0.143	75	-10.7	%	0.1589
Snowfall fraction	Annual	-0.067	75	-5.1	%	0.0316
Snowmelt	Annual	-0.152	75	-11.4	%	0.0633
Day of 50% melt	Annual	-0.220	75	-16.5	day	0.0001
Peak weekly melt rate	Annual	0.183	75	13.7	mm/wk	0.1074

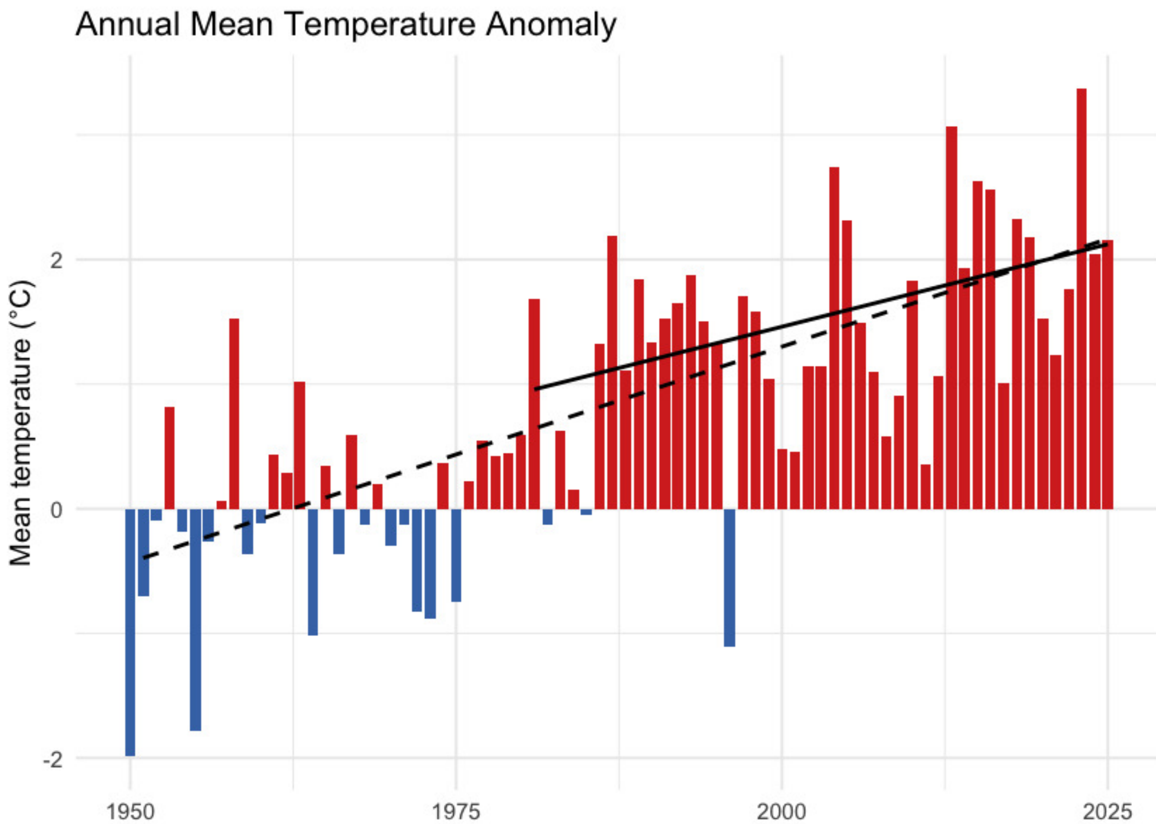

Figure 5.2: Annual mean temperature anomaly relative to the 1951-1980 baseline. Bars are degrees above (red) or below (blue) the pre-warming average; the two trend lines are the 75-year (1951-present, dashed) and 45-year (1981-present, solid) Theil-Sen slopes.

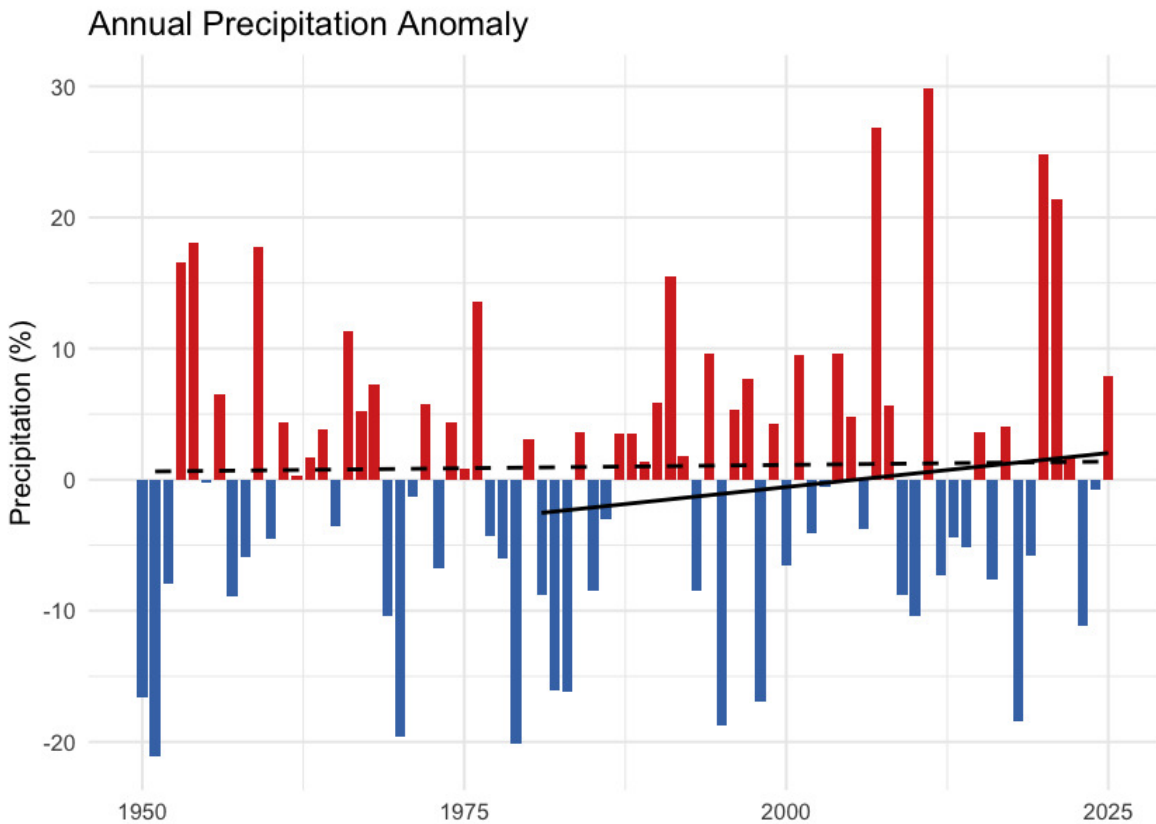

Figure 5.3: Annual precipitation anomaly (% of the 1951-1980 baseline). The long-term trend is essentially flat — regional precipitation has not shifted meaningfully against the natural year-to-year variability.

One pattern in the trend table is worth noting: the 45-year (since 1981) tmean slope is **shallower** than the 75-year slope — 0.27 versus 0.35 °C per decade. The same direction holds for tmax, tmin, and VPD. Warming since 1981 has not been faster than the longer-term average rate; the cumulative shift since 1951 reflects steady, significant warming rather than a recent acceleration.

Daytime highs and overnight lows

Daytime maximum (tmax) and overnight minimum (tmin) temperatures carry distinct information. Overnight minimums warming faster than daytime maximums — the "day-night asymmetry" — is one of the textbook fingerprints of greenhouse warming. Karl et al. (1993) documented this first at the global scale, finding overnight minimums rose at roughly three times the rate of daytime maximums between 1951 and 1990.

In this study area, the asymmetry is present. Daytime maximums warmed 0.32 °C per decade since 1951 (+1.9 °C cumulative); overnight minimums warmed 0.37 °C per decade (+2.2 °C cumulative). The overnight side warmed about 0.3 °C more than the daytime side over the recent decade, narrowing the diurnal temperature range by the same amount (Figure 5.6). The asymmetry is present but weaker than the canonical signal — the local tmin/tmax warming ratio is about 1.2:1 versus the ~3:1 ratio Karl et al. reported at the global scale between 1951 and 1990.

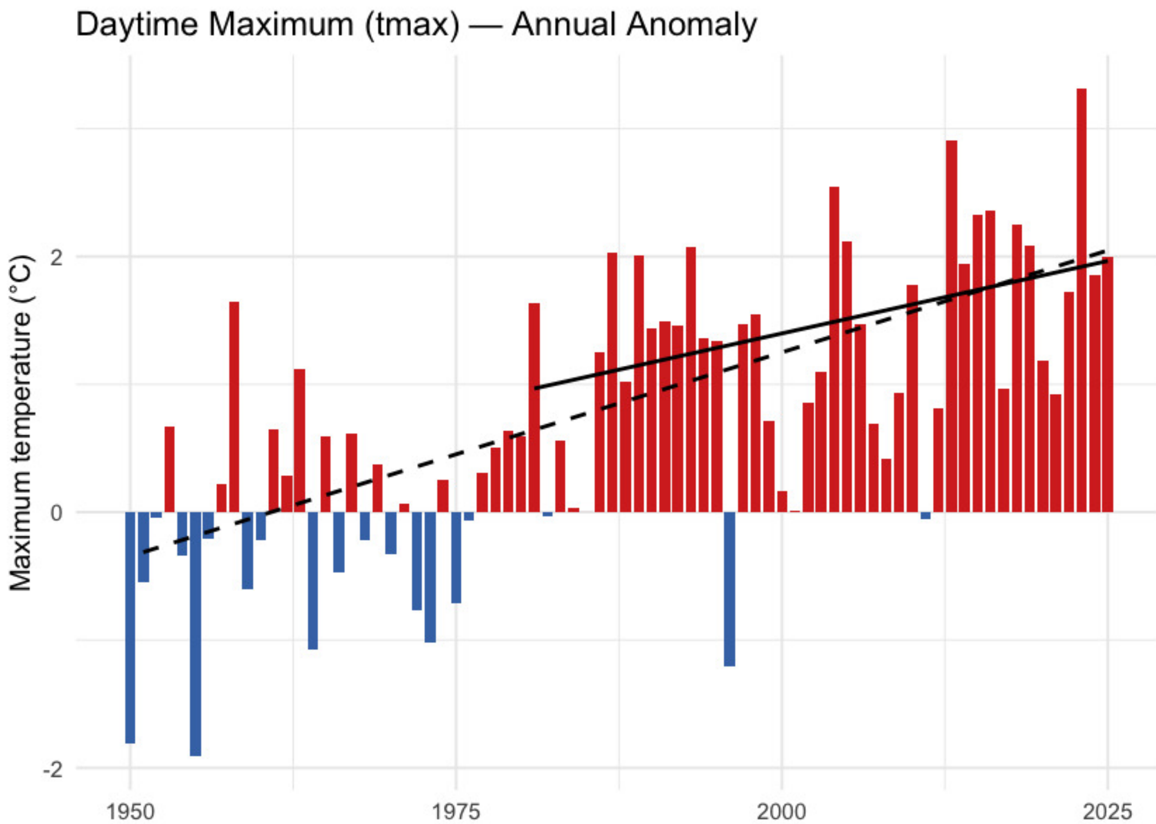

Figure 5.4: Annual daytime maximum temperature (tmax) anomaly relative to the 1951-1980 baseline.

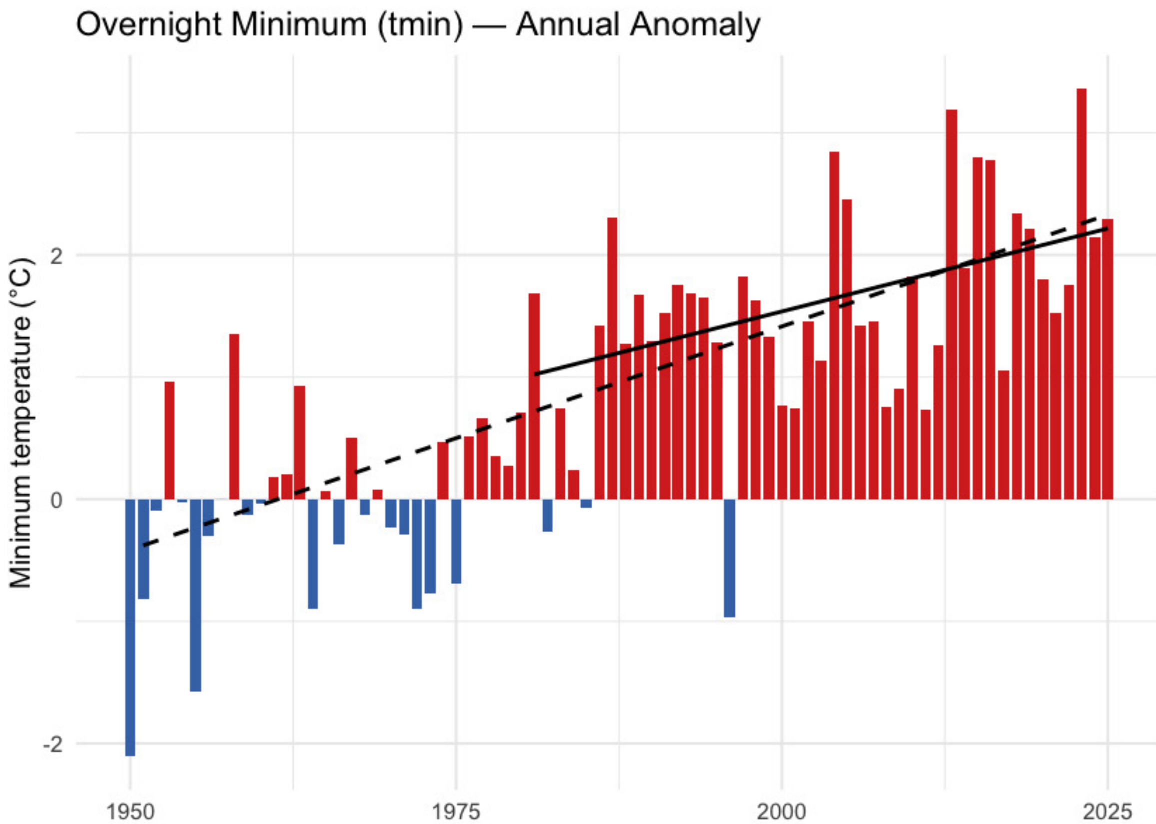

Figure 5.5: Annual overnight minimum temperature (tmin) anomaly relative to the 1951-1980 baseline.

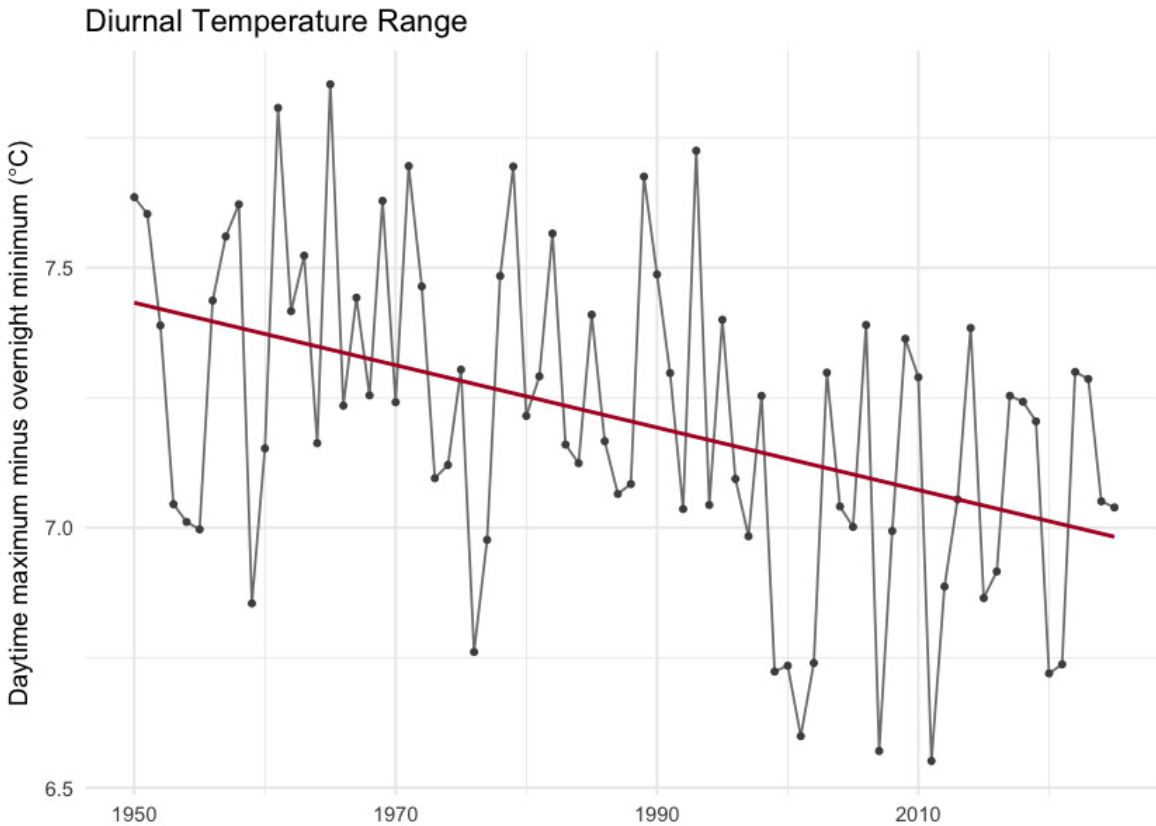

Figure 5.6: Diurnal temperature range (daytime maximum minus overnight minimum) annual mean. The downward slope is the day-night asymmetry — overnight lows are warming faster than daytime highs.

Snowpack

Snowpack drives the seasonal water supply across British Columbia: winter precipitation falls as snow, accumulates on the ground, and releases as meltwater across spring and summer. That seasonal storage is the difference between a late-summer creek that still flows and one that does not. It also sets the timing of the spring freshet — the annual flood pulse that salmonids time their up-river migrations to.

In this study area the snowpack signal is one of **earlier melt, not less snow**. Total annual snowfall fell by about 8 % but neither tests confirm a real change against year-to-year variability. Where the signal is unambiguous is in *when* the snow leaves the ground. The midpoint day of annual snowmelt shifted 15 days earlier in the calendar over the same window ($p < 0.001$), and snow cover dropped 6.5 percentage points ($p = 0.002$).

Seasonal snowpack curve

Aggregated to the four standard meteorological seasons — winter (December–February), spring (March–May), summer (June–August), and fall (September–November) — the table below compares the recent decade (2015–2025) directly against the pre-warming reference (1951–1980) for each of the four monthly-native snow variables. The seasonal redistribution is the headline. Statistically the load-bearing signals are on the summer side — summer SWE has collapsed ($p \approx 0.01$) and summer snowmelt has fallen by a comparable amount ($p \approx 0.02$). The counterbalancing winter and spring snowmelt rises are directionally consistent but noisier season-by-season (Welch $p \approx 0.28$ and 0.06 respectively) — see the *What this means* paragraph below for the full reading.

Table 5.4: Seasonal snowpack: recent decade versus pre-warming reference. Summer SWE collapse (-52 %, $p \approx 0.01$) and summer snowmelt fall (-37 %, $p \approx 0.02$) are the statistically significant signals. Winter (+45 %) and spring (+18 %) snowmelt rises are directionally consistent with the redistribution but do not individually clear statistical significance (Welch $p \approx 0.28$ and 0.06).

Variable	Season	Pre-warming (1951–1980)	Recent (2015–2025)	Δ %
SWE (mm)	winter	459.4	405.6	-11.7
SWE (mm)	spring	717.6	611.0	-14.8
SWE (mm)	summer	181.9	72.2	-60.3
SWE (mm)	fall	69.7	51.3	-26.4
Snowfall (mm)	winter	409.8	393.3	-4.0
Snowfall (mm)	spring	197.2	170.7	-13.4
Snowfall (mm)	summer	8.9	3.1	-65.3
Snowfall (mm)	fall	258.1	238.1	-7.8
Snowmelt (mm)	winter	1.5	2.3	46.4
Snowmelt (mm)	spring	296.4	449.1	51.5
Snowmelt (mm)	summer	548.8	324.3	-40.9
Snowmelt (mm)	fall	51.5	37.2	-27.9

Annual snowpack signals

Four derived annual scalars capture the snowpack signals the literature treats as headline metrics for snow hydrology: peak snowpack (Mote et al. 2018, 2005), the date of melt midpoint (Stewart et al. 2005; Cayan et al. 2001), freshet flashiness, and the fraction of precipitation falling as snow (Knowles et al. 2006).

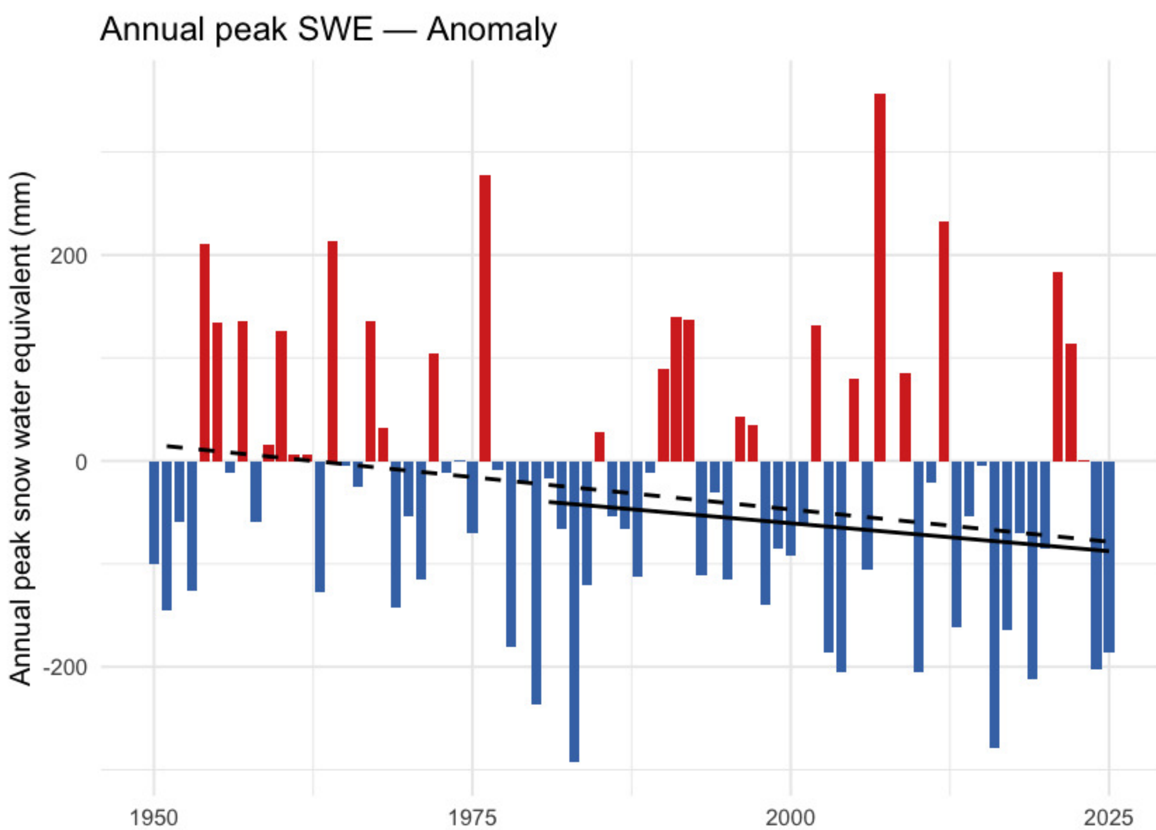

Figure 5.7: Annual peak snow water equivalent (swe_max). ERA5-Land mm SWE, regional spatial mean. The downward trend is present but does not clear statistical significance against year-to-year variability at the regional scale.

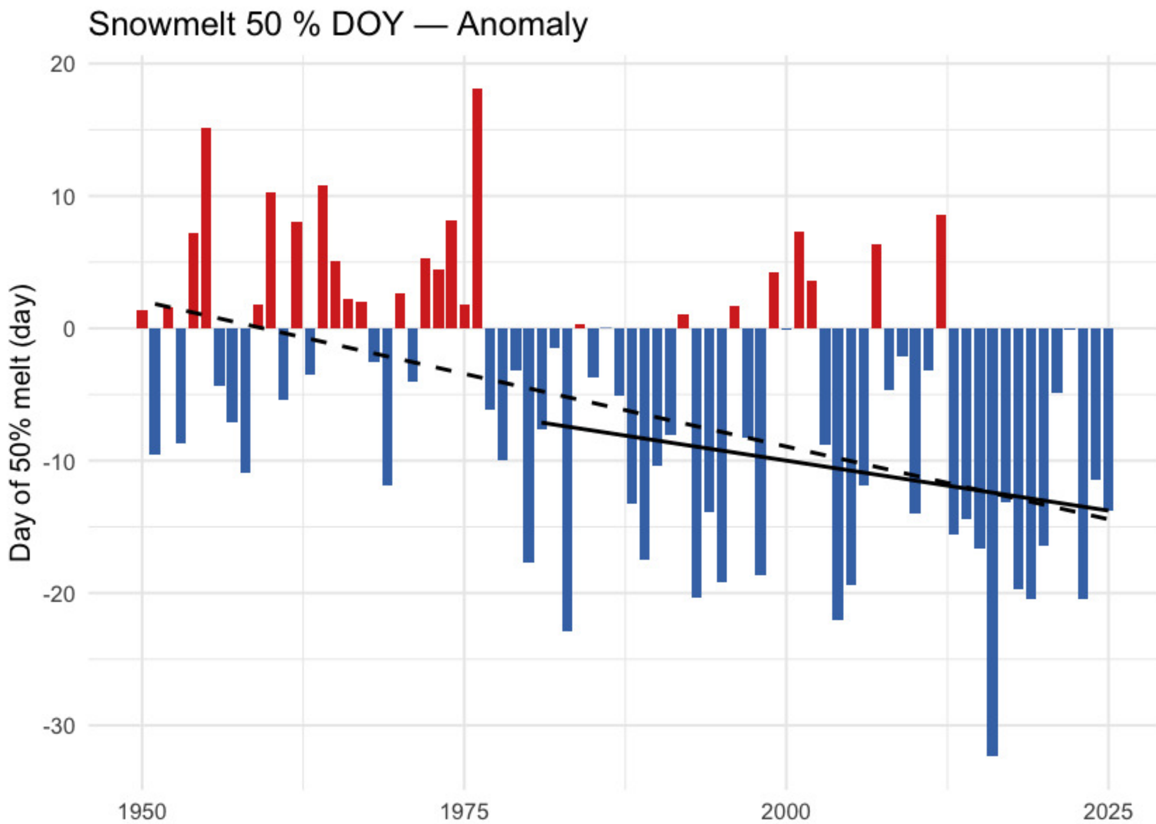

Figure 5.8: Day of year when half the annual snowmelt has accumulated (snowmelt_doy_50). Earlier dates indicate an earlier freshet centroid — directly relevant to crossing-design hydrology and to migration timing. The trend is highly significant ($p < 0.001$ at the regional scale and in every ecoregion).

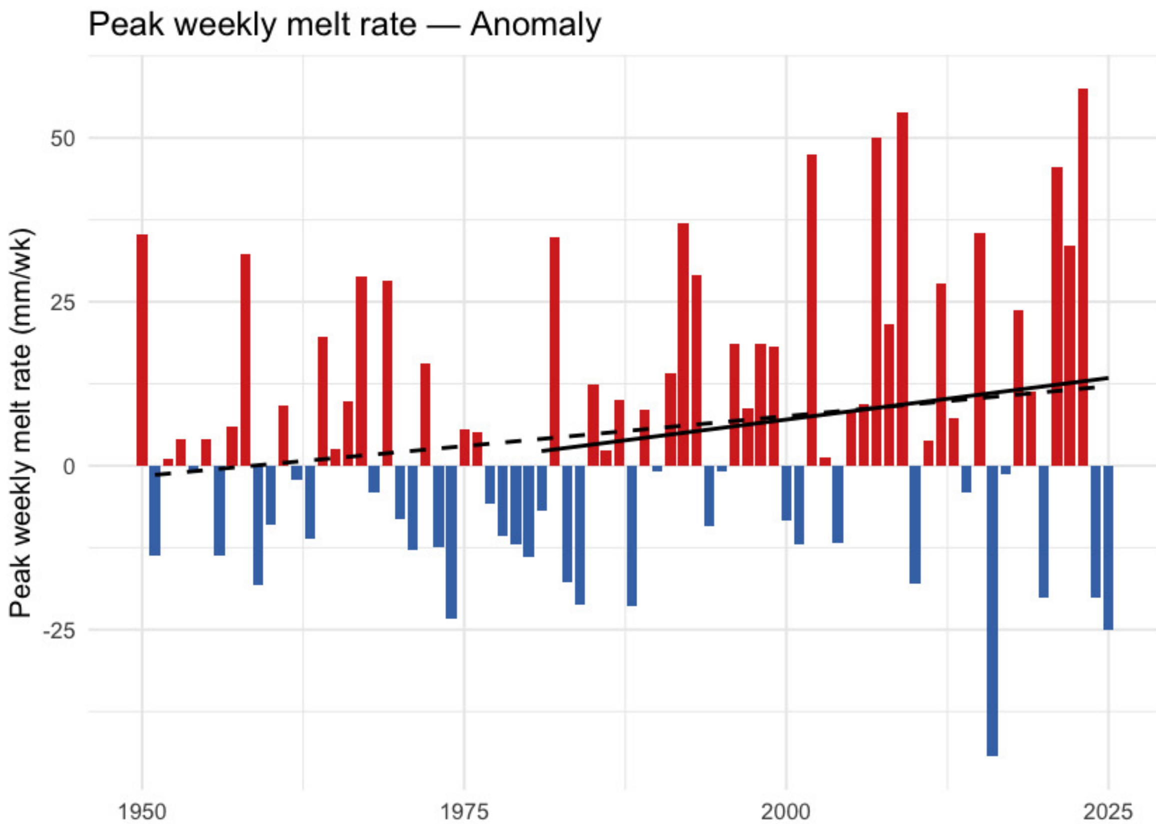

Figure 5.9: Annual maximum of 7-day rolling daily snowmelt (snowmelt_rate_peak). Peak melt intensity has not shifted meaningfully — the freshet is arriving earlier, but its peak rate has not sharpened.

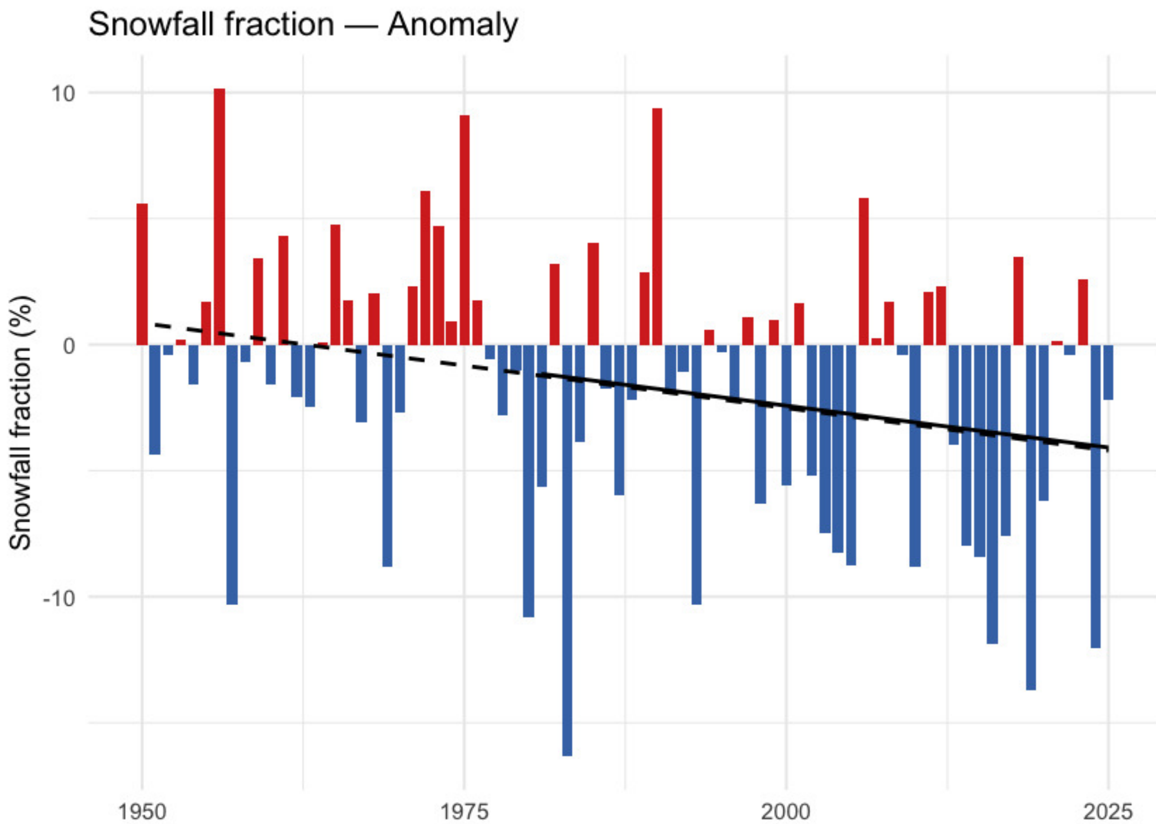

Figure 5.10: Annual snowfall fraction: the percent of annual precipitation that fell as snow. Anomaly is in percentage points relative to the 1951-1980 baseline. The downward trend sits just above statistical significance (Mann-Kendall $p \approx 0.05$) — directionally a shift toward rain-dominant precipitation, but not formally significant on either the trend or the recent-vs-baseline window test.

What this means

Three findings carry the snowpack story.

Snow is leaving the region earlier each year, not falling less. Annual snowfall fell about 8 % and annual SWE about 20 %, but neither change clears statistical significance against year-to-year variability. The clear signals are in *timing*: the snowmelt midpoint shifted nearly two weeks earlier ($p < 0.001$ regionally and in every ecoregion), snow cover declined 3.8 percentage points ($p = 0.002$), and snowfall fraction edged down 3.7 pp (Welch and Mann-Kendall p both just above 0.05 — directionally consistent with a shift toward rain-dominant precipitation, but just shy of statistical significance). This matches in direction the broader Pacific-Northwest signal documented since Mote et al. (2005).

Melt is shifting earlier. The unambiguous signals are on the summer side and at the midpoint: summer snowmelt fell 41 % ($p \approx 0.02$), summer SWE collapsed 60 % ($p \approx 0.01$), and the annual snowmelt midpoint moved ~12 days earlier ($p < 0.001$). Apparent rises in spring snowmelt (+52 %, Welch $p \approx 0.06$) and winter snowmelt (+46 %, Welch $p \approx 0.28$) mirror this redistribution but are noisier season- by-season — winter and spring have smaller absolute snowmelt volumes, so the same shifts produce larger relative-change variance. The total annual melt has not changed; its centre of mass has moved earlier. This is consistent in direction with the 1–4 week earlier streamflow timing across western North America documented by Stewart et al. (2005) and with the ~10-day earlier Fraser freshet at Hope Kang et al. (2016).

Summer SWE has dropped sharply. Summer SWE fell 60 %. The high-elevation snowpack that historically lingered through the warmest weeks of the year no longer carries the same volume — reducing the late-season cold-water input to streams during the window when stream-temperature stress on cold-water salmonids is highest.

Spatial pattern of warming

The regional zonal mean reduces an AOI this size to a single number; the spatial pattern carries the rest of the story.

Warming across this AOI is **close to spatially uniform**. Individual grid cells range from about +1.8 to +2.4 °C, and ecoregion means span only +1.95 to +2.15 °C — a spread of 0.20 °C across the whole study area (Figure 5.11). No part of the AOI escapes warming of roughly two degrees.

What structure there is runs **west-warm to east-cool** (correlation with longitude -0.79), with a secondary south-to-north component (0.61) consistent with high-latitude amplification. The warmest ecoregion is the Nass Ranges at +2.15 °C, the coolest the Fraser Basin at +1.95 °C.

The implication for fish passage is that the thermal question is an elevation and reach-scale one. The warmest and coolest watershed groups — Kalum River (+2.21 °C) and Kispiox River (+2.13 °C) against Morice River (+1.98 °C) — differ by less than 0.2 °C, so the differences that separate one barrier reach from another are local: elevation, aspect, riparian cover and groundwater contribution.

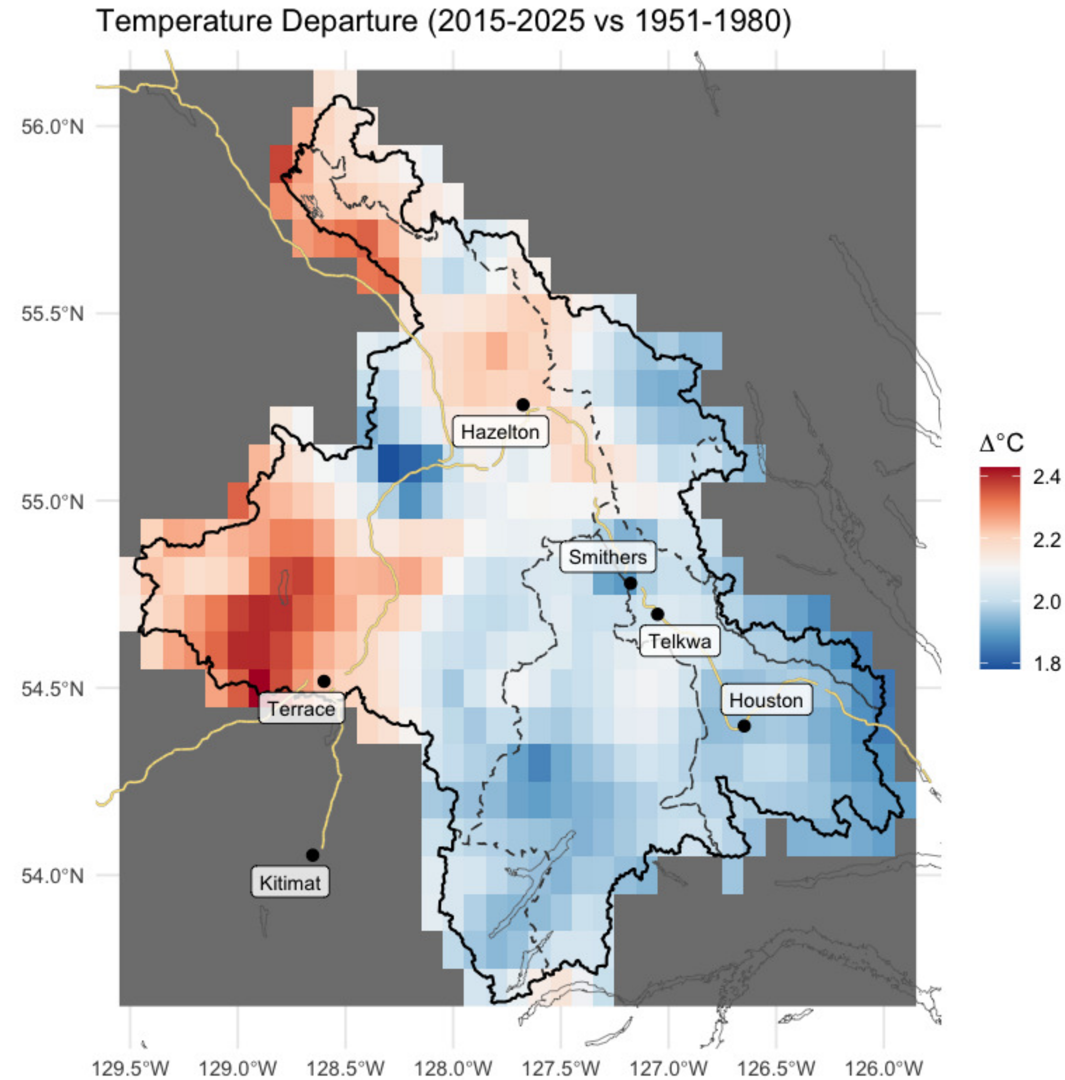

Figure 5.11: Spatial pattern of annual mean temperature departure across the AOI (2015-2025 mean minus 1951-1980 mean, degrees C). Warming is close to uniform, with a weak west-warm to east-cool axis and a secondary south-to-north amplification.

Per-ecoregion variation

The regional zonal mean averages over six ecoregions with different elevations and exposures. Running the same pipeline on each ecoregion polygon tests whether the regional story holds within each one, and where it does not.

All six ecoregions warmed significantly (Mann-Kendall $p < 0.001$ for tmean in every ecoregion), and the range between them is narrow: Nass Ranges (+2.70 °C) and Skeena Mountains (+2.62 °C) lead, Fraser Basin trails at +2.42 °C, and the whole spread is 0.28 °C of cumulative warming since 1951. The mountain ecoregions warm marginally faster than the interior plateau, though the difference is small enough to carry little practical weight. The day-night asymmetry (tmin > tmax) holds in every ecoregion. Vapour pressure deficit rose significantly ($p < 0.05$) in every ecoregion.

Precipitation is the variable that does not move. Annual precipitation shows no significant trend in any of the six ecoregions (Mann-Kendall p ranges 0.35 to 0.95). The climate signal for this study area is unambiguously about temperature and snowpack timing, not about precipitation magnitude.

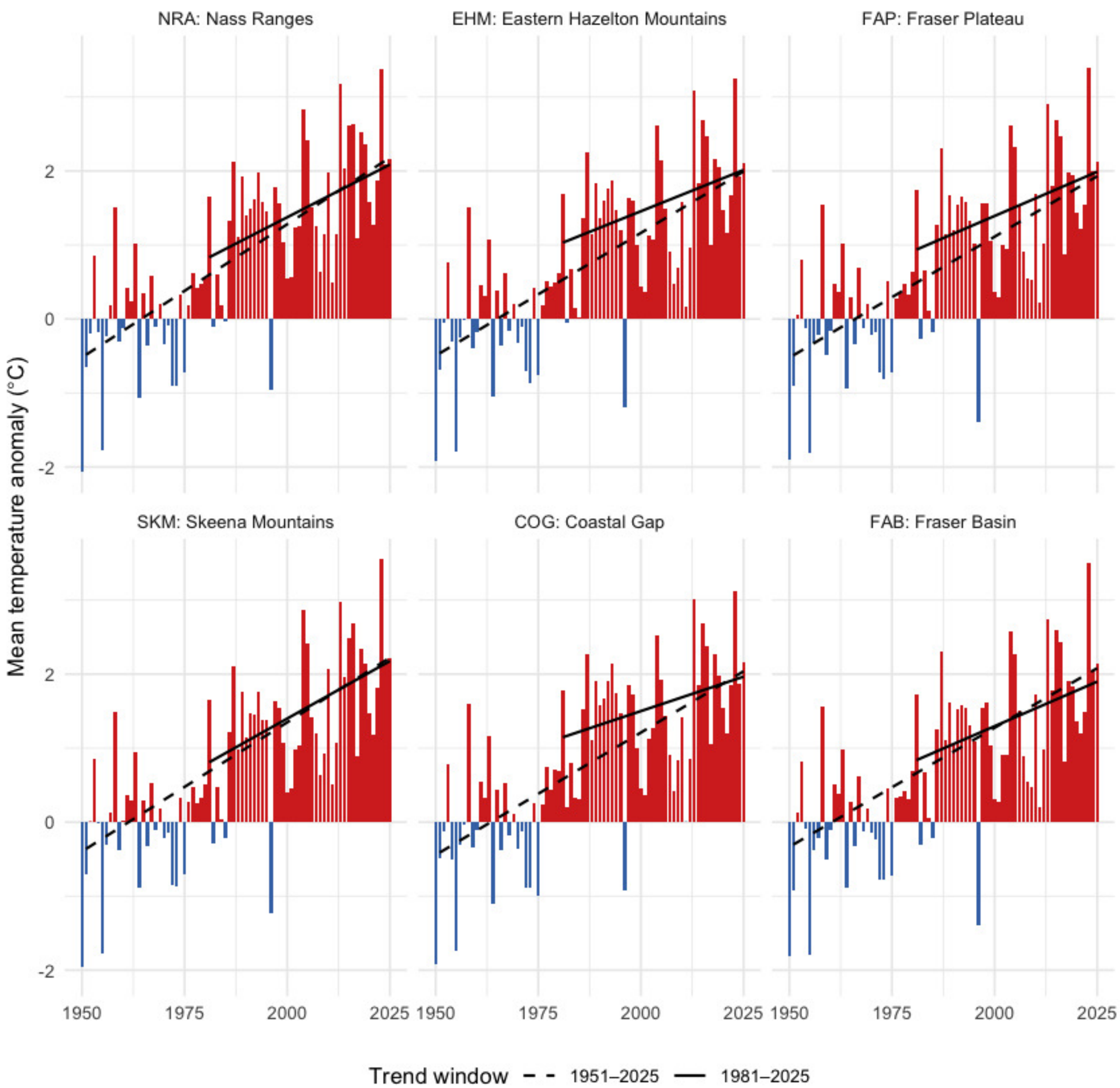

Figure 5.12: Annual mean temperature anomaly relative to the 1951-1980 baseline, by ecoregion. Dashed line is the 75-year Theil-Sen trend (1951-present); solid line is the 45-year trend (1981-present). A solid line steeper than the dashed line indicates accelerating warming — note that across this AOI, the 45-year slope is generally shallower than the 75-year slope, meaning recent warming has not been faster than the long-term rate.

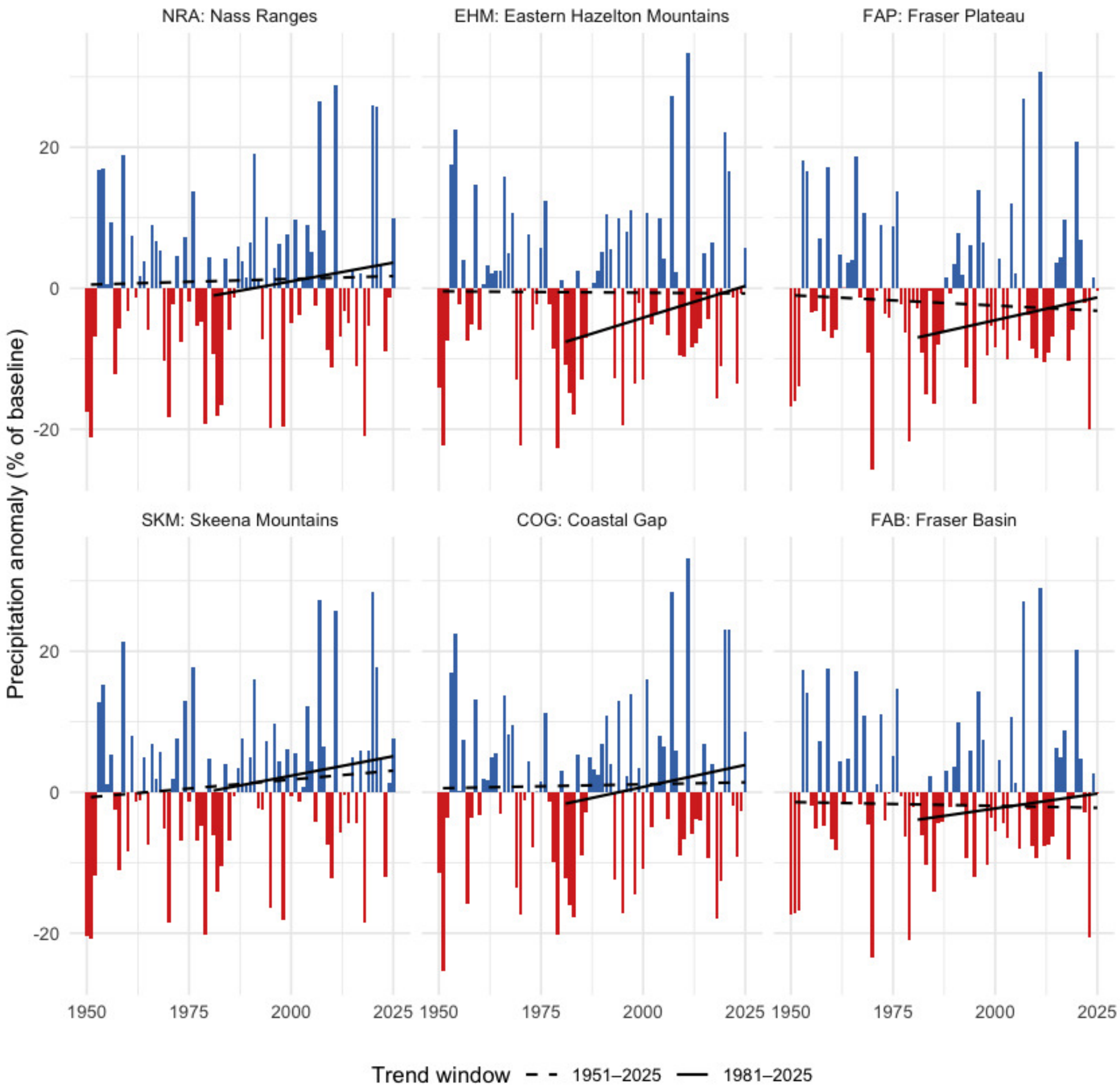

Figure 5.13: Annual precipitation anomaly (% of the 1951-1980 baseline) by ecoregion. None of the eight ecoregions shows a statistically significant precipitation trend over the full record.

Snow by ecoregion

Two snow metrics are worth viewing per-ecoregion: peak SWE (annual snowpack magnitude) and DOY-50 (the day-of-year by which half the year's snowmelt has already happened — earlier values mean an earlier freshet). The peak-SWE signal is noisy at the per- ecoregion scale — none of the eight ecoregions clears statistical significance for the long-term swe_max trend. The DOY-50 signal, in contrast, is highly significant in every ecoregion (Mann- Kendall $p < 0.01$ in each), with shifts ranging from -1.4 to -2.1 days per decade. Earlier melt is universal across the AOI.

Per-ecoregion variation

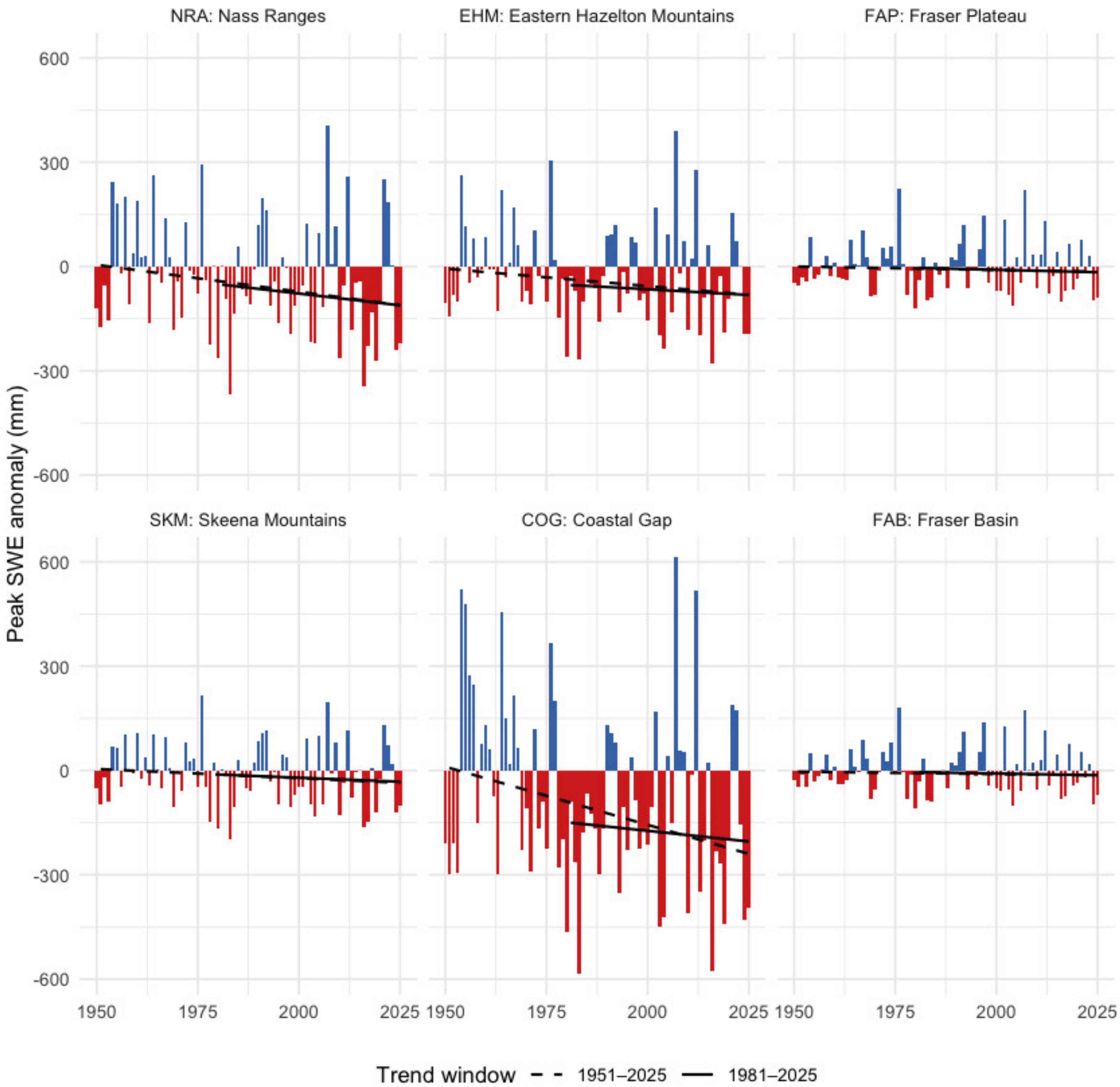

Figure 5.14: Annual peak SWE anomaly by ecoregion, relative to the 1951-1980 baseline. Year-to-year variability is large; the long-term trend is not statistically significant in any single ecoregion.

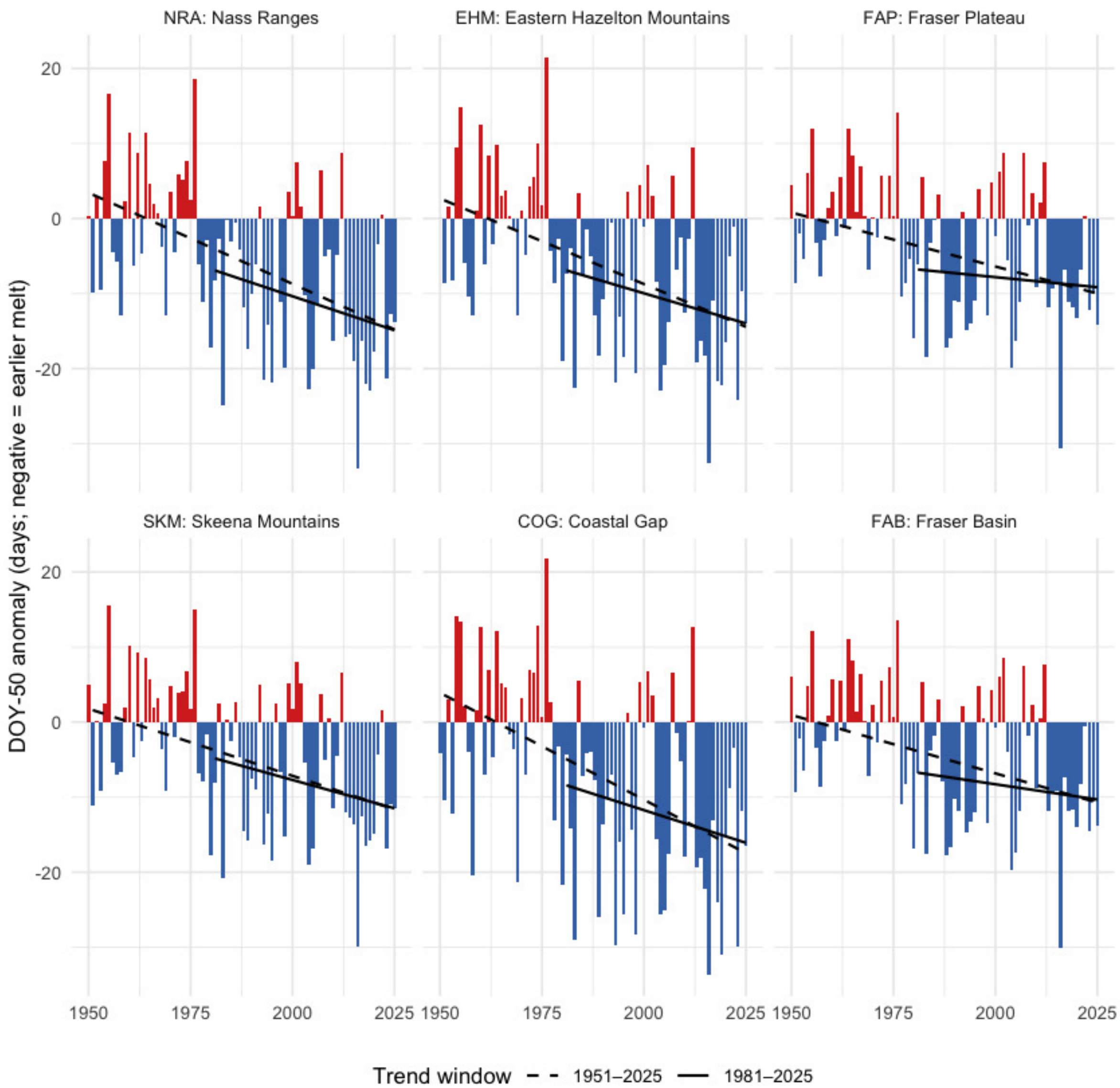

Figure 5.15: Snowmelt 50 % day-of-year (DOY-50) anomaly by ecoregion, relative to the 1951-1980 baseline. Negative values (red) mean the freshet midpoint shifted earlier in the year. The trend is highly significant in every ecoregion.

Table 5.5: Per-ecoregion roll-up over the 75-year window (1951-present): annual mean / daytime maximum / overnight minimum temperature trends (°C per decade); annual precipitation trend (mm per year) and Mann-Kendall p-value; annual vapour pressure deficit trend (hPa per decade) and p-value; and recent (2015-2025) vs pre-warming (1951-1980) percent change for precipitation and soil moisture. All temperature p-values are below 0.001. A tmin slope greater than the tmax slope indicates the day-night asymmetry.

Ecoregion	tmean °C/dec	tmax °C/dec	tmin °C/dec	prcp mm/yr	prcp p	vpd hPa/dec	vpd p	prcp % change	soil moisture % chg
NRA	0.36	0.32	0.39	0.016	0.826	0.058	0	2.0	-0.3
EHM	0.33	0.30	0.34	-0.004	0.949	0.057	0	0.9	-1.8
FAP	0.33	0.31	0.34	-0.029	0.687	0.072	0	0.7	-1.8
SKM	0.35	0.32	0.37	0.051	0.351	0.051	0	3.8	-0.3

Ecoregion codes: NRA — Nass Ranges; EHM — Eastern Hazelton Mountains; FAP — Fraser Plateau; SKM — Skeena Mountains; COG — Coastal Gap; FAB — Fraser Basin.

Watershed groups across ecoregions

The study area is reported and managed at the watershed-group scale — the British Columbia Freshwater Atlas hydrological reporting unit, and the unit that drives barrier prioritisation in this report. The five watershed groups (BULK, KISP, KLUM, MORR, ZYMO) span the six ecoregions in different ways: some sit cleanly within one or two, others split across several (Figure 5.16). The table that follows gives, for each watershed group, the share of its area falling in each ecoregion — the cross-reference that maps the regional and ecoregion climate signals onto the per-watershed prioritisation unit.

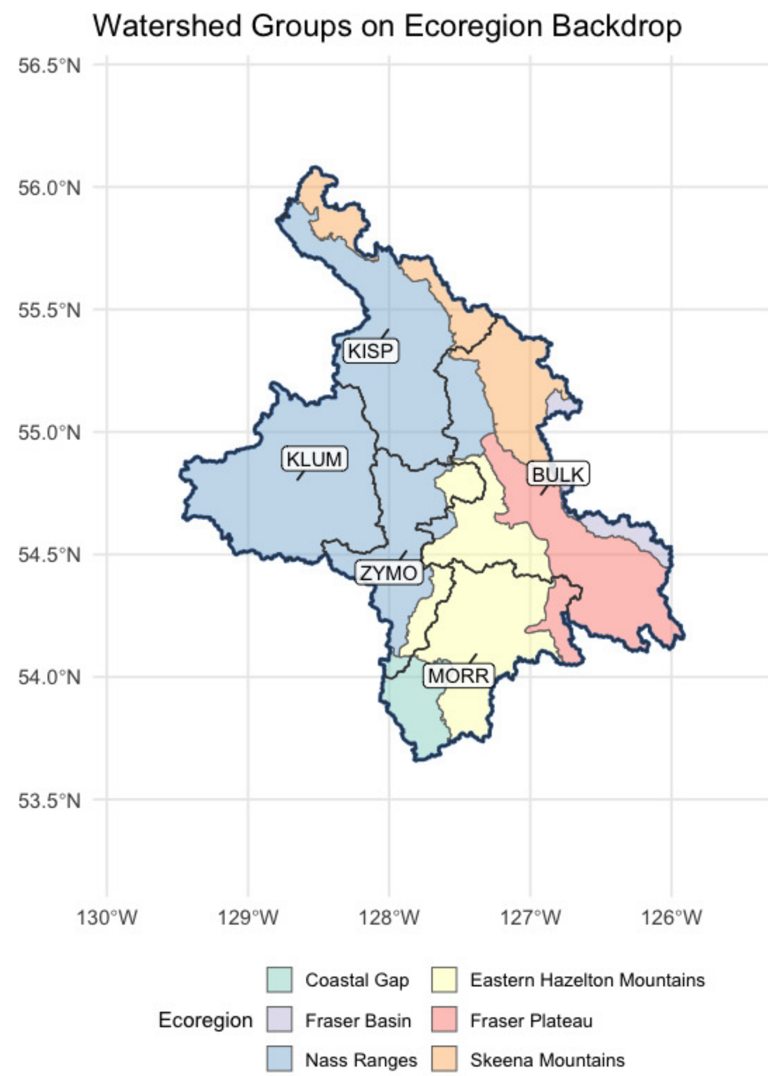

Figure 5.16: Watershed groups in the study area, labelled with their codes, on top of ecoregion fills. Kalum sits entirely within the Nass Ranges; Kispiox and Zymoetz are dominantly Nass Ranges; Morice is dominantly Eastern Hazelton Mountains reaching into the Coastal Gap; Bulkley is the only group spread across five ecoregions without a majority.

Table 5.6: Watershed groups in the study area with the percent of each group's area falling in each ecoregion. Rows where one ecoregion is at or near 100 % indicate watershed groups contained within a single climate-physiography zone; rows split across two or more ecoregions indicate watershed groups whose climate departure can pull from more than one regional pattern. BULK in particular spans five ecoregions with no majority among them, so it cannot be characterised by a single climate-physiography zone.

Code	Name	km²	NRA %	EHM %	FAP %	SKM %	COG %	FAB %
BULK	Bulkley River	7758	11.6	18.8	40.1	22.5	0.0	7.1
KISP	Kispiox River	5212	78.1	0.1	0.0	21.8	0.0	0.0
KLUM	Kalum River	4955	100.0	0.0	0.0	0.0	0.0	0.0
MORR	Morice River	4377	0.0	67.5	9.7	0.0	22.8	0.0
ZYMO	Zymoetz River	3026	73.4	23.3	0.0	0.0	3.3	0.0

Interpretation for fish passage

Four findings tie the regional climate-departure analysis to per-watershed barrier prioritisation.

Warming is broad and significant, and barely varies across the AOI. All six ecoregions warmed +2.42 to +2.70 °C since 1951 — a spread of only 0.28 °C between the most and least affected. Temperature does not *separate* watershed groups within this AOI at all; what it sets is the regional context. Every reach in the study area is operating in air 2.1 °C warmer than the conditions the channel, floodplain, and fish community evolved under. The within-AOI thermal differentiation comes from elevation, aspect, riparian cover, and groundwater contribution — not from climate-departure magnitude.

Precipitation has not moved. Across every ecoregion, the long-term precipitation trend is not statistically distinguishable from natural year-to-year variability. The climate-driven hydrology story for these five watershed groups is dominated by *atmospheric drying* (rising VPD, significant in every ecoregion) and *snowpack timing*, not by precipitation magnitude.

Snow is leaving the AOI earlier, not falling less. The snowmelt midpoint shifted 15 days earlier ($p < 0.001$ regionally and in every ecoregion), summer SWE collapsed by 60 % ($p < 0.001$) and summer snowmelt fell 41 % ($p < 0.01$). The counterbalancing rise is concentrated in spring, which gained 52 % and is highly significant; the winter rise is of similar magnitude but does not clear significance on its own. The total annual melt has not changed; its centre of mass has moved earlier. The freshet — the dominant high-flow event that mobilises spawning gravels, refills off-channel rearing habitat, and that crossing structures are sized against — is arriving weeks earlier across all five watershed groups, regardless of which ecoregion they sit in. Crossing-structure design assumptions and peak-flow-timing baselines baked into the assessment scoring need to track this shift. The late-summer cold-water input that high-elevation snowpack historically provided during the warmest, most thermally stressful weeks of summer is dropping in parallel.

There is no geographic gradient worth prioritising on. The west-warm to east-cool axis is present but weak: watershed-group mean departures span +1.98 to +2.21 °C, a difference of 0.23 °C across the whole study area. Kalum River sits at the warm end and Morice River at the cool end, but the separation is small enough to be immaterial next to within-watershed variation in elevation, aspect and groundwater contribution.

Warming this even means barrier prioritisation rests on reach-scale thermal and habitat evidence rather than on where a crossing sits in the region.

For the cold-water salmonids the study area supports — steelhead, coho, chinook, bull trout, Dolly Varden and rainbow trout — the implication for barrier prioritisation is that the two competing directions play out **within** watershed groups rather than between them. Headwater reaches that were cold-limited can gain growing-degree-days from warming, raising the intrinsic potential of habitat above the barrier and *increasing* the recovery value of restoring passage. Reaches already sitting near the upper edge of the cold-water thermal niche face the opposite direction — the same warming reduces usable habitat and lowers the payoff. Because the regional departure is near-uniform, which of those applies is an elevation and reach-scale question, not one the watershed-group ecoregion mapping can answer.

The one signal that does apply evenly, and therefore to every crossing in the study area, is timing: the freshet is arriving more than two weeks earlier than the record it is designed against, and late-summer snowpack contribution is falling. Those affect structure sizing and the thermal window for passage regardless of which watershed group a barrier sits in.

Appendix - Floodplain Delineation

This appendix details the floodplain delineation for the Bulkley River Watershed Group. The rationale for mapping floodplain extent alongside the fish passage barrier inventory is described in the report background — in brief, the off-channel habitats that floodplains sustain are often as important to fish productivity as the mainstem passage that crossing remediations restore, and infrastructure corridors can compromise floodplain function in ways that structure replacement alone does not address.

Floodplain footprints were delineated using the `flooded` R package, which implements the Valley Confinement Algorithm of Nagel et al. (2014). The algorithm combines a digital elevation model with the stream network and a modelled bankfull flood surface to distinguish valley bottoms from confined canyon reaches. Bankfull depth — the flow depth at which a stream begins to spill onto its floodplain — is estimated from upstream watershed area and mean annual precipitation using the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams.

We mapped the **functional floodplain**: the area a river actively inundates and reworks during regular high-flow events. This sits between two other scales — the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain matters most for habitat planning because it is the ground where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Inputs to the published delineation were the MRDEM-30 30 m DEM, a coho accessible stream network of stream order ≥ 3 built with `link` over the BC Freshwater Atlas, and Freshwater Atlas lakes and wetlands on that network. Waterbodies were included so the valley confinement model fills cells that gradient and cost-distance masks would otherwise exclude.

The stream lengths reported below are measured against the coho accessible network from bcfishpass, which is the network used throughout the rest of this report. It is not the network the delineation itself was built on, and the two are aligned by intent rather than by construction, so the lengths should be read as close estimates rather than exact intersections.

The delineation shown here was not generated for this report. It is published as item `bulk_co_ff04` in the `stac-floodplains-bc` collection — see stac_floodplains_bc for the publishing pipeline and how to query the collection — and is read from there, so a single modelled product serves every report that needs it and the repository does not carry its own copy of a layer that already exists elsewhere. The Bulkley is the first watershed group in this study area delivered this way; Morice and Kispiox are already published and can follow without re-running the model.

Table 5.7: Bulkley River Watershed Group — modelled functional floodplain and the off-channel habitat units within it.

Metric	Value
Watershed group area (km²)	7,762
Floodplain area (ha)	48,344
Floodplain as % of watershed group	6.2
Coho accessible streams, order 3+ (km)	2,206
Streams within floodplain (km)	1,587
Lakes within floodplain (count)	180
Lakes within floodplain (ha)	3,912
Wetlands within floodplain (count)	345
Wetlands within floodplain (ha)	4,171

The Bulkley River Watershed Group covers roughly 7,762 km². Modelled functional floodplain (Table 5.7) covers about 48,344 ha, or 6.2 % of the watershed group, attached to 1,587 km of the 2,206 km coho accessible order-3+ network. Within that footprint sit 180 lakes and 345 mapped wetland polygons — the off-channel habitat units that lateral floodplain connectivity supports.

The watershed-wide footprint is shown in Figure 5.17; a detail view of the valley immediately south of Smithers is shown in Figure 5.18.

Bulkley River Watershed Group
functional floodplain, accessible order 3+

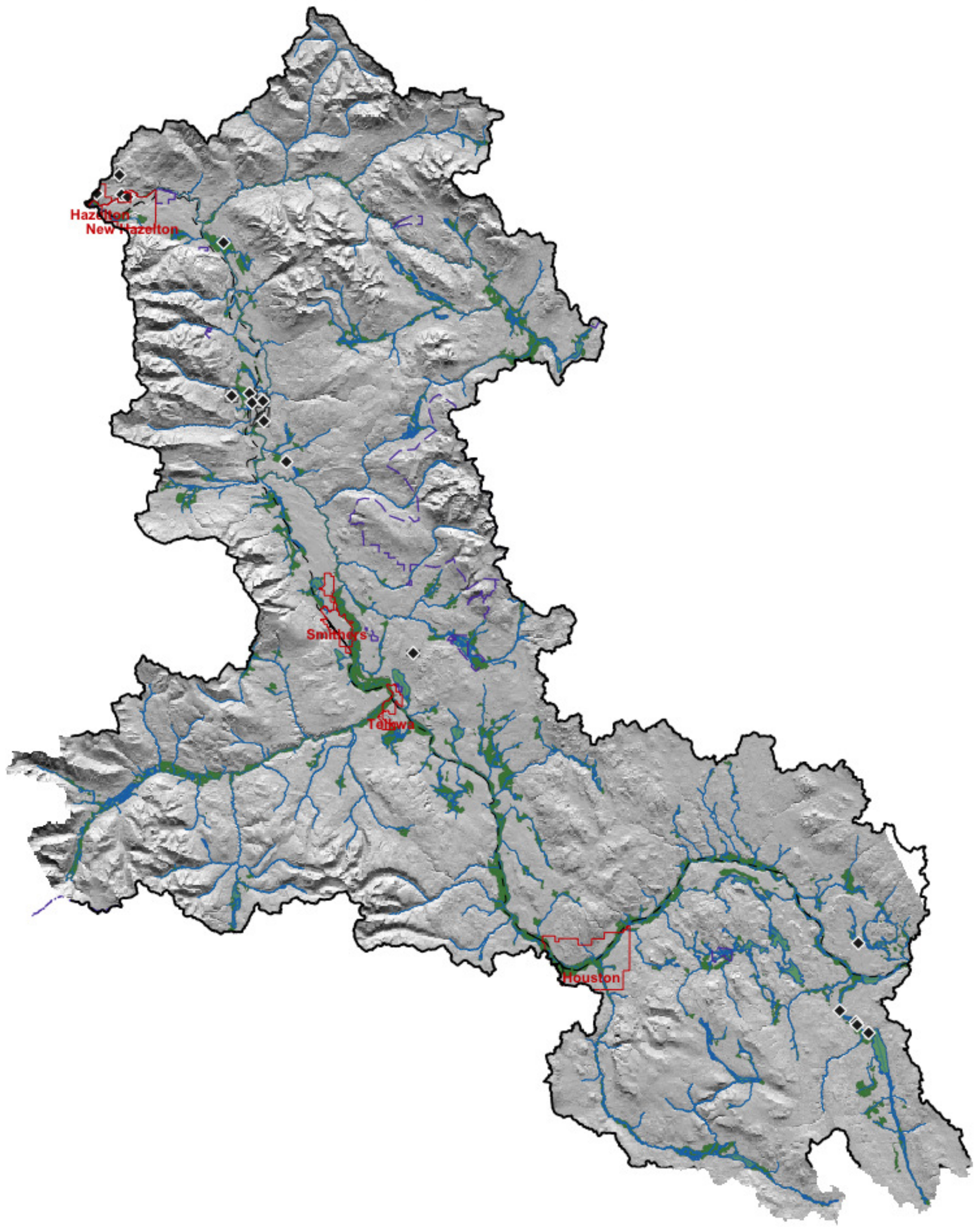

Figure 5.17: Bulkley River Watershed Group functional floodplain (green) over hillshaded terrain. Coho accessible order-3+ streams in blue; lakes and wetlands in light blue; parks and protected areas outlined purple dashed; First Nations reserves light grey with diamond markers; municipalities outlined and labelled in red; forest service / resource roads grey; railways black dashed; watershed boundary heavy black. Reserves are labelled on the detail map, where the labels are legible.

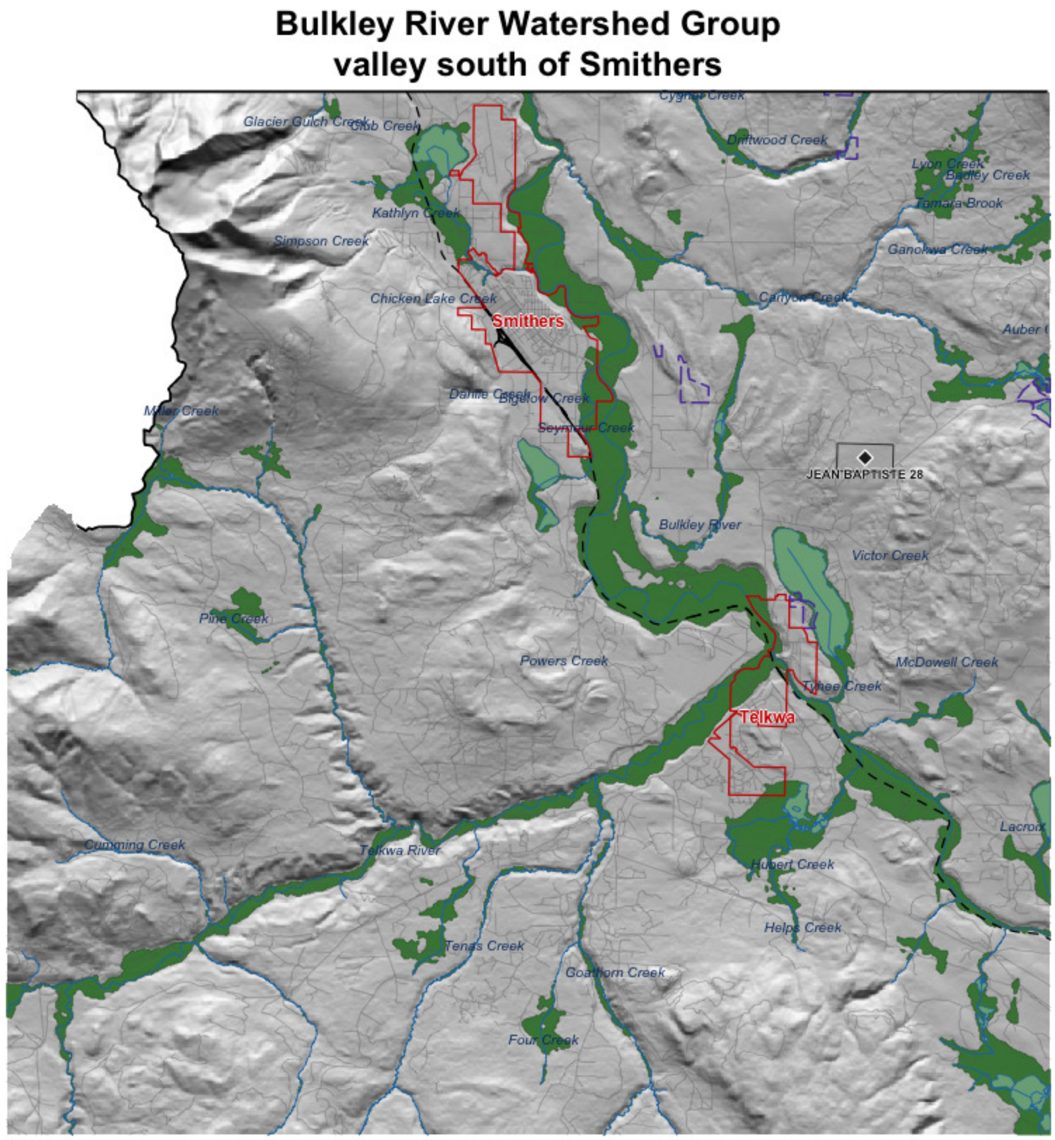

Figure 5.18: Detail view of the Bulkley River valley immediately south of Smithers. Symbology as Figure 5.17; named streams labelled in italic blue; First Nations reserves labelled; municipalities outlined and labelled in red.

The watershed-wide view shows floodplain tracking the trunk Bulkley River through broad valley reaches and tributary confluences. The detail view south of Smithers resolves how Highway 16, the Canadian National Railway and agricultural conversion through Telkwa intersect the floodplain footprint at reach scale — context that informs whether crossing remediation at a given site should account for restoring lateral connectivity beyond the structure itself. We plan to extend this mapping to additional watershed groups in coming years so that floodplain context can inform barrier prioritisation and restoration scoping across the region.

Appendix - Assessment Data Summary

Fish passage assessment procedures conducted through SERNbc since 2020 are amalgamated here — assessments, habitat confirmations, design, remediation, monitoring, fish sampling, eDNA sampling and drone imagery. The detailed per-site table is interactive and is provided online within this appendix here.

Table 5.8: Summary of fish passage assessment procedures conducted in northern British Columbia through SERNbc. **NOTE: To view all columns in the table - please click on one of the sort arrows within column headers before scrolling to the right.**

Table 5.9: Details of fish passage assessment procedures conducted in northern British Columbia through SERNbc.

Appendix - UAV Imagery

Since 2020, orthoimagery and elevation model rasters have been generated and stored as Cloud Optimized Geotiffs on a cloud service provider (AWS) with select imagery linked to in the collaborative GIS project. Additionally - a tile service has been set up to facilitate viewing and downloading of individual images, provided athttps://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting//results-and-discussion.html.

Appendix - Collaborative GIS Layers

In addition to numerous layers documenting fieldwork activities since 2020, a summary of background information spatial layers and tables loaded to the collaborative GIS project (sern_skeena_2023) at the time of writing (2026-08-06) are listed below. The layer table is interactive and is provided online within this appendix here.

Appendix - Environmental DNA Results

This appendix carries the site-level detail behind the headline counts in the Results chapter. Field blanks are reported separately because they test the sampling protocol rather than a location. Office blanks are excluded entirely — they are distilled-water controls filtered at the end of a session at accommodation, so although they carry real GPS coordinates, those coordinates record where the filtering happened rather than a stream site. The interactive map is linked at the end.

Per-site detection results

Format: `CODE(max_droplets)`, with ∗ flagging samples that UNBC reran. Detected = ≥4 positive droplets, the call threshold applied by the laboratory. Everything below that threshold is reported as not detected, including 2-3 droplet results that were retested without reaching the cutoff.

Table 5.10: Site-level eDNA detection results across real environmental samples in the project area. UNBC ID is the lab sample identifier; Pos. control = yes indicates the sample was collected at a site where the target species was also confirmed by electrofishing.

Site ID	Stream	UNBC ID	Pos. control	Detected	Not detected
197360_ds_ed	Riddeck Creek	S61182		RAIN(36)*	BULT(0), *CHIN(1)*, COHO(1)*
197378_Us_ed1	Tributary To Owen Creek	S61184		RAIN(512)	BULT(0), CHIN(0), COHO(1)
197378_ds_ed1b	Tributary To Owen Creek	S61185		RAIN(679)	BULT(0), CHIN(0), COHO(0)
197379_ds_ed	Avin Creek	S61186		RAIN(30)	BULT(0), CHIN(0), COHO(0)
197379_us_ed1	Avin Creek	S31493		RAIN(111)	BULT(0), COHO(1), SOCK(0)
197960_ds_ed1	Corya Creek	S50879		BULT(113), COHO(12)*	CHIN(0), SOCK(0)
197962_ds_ed	Peacock Creek	S60678		COHO(8), RAIN(124)	CHIN(0), SOCK(0)
197962_us_ed	Peacock Creek	S60880		BULT(84), COHO(6)	CHIN(0), SOCK(0)
58067_ds_ed1	Tributary To Gramaphone Creek	S61187		RAIN(96)	BULT(0), CHIN(0), COHO(0)
8525_ds_ed1	Tributary To Coal Creek	S61188		BULT(128)	CHIN(0), COHO(0), RAIN(0)
8525_us_ed1	Tributary To Coal Creek	S61189	yes	BULT(110)	CHIN(0), COHO(0), RAIN(0)
8525_us_ed2	Tributary To Coal Creek	S61190		BULT(18)	CHIN(0), COHO(0), RAIN(0)
8530_ds_ed1	Sandstone Creek	S61191		BULT(63), RAIN(59)	CHIN(0), COHO(0)
8530_us_ed1	Sandstone Creek	S61192		BULT(66), RAIN(57)	CHIN(0), COHO(0)
8547_ds_ed1	Tributary To Mcdonnell Lake	S60881		—	BULT(2)*, CHIN(0), COHO(0), SOCK(0)
8547_ds_ed2	Tributary To Mcdonnell Lake	S60377		—	BULT(0), COHO(0), RAIN(0), SOCK(0)

Site ID	Stream	UNBC ID	Pos. control	Detected	Not detected
8547_us_ed1	Tributary To Mcdonnell Lake	S60274		—	BULT(0), CHIN(0), COHO(0), SOCK(0)
bulkley_falls_ds_ed1	Bulkley River	S60275		—	BULT(0), CHIN(0), COHO(0), SOCK(0)
bulkley_falls_ds_ed2	Bulkley River	S60276		—	BULT(1), CHIN(0), COHO(1)*, SOCK(0)

Field blanks

Field blanks filter distilled water at the field site to test for protocol contamination during collection and processing. **Detections in blanks indicate potential cross-contamination, not species presence at the location** — these rows are presented for QA transparency only, not as eDNA detections.

Table 5.11: Field blanks collected during the `r params$project_year` eDNA program in the project area. Detections in blanks reflect protocol-side contamination, not site eDNA.

Site ID	Stream	UNBC ID	Detected	Not detected
197378 ds_ed1a	–	S61183	—	BULT(0), *CHIN(0)*, COHO(0), *RAIN(0)*

Retests

UNBC reran a subset of samples to confirm borderline results. The table below lists every site × species combination that was rerun, including reruns of both real environmental samples and field blanks.

Table 5.12: UNBC retests for the `r params$project_year` eDNA samples. Runs is the total number of times the sample x species combination was assayed; Max droplets is the highest positive-droplet count observed across all runs.

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
197360_ds_ed	Riddeck Creek	Bull Trout	BULT	environmental	6	0	Not detected
197360_ds_ed	Riddeck Creek	Chinook Salmon	CHIN	environmental	6	1	Not detected
197360_ds_ed	Riddeck Creek	Coho Salmon	COHO	environmental	6	1	Not detected
197360_ds_ed	Riddeck Creek	Rainbow Trout	RAIN	environmental	6	36	Detected (≥4 droplets)
197378 ds_ed1a	–	Bull Trout	BULT	field blank	4	0	Not detected
197378 ds_ed1a	–	Chinook Salmon	CHIN	field blank	4	0	Not detected
197378 ds_ed1a	–	Coho Salmon	COHO	field blank	4	0	Not detected

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
197960_ds_ed1	Corya Creek	Coho Salmon	COHO	environmental	4	12	Detected (≥4 droplets)
8547_ds_ed1	Tributary To Mcdonnell Lake	Bull Trout	BULT	environmental	4	2	Not detected
bulkley_falls_ds_ed2	Bulkley River	Coho Salmon	COHO	environmental	4	1	Not detected

Interactive map

The interactive eDNA map shows site-level detection results with toggleable per-species layers and an opt-in Controls layer for field blanks. See the interactive eDNA map.

Waterfall Creek - 124421 - Monitoring Appendix

PSCIS crossing 124421 on Waterfall Creek is located on 11th Avenue within the Bulkley River watershed group. The culvert at this crossing has been removed by the Gitksan Watershed Authorities, and the site was visited in 2025 under the effectiveness monitoring program.

Crossing Characteristics

Before removal, the structure was a round culvert ranked a barrier to upstream migration under the provincial protocol (MoE 2011). The September 2021 assessment recorded the pipes bent in the middle from road weight and an inlet drop of 0.10 m on both, in a reach described as slow, steady flow through almost wetland habitat above and below the crossing (Figure 5.19).

The culvert has since been removed. At the 2025 visit the riprap at the site had been buried with soil and seeded, with good regrowth of grasses established, and shrub planting was planned for the fall (Figure 5.20).

Figure 5.19: Culvert on Waterfall Creek at PSCIS crossing 124421 on 11th Avenue, photographed in September 2021 before removal.

Figure 5.20: Waterfall Creek at PSCIS crossing 124421 on 11th Avenue, photographed on 2025-10-04 following removal of the culvert.

Figure 5.21: Revegetation of the buried riprap at PSCIS crossing 124421, recorded during the 2025 effectiveness monitoring visit.

Monitoring Form

2025 effectiveness monitoring form results are summarized in Table 5.13.

Table 5.13: Summary of effectiveness monitoring form results for site 124421 in 2025.

Variable	Value
Pscis Crossing Id	124421
Stream Name	Waterfall Creek
Road Name	11th Avenue
Crossing Subtype	FORD
Span	–
Width	–

Variable	Value
Assessment Comment	Culvert was removed by Gitksan Watershed Authorities. Site riprap was buried with dirt and seated with good regrowth of grasses. Shrub planting is planned for this fall.
Dewatering Notes	Water is flowing well through construction footprint with no evidence of dewatering
Velocity Notes	Velocities through the construction footprint were comparable to riffle-type morphology sections upstream and downstream. Placement of riprap boulders within the wetted channel created localized pockets of reduced velocity suitable for resting.
Constriction Notes	Channel widths ranged from 4.5–5m, with wetted width at the time of assessment approximately 4m. The channel did not appear constricted through the construction footprint.
Substrate Notes	Channel substrate was composed primarily of cobbles with some strategically placed boulders. Overall composition appeared natural and suitable for the stream.
Riparian Notes	Riparian removal to accommodate construction appeared minimal, with shrub planting planned for the fall off 2025. The site was re-vegetating well with a reclamation grass mix and mulch, and much of the armoring rock had been covered with quality topsoil.
Flow Depth Notes	Flow depths through the construction footprint appeared suitable, ranging from 5–15cm. Over time, natural stone lines were expected to develop, creating deeper areas as the channel continued to mature.
Stability Notes	Site appears stable with ample rock to prevent washouts.
Revegetation Notes	Reclamation grasses that were seeded appeared to be growing well, supported by coconut matting and areas where rock had been buried with topsoil. Shrub planting was reportedly planned for the fall 2025.
Cover Notes	Strategically placed boulders provided some instream cover, and one piece of woody debris was present at the downstream end of the site. Overhead vegetation within the footprint is expected to provide additional cover as it matures.
Maintenance Notes	No maintenance required at the time of assessment however, revegetation activities were not yet complete.

Habitat

Habitat confirmation assessments were completed at this site in September 2021, before the culvert was removed. Both reaches were rated moderate habitat value, with 120 m surveyed upstream and 50 m downstream, and approximately 1.3 km of upstream habitat length estimated. The site was ranked a moderate priority for remediation.

The upstream reach was described as a slow-moving wetland-type stream with deep glides throughout, instream vegetation, overhanging vegetation and undercut banks, with gradient increasing slightly and substrate changing from fines and organics to small gravel near the top of the survey. The downstream reach was a defined channel widening into a small wetland complex dominated by tall grasses, with high amounts of instream vegetation and muddy substrate.

No fish sampling was conducted at this site in 2021, and no environmental DNA samples were collected here in 2025.

Aerial Imagery

An aerial survey was conducted with a remotely piloted aircraft and the resulting imagery was processed into an orthomosaic available to view and download here.

Discussion

Removing a crossing outright, rather than replacing it, is the most complete form of remediation available and it is uncommon in this program. The 2025 visit found the site stabilising as intended — riprap buried and seeded, grasses established, shrub planting to follow.

What the site does not yet have is any direct evidence of fish use, before or after. The 2021 assessment rated the habitat moderate on the basis of channel and cover characteristics rather than observed fish, no electrofishing was conducted, and the site was not part of the 2025 environmental DNA program. The reach is wetland-type habitat with deep glides, instream vegetation and undercut banks, which the 2021 survey noted transitions to small gravel toward the upper end.

Recommendations for continued monitoring at this site:

- Collect environmental DNA upstream and downstream of the former crossing. This is the lowest-cost way to establish which species use the reach, and there is currently no fish presence record here at all.
- Continue photographic monitoring of the revegetating riprap and the channel through the former crossing, so recovery can be tracked against the shrub planting.
- Confirm with the Gitksan Watershed Authorities what the planned shrub planting involved and when it was completed, so the revegetation record reflects the work actually done.

Peacock Creek - 197962 - Monitoring Appendix

PSCIS crossing 197962 on Peacock Creek is located on the Morice Forest Service Road within the Morice River watershed group. The culvert at this crossing was replaced with a bridge in 2021, and the site was revisited in 2025 under the effectiveness monitoring program.

Crossing Characteristics

Before replacement, the structure was a corrugated metal culvert with a concrete headwall, ranked a barrier to upstream migration under the provincial protocol (MoE 2011). The assessment recorded a partial inlet drop of varying height caused by large woody debris and boulders. Photographs taken during the September 2021 assessment show the outlet perched above a plunge pool, with the drop and the pool visible below the headwall (Figure 5.22).

The replacement structure is an open-span bridge. Photographs from the 2025 monitoring visit show an unconstricted channel through the crossing, with the bed continuous through the opening and no residual outlet drop (Figure 5.23).

Figure 5.22: Culvert on Peacock Creek at PSCIS crossing 197962 on the Morice FSR, photographed on 2021-09-07 before replacement. The outlet is perched above a plunge pool.

Figure 5.23: Bridge on Peacock Creek at PSCIS crossing 197962 on the Morice FSR, photographed on 2025-09-29 following replacement.

Figure 5.24: Substrate condition through the crossing at PSCIS crossing 197962, recorded during the 2025 effectiveness monitoring visit.

Monitoring Form

2025 effectiveness monitoring form results are summarized in Table 5.14.

Table 5.14: Summary of effectiveness monitoring form results for site 197962 in 2025.

Variable	Value
Pscis Crossing Id	197962
Stream Name	Peacock Creek
Road Name	Morice FSR
Crossing Subtype	BRIDGE
Span	20
Width	10

Variable	Value
Assessment Comment	Bridge installed in 2020 or 2021. Bridge span estimated from photos.
Dewatering Notes	No dewatering. Stream flows nicely through construction footprint.
Velocity Notes	Velocities within the construction footprint are comparable to upstream and downstream.
Constriction Notes	Although there is no additional constriction to the channel within the construction footprint, it appears as though the channel upstream was channelized somewhat to align with the inlet of the previous structure and was not widened as part of the project. Underneath the bridge, there is no constriction at all with generous room for channel migration.
Substrate Notes	Substrate within the channel footprint is comparable to upstream and downstream with some boulders strategically placed to promote stone lines and break up some of the steeper sections.
Riparian Notes	Riparian within the construction footprint is primarily herbaceous with no evidence of shrub or tree planting related to the project.
Flow Depth Notes	Natural in filling of the channel has produced stone lines and placed boulders have produced shallow pools suitable for refuge.
Stability Notes	Structure appears stable with no evidence of slumping or erosion.
Revegetation Notes	Revegetation appears to be primarily herbaceous likely from a clover and grass seed mix
Cover Notes	Boulders placed within the construction footprint and natural stone lines have produced areas of refuge with shallow pool cover. There is no overhanging vegetation and no large wood debris with very sparse small wood debris present.
Maintenance Notes	No maintenance required.

Habitat and Fish Presence

Habitat confirmation assessments were completed at this site in September 2021, before the culvert was replaced. Both the reach upstream and the reach downstream of the crossing were rated high habitat value, with 750 m surveyed upstream and 600 m downstream. The upstream reach was described as complex habitat with undercut banks, large and small woody debris and pools; the downstream reach as carrying consistent undercut banks with woody debris and overhanging vegetation providing cover. Approximately 4.8 km of upstream habitat length was estimated, and the site was ranked a high priority for remediation.

The site was electrofished on 2021-09-17 at three closed-site multi-pass locations upstream of the crossing and three downstream, capturing 96 fish across four species (Table 5.15). Dolly Varden and rainbow trout were caught on both sides of the crossing. Coho salmon and cutthroat trout were caught only downstream. Fork lengths ranged from 48 to 148 mm across all species, with coho between 63 and 85 mm.

Table 5.15: Fish captured by multi-pass closed-site electrofishing at PSCIS crossing 197962 on 2021-09-17, before the culvert was replaced. Three sites upstream and three downstream.

Species	Downstream	Upstream	Total
Coho Salmon	10	0	10
Cutthroat Trout	1	0	1
Dolly Varden	16	40	56
Rainbow Trout	26	3	29
Total	53	43	96

Replacement of the culvert began three days later, on 2021-09-20, so the entire 2021 record — assessment, habitat confirmation and fish sampling — describes the site in its pre-remediation condition.

In 2025, environmental DNA was collected both upstream and downstream of the crossing. Downstream, rainbow trout and coho were detected. Upstream, bull trout and coho were detected. Chinook and sockeye were not detected at either location (Table 5.16).

Table 5.16: Environmental DNA detection results at PSCIS crossing 197962, upstream and downstream of the crossing.

Site ID	UNBC ID	Detected	Not detected
197962_ds_ed	S60678	COHO(8), RAIN(124)	CHIN(0), SOCK(0)
197962_us_ed	S60880	BULT(84), COHO(6)	CHIN(0), SOCK(0)

Aerial Imagery

An aerial survey was conducted with a remotely piloted aircraft and the resulting imagery was processed into an orthomosaic available to view and download here.

Discussion

The site carries an uncommonly strong record. The pre-remediation habitat confirmation rated both reaches high value and estimated 4.8 km of habitat upstream, multi-pass electrofishing established what was present on each side of the culvert before it was replaced, and the 2025 environmental DNA re-establishes species presence four years after replacement.

Coho and cutthroat trout were captured only downstream of the culvert in 2021. In 2025, coho was detected on both sides of the bridge.

Recommendations for continued monitoring at this site:

- Resample for environmental DNA at both locations across multiple points in the season. The 2025 samples were collected on one day at the end of September.
- Assay the same species panel at both locations so the two are directly comparable. In 2025 the upstream and downstream samples were assayed for different sets of targets.
- Repeat the electrofishing conducted in 2021 at the same closed-site locations. Pre-remediation multi-pass data exist for three sites upstream and three downstream, which is an uncommonly strong baseline and the most direct available measure of whether the replacement changed fish use rather than only passage condition.
- Continue photographic monitoring of substrate, cover, bank stability and revegetation through the crossing, so channel recovery following construction can be tracked over time.

References

Amazon Web Services. 2025. *Aws/Aws-Cli: Universal Command Line Interface for Amazon Web Services*. V. 2.18.15. Released. https://github.com/aws/aws-cli.

Bell, M. C. 1991. *Fisheries Handbook of Engineering Requirements and Biological Criteria*. https://www.fs.fed.us/biology/nsaec/fishxing/fplibrary/Bell_1991_Fisheries_handbook_of_engineering_requirements_and.pdf.

Bourne, Christina, Dan Kehler, Yolanda Wiersma, and David Cote. 2011. "Barriers to Fish Passage and Barriers to Fish Passage Assessments: The Impact of Assessment Methods and Assumptions on Barrier Identification and Quantification of Watershed Connectivity."*Aquatic Ecology* 45: 389–403. https://doi.org/10.1007/s10452-011-9362-z.

Bramblett, Robert G., Mason D. Bryant, Brenda E. Wright, and Robert G. White. 2002. "Seasonal Use of Small Tributary and Main-Stem Habitats by Juvenile Steelhead, Coho Salmon, and Dolly Varden in a Southeastern Alaska Drainage Basin."*Transactions of the American Fisheries Society* 131 (3): 498–506. https://doi.org/10.1577/1548-8659(2002)131<0498:SUOSTA>2.0.CO;2.

Busch, D.Shallin, Mindi Sheer, Kelly Burnett, Paul McElhany, and Tom Cooney. 2011. "Landscape-Level Model to Predict Spawning Habitat For Lower Columbia River Fall Chinook Salmon (*Oncorhynchus Tshawytscha*): Intrinsic Potential Model for Spawning Fall Chinook Salmon."*River Research and Applications* 29 (3): 297–312. https://doi.org/10.1002/rra.1597.

Bustard, D, and C Schell. 2002. *Conserving Morice Watershed Fish Populations and Their Habitat*. Community Futures Development Corporation of Nadina. https://waves-vagues.dfo-mpo.gc.ca/Library/315091.pdf.

Calvin, Katherine, Dipak Dasgupta, Gerhard Krinner, et al. 2023. *IPCC: Summary for Policymakers. In: Climate Change 2023: Synthesis Report*. First. Intergovernmental Panel on Climate Change (IPCC). https://doi.org/10.59327/IPCC/AR6-9789291691647.

Canada, Environment and Climate Change. 2024. "National Water Data Archive: HYDAT." Service description. Aem. https://www.canada.ca/en/environment-climate-change/services/water-overview/quantity/monitoring/survey/data-products-services/national-archive-hydat.html.

Cayan, Daniel R., Michael D. Dettinger, Susan A. Kammerdiener, Joseph M. Caprio, and David H. Peterson. 2001. "Changes in the Onset of Spring in the Western United States."*Bulletin of the American Meteorological Society* 82 (3): 399–415. https://doi.org/10.1175/1520-0477(2001)082<0399:citoos>2.3.co;2.

Clarkin, K, A Connor, M Furniss, et al. 2005. *National Inventory and Assessment Procedure For Identifying Barriers to Aquatic Organism Passage at Road-Stream Crossings*. United States Department of Agriculture, Forest Service, National Technology; Development Program. https://www.fs.fed.us/biology/nsaec/fishxing/publications/PDFs/NIAP.pdf.

cogeotiff. (2018) 2025. *Cogeotiff/Rio-Cogeo*. March 9; Cogeotiff, released March 19. https://github.com/cogeotiff/rio-cogeo.

Cooney, Thomas, and Damon Holzer. 2006. *Appendix C: Interior Columbia Basin Stream Type Chinook Salmon and Steelhead Populations: Habitat Intrinsic Potential Analysis*. 21.

Cote, David, P Frampton, M Langdon, and R Collier. 2005. *Fish Passage and Stream Habitat Restoration in Terra Nova National Park Highway Culverts*.

Department of Fisheries and Oceans. 1991. *Fish Habitat Inventory and Information Program SISS Stream Summary Catalogue. Subdistrict 4D, Smithers (Volume 2). Bulkley.*https://waves-vagues.dfo-mpo.gc.ca/library-bibliotheque/40602369.pdf.

Development Seed. (2019) 2025. *Developmentseed/Titiler*. V. 0.21.1. June 28; Development Seed, released. https://github.com/developmentseed/titiler.

Diebel, M. W., M. Fedora, S. Cogswell, and J. R. O'Hanley. 2015. "Effects of Road Crossings on Habitat Connectivity for Stream-Resident Fish: STREAM-RESIDENT FISH HABITAT CONNECTIVITY." *River Research and Applications* 31 (10): 1251–61. https://doi.org/10.1002/rra.2822.

Dyson, J. B. 1949. *Bulkley Falls Investigation Report*. https://data.skeenasalmon.info/dataset/0af0ecf8-0d55-4d48-9bde-d869db0fb71a/resource/b3360add-f7df-4906-8cb5-c2cb4a3e7fa9/download/bulkley-falls-investigation-dfo-1949.pdf.

ECCC. 2016. *Climate Data and Scenarios for Canada: Synthesis of Recent Observation and Modelling Results*. Environment; Climate Change Canada (ECCC). https://publications.gc.ca/collections/collection_2016/eccc/En84-132-2016-eng.pdf.

Elmer, Laura. 2021. *Initial Assessment of Fish Habitat Usage and Accessibility in Clear Creek Eastern Side Channel*. https://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=61024.

Fernandes, I. M., Y. F. Bastos, D. S. Barreto, L. S. Lourenço, and J. M. Penha. 2017. "The Efficacy of Clove Oil as an Anaesthetic and in Euthanasia Procedure for Small-Sized Tropical Fishes." *Brazilian Journal of Biology = Revista Brasleira De Biologia* 77 (3): 444–50. https://doi.org/10.1590/1519-6984.15015.

Finnegan, Noah J., Gerard Roe, David R. Montgomery, and Bernard Hallet. 2005. "Controls on the Channel Width of Rivers: Implications for Modeling Fluvial Incision of Bedrock." *Geology* 33 (3): 229–32. https://doi.org/10.1130/G21171.1.

Fish Passage Technical Working Group. 2011. *A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation Fo a Stream Crossing*. https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/checklist-for-fish-habitat-confirmation-201112.pdf.

Fisheries and Oceans Canada. n.d. "NuSEDS-New Salmon Escapement Database System - Open Government Portal." Accessed April 17, 2025. https://open.canada.ca/data/en/dataset/c48669a3-045b-400d-b730-48aafe8c5ee6.

FLNRO. 2013a. *Bulkley River Angling Management Plan*. Ministry of Forests, Lands; Natural Resource Operations (FLNRO). https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fishery-resources/region-6-skeena/skeena-amp-bulkley-river-2013.pdf.

FLNRO. 2013b. *Overview of Angling Management Plans for the Skeena Watershed*. Ministry of Forests, Lands, Natural Resource Operations (FLNRO). https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fishery-resources/region-6-skeena/skeena-amp-overview-2013.pdf.

FLNRORD. 2013. *Zymoetz River Class I Section Angling Management Plan*. https://www.env.gov.bc.ca/skeena/fish/AMPs/Zymoetz1_AMP.pdf.

FLNRORD. 2017. *Natural Resource Stewardship Monitoring and Assessment Report for the Wet'suwet'en Hereditary Territory*. https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/forestry/integrated-monitoring/nrsmonitoringandassessmentreport-wetsuweten.pdf.

FLNRORD. 2019. *Freshwater Fishing Regulations Synopsis*. Ministry of Forests, Lands, Natural Resource Operations & Rural Development (FLNRORD). https://www2.gov.bc.ca/assets/gov/sports-recreation-arts-and-culture/outdoor-recreation/fishing-and-hunting/freshwater-fishing/region_6_skeena.pdf.

FLNRORD. 2020a. "Digital Road Atlas (DRA) - Master Partially-Attributed Roads - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development. https://catalogue.data.gov.bc.ca/dataset/digital-road-atlas-dra-master-partially-attributed-roads.

FLNRORD. 2020b. "Forest Tenure Road Section Lines - Data Catalogue." Published by Ministry of Forests, Lands, Natural Resource Operations and Rural Development (FLNRORD). https://

catalogue.data.gov.bc.ca/dataset/forest-tenure-road-section-lines.
Forest Investment Account. 2003. *Guidelines for in-Stream and Off-Channel Routine Effectiveness Evaluation*. British Columbia Ministry of Forests. http://env.gov.bc.ca/wld/documents/fia_docs/ree_guidelines.pdf.
“Gitxsan Huwilp Government.” n.d. Accessed May 29, 2024. https://gitxsan.ca/.
Gottesfeld, Allen, Ken Rabnett, and Peter Hall. 2002. *Conserving Skeena Fish Populations and Their Habitat - Skeena Stage I Watershed-based Fish Sustainability Plan*. Skeena Fisheries Commission. https://www.psf.ca/sites/default/files/Skeena%20WFSP%2012%20%28low%20res%29.pdf.
Gottesfeld, A, and K Rabnett. 2007. *Skeena Fish Populations and Their Habitat*. Skeena Fisheries Commission.
“Groups Challenge Rush Start of Construction on Pipeline.” 2025. Ecojustice, March 24. https://ecojustice.ca/news/groups-in-court-to-challenge-the-bending-of-permit-requirements-to-rush-start-of-construction-on-900-km-pipeline/.
Hall, Jason E., Diane M. Holzer, and Timothy J. Beechie. 2007. “Predicting River Floodplain and Lateral Channel Migration for Salmon Habitat Conservation.”*Journal of the American Water Resources Association* 43 (3): 786–97. https://doi.org/10.1111/j.1752-1688.2007.00063.x.
Hansen, James, Makiko Sato, and Reto Ruedy. 2012. “Perception of Climate Change.”*Proceedings of the National Academy of Sciences* 109 (37). https://doi.org/10.1073/pnas.1205276109.
Healthy Watersheds Initiative. 2021. “Riparian Restoration Surveys on Streams Near Terrace - Healthy Watersheds Initiative.” June 8. https://healthywatersheds.ca/project/riparian-restoration-surveys-on-streams-near-terrace/.
Hill, N. E., J. L. Thorley, and A. Irvine. 2025. *Spatial Stream Network Analysis of Skeena Watershed Stream Temperatures 2025*. Analytic Appendix. Poisson Consulting. https://www.poissonconsulting.ca/f/1130667589.
Hill, N. H., J. L. Thorley, and A. Irvine. 2024. *Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b*. Poisson Consulting Analysis Appendix. https://www.poissonconsulting.ca/f/1295467017.
Hobbs, Jared, Caren C. Helbing, and Caren Goldberg. 2025. *Environmental DNA Protocol for Freshwater Aquatic Ecosystems: Version 3.0*. BC Ministry of Environment, Ecosystems Branch.
IBM Business Consulting Services. 2006. “Valuation of the Wild Salmon Economy of the Skeena River Watershed.”https://www.psf.ca/sites/default/files/IBM_skeena_report_061.pdf.
ILMB. 2007. *Morice Land and Resource Management Plan*. {Ministry of Agriculture; Lands - Integrated Land Management Bureau (ILMB). https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/natural-resource-use/land-water-use/crown-land/land-use-plans-and-objectives/skeena-region/morice-lrmp/morice_lrmp_july2007.pdf.
Irvine. 2021. *Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning*. https://newgraphenvironment.github.io/fish_passage_bulkley_2020_reporting/.
Irvine. (2021) 2022. *Bulkley River and Morice River Watershed Groups Fish Passage Restoration Planning 2021*. https://github.com/NewGraphEnvironment/fish_passage_skeena_2021_reporting.
Irvine. 2023. *Bulkley Watershed Fish Passage Restoration Planning 2022*. https://www.newgraphenvironment.com/fish_passage_bulkley_2022_reporting/.
Irvine. (2025) 2025a. *NewGraphEnvironment/Stac_uav_bc*. January 29, released. https://github.com/NewGraphEnvironment/stac_uav_bc.
Irvine, Allan. (2025) 2025b. *NewGraphEnvironment/Water-Temp-Bc*. May 21, released September 5. https://github.com/NewGraphEnvironment/water-temp-bc.

Irvine, Al, and Lucy Schick. 2024. *Restoring Fish Passage in the Fraser Region - 2023*. New Graph Environment Ltd. Prepared for Ministry of Transportation and Infrastructure and Society for Ecosystem Restoration in Northern BC (SERNbc). https://newgraphenvironment.com/fish_passage_fraser_2023_reporting/.

Irvine, A, and Lucy Schick. 2023. *Skeena Watershed Fish Passage Restoration Planning 2023*. https://www.newgraphenvironment.com/fish_passage_skeena_2023_reporting/.

Irvine, A, and Lucy Schick. 2025. *Skeena Watershed Fish Passage Restoration Planning 2024*. https://www.newgraphenvironment.com/fish_passage_skeena_2024_reporting/.

Irvine, A, and Wintersheidt. 2023. *Skeena Watershed Fish Passage Restoration Planning 2022*. https://www.newgraphenvironment.com/fish_passage_skeena_2022_reporting/.

Kang, Do Hyuk, Huilin Gao, Xiaogang Shi, Siraj ul Islam, and Stephen J. Déry. 2016. "Impacts of a Rapidly Declining Mountain Snowpack on Streamflow Timing in Canada's Fraser River Basin."*Scientific Reports* 6 (1). https://doi.org/10.1038/srep19299.

Karl, Thomas R., Richard W. Knight, Kevin P. Gallo, et al. 1993. "A New Perspective on Recent Global Warming: Asymmetric Trends of Daily Maximum and Minimum Temperature."*Bulletin of the American Meteorological Society* 74 (6): 1007–23. https://doi.org/10.1175/1520-0477(1993)074<1007:ANPORG>2.0.CO;2.

Kemp, P. S., and J. R. O'Hanley. 2010. "Procedures for Evaluating and Prioritising the Removal of Fish Passage Barriers: A Synthesis: EVALUATION OF FISH PASSAGE BARRIERS."*Fisheries Management and Ecology*, no–. https://doi.org/10.1111/j.1365-2400.2010.00751.x.

Kirsch, J M, Joseph D Buckwalter, and Daniel J Reed. 2014. *Fish Inventory and Anadromous Cataloging in the Susitna River, Matanuska River, and Knik River Basins, 2003 and 2011.* 149.

Kitsumkalum Band. n.d. "Our Culture." Kitsumkalum, a Galts'ap (community) of the Tsimshian Nation. Accessed April 15, 2024. https://kitsumkalum.com/about/our-culture/.

Knowles, Noah, Michael D. Dettinger, and Daniel R. Cayan. 2006. "Trends in Snowfall Versus Rainfall in the Western United States."*Journal of Climate* 19 (18): 4545–59. https://doi.org/10.1175/jcli3850.1.

"LNG Pipelines | Skeena Watershed Conservation Coalition." n.d. Accessed April 15, 2025. https://skeenawatershed.com/lng-pipelines/.

McElhanney. 2022. *Kitsumkalum River Flood Mitigation Plan*. https://www.terrace.ca/sites/default/files/docs/Kitsumkalum%20River%20Flood%20Mitigation%20Plan_no%20appendices.pdf.

MoE. 2011. *Field Assessment for Determining Fish Passage Status of Closed Bottom Structures*. BC Ministry of Environment (MoE). https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/land-based-investment/forests-for-tomorrow/field-assessment-for-determining-fish-passage-status-of-cbs.pdf.

MoE. 2024a. "Known BC Fish Observations and BC Fish Distributions."https://catalogue.data.gov.bc.ca/dataset/known-bc-fish-observations-and-bc-fish-distributions.

MoE. 2024b. "Provincial Obstacles to Fish Passage - Data Catalogue."https://catalogue.data.gov.bc.ca/dataset/provincial-obstacles-to-fish-passage.

Mote, Philip W., Alan F. Hamlet, Martyn P. Clark, and Dennis P. Lettenmaier. 2005. "DECLINING MOUNTAIN SNOWPACK IN WESTERN NORTH AMERICA*."*Bulletin of the American Meteorological Society* 86 (1): 39–50. https://doi.org/10.1175/bams-86-1-39.

Mote, Philip W., Sihan Li, Dennis P. Lettenmaier, Mu Xiao, and Ruth Engel. 2018. "Dramatic Declines in Snowpack in the Western US."*Npj Climate and Atmospheric Science* 1 (1). https://doi.org/10.1038/s41612-018-0012-1.

Muñoz Sabater, Joaquín. 2019. *ERA5-Land Hourly Data from 1950 to Present*. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). https://doi.org/10.24381/cds.e2161bac.

Nagel, David E., John M. Buffington, Sharon L. Parkes, Seth Wenger, and John R. Goode. 2014. *A Landscape Scale Valley Confinement Algorithm: Delineating Unconfined Valley Bottoms for Geomorphic, Aquatic, and Riparian Applications*. Gen. Tech. Rep. RMRS-GTR-321. U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station. https://www.fs.usda.gov/rmrs/publications/landscape-scale-valley-confinement-algorithm-delineating-unconfined-valley-bottoms.

Norris, Simon. (2020) 2024. *Smnorris/Bcfishpass*. November 7, released January 12. https://github.com/smnorris/bcfishpass.

Office of the Wet'suwet'en. 2013. *Wet'suwet'en Title and Rights Regarding Canada Department of Fisheries & Oceans And Pacific Trails Pipeline.*http://www.wetsuweten.com/files/PTP_FHCP_Response_to_DFO-25Nov13-Final.pdf.

"Office of the Wet'suwet'en." 2021. http://www.wetsuweten.com/.

Oliver, Allison. 2018. *Analysis of Water Quality Monitoring in the Morice Water Management Area*. https://data.skeenasalmon.info/dataset/analysis-of-water-quality-monitoring-in-the-morice-water-management-area/resource/17125deb-57c3-4a91-b64c-377735bc2c32.

OpenDroneMap Authors. (2014) 2025. *OpenDroneMap/ODM*. V. 3.5.4. September 25; OpenDroneMap, released. https://github.com/OpenDroneMap/ODM.

Porter, Marc, Darcy Pickard, Katherine Wieckowski, and Katy Bryan. 2008. *Developing Fish Habitat Models for Broad-Scale Forest Planning in the Southern Interior of B.C.*ESSA Technologies Ltd. and B.C. Ministry of the Environment (MOE) for B.C. Forest Science Program. https://www.for.gov.bc.ca/hfd/library/FIA/2008/FSP_Y081231.pdf.

Powell, J. V., Vickie D. Jensen, and Anne-Marie Pedersen. 2018. "Gitxsan." The Canadian Encyclopedia. https://www.thecanadianencyclopedia.ca/en/article/gitksan.

PySTAC Authors. (2017) 2025. *Stac-Utils/Pystac*. V. 1.12.2. November 8; Stac-utils, released. https://github.com/stac-utils/pystac.

Radiantearth. 2024. "STAC: SpatioTemporal Asset Catalogs."https://stacspec.org/en/.

Resource Inventory Standards Committee. 2001. *Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Standards and Procedures*. https://www2.gov.bc.ca/assets/gov/environment/natural-resource-stewardship/nr-laws-policy/risc/recce2c.pdf.

Roberge, M, J M B Hume, C K Minns, and T Slaney. 2002. *Life History Characteristics of Freshwater Fishes Occurring in British Columbia and the Yukon, with Major Emphasis on Stream Habitat Characteristics*. 262.

Rosenfeld, Jordan, Marc Porter, and Eric Parkinson. 2000. *Habitat Factors Affecting the Abundance and Distribution of Juvenile Cutthroat Trout (Oncorhynchus Clarki) and Coho Salmon (Oncorhynchus Kisutch)*. 57: 9.

Schell, C. 2003. *A Brief Overview of Fish, Fisheries and Aquatic Habitat Resources in the Morice TSA*. https://data.skeenasalmon.info/dataset/a-brief-overview-of-fish-fisheries-and-aquatic-habitat-resources-in-the-morice-tsa.

Simoes, Rolf, Felipe Souza, Matheus Zaglia, Gilberto Ribeiro de Queiroz, Rafael dos Santos, and Karine Ferreira. 2021. "Rstac: An R Package to Access Spatiotemporal Asset Catalog Satellite Imagery."*2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 7674–77. https://doi.org/10.1109/IGARSS47720.2021.9553518.

Slaney, P A, and D Zaldokas. 1997. *Fish Habitat Rehabilitation Procedures*. https://www.for.gov.bc.ca/hfd/library/FFIP/Slaney_PA1997_A.pdf.

Sloat, Matthew R., Gordon H. Reeves, and Kelly R. Christiansen. 2017. "Stream Network Geomorphology Mediates Predicted Vulnerability of Anadromous Fish Habitat to Hydrologic Change in Southeast Alaska."*Global Change Biology* 23 (2): 604–20. https://doi.org/10.1111/gcb.13466.

stac-utils. 2025. *Stac-Utils/Stac-Fastapi-Pgstac: PostgreSQL Backend for Stac-Fastapi Using Pgstac (https://github.com/stac-utils/pgstac)*. V. 4.0.2. Released. https://github.com/stac-utils/stac-fastapi-pgstac.

Stewart, Iris T., Daniel R. Cayan, and Michael D. Dettinger. 2005. "Changes Toward Earlier Streamflow Timing Across Western North America."*Journal of Climate* 18 (8): 1136–55. https://doi.org/10.1175/jcli3321.1.

Stokes, J. 1956. *Upper Bulkley River Survey 1956*. https://data.skeenasalmon.info/dataset/6d9cc7a6-683e-4de5-879e-77b592882a35/resource/9ff7c750-1a1d-40b3-9ee1-6bbb46bdf38e/download/dfo-ubulkley-hab-survey-1956.pdf.

Swales, S., and C. D. Levings. 1989. "Role of Off-Channel Ponds in the Life Cycle of Coho Salmon (*Oncorhynchus Kisutch*) and Other Juvenile Salmonids in the Coldwater River, British Columbia."*Canadian Journal of Fisheries and Aquatic Sciences* 46 (2): 232–42. https://doi.org/10.1139/f89-032.

Tamblyn, Gregory C. 2005. *A Plan to Conserve and Protect Morice Watershed Fish Populations*. 78.

Thompson, Richard. 2013. *Assessing Fish Passage at Culverts – the Method, Its Metrics and Preliminary Findings from over 4,000 Assessments*.https://www2.gov.bc.ca/assets/gov/environment/plants-animals-and-ecosystems/fish-fish-habitat/fish-passage/assessing_fish_passage_at_culverts.pdf.

Thorley, J. L., S. Norris, and A. Irvine. 2021. *Channel Width 2021b*. https://www.poissonconsulting.ca/f/859859031.

Thorley, J, and A Irvine. 2021. "Channel Width 2021."https://www.poissonconsulting.ca/f/1792764180.

Toffolon, Marco, and Sebastiano Piccolroaz. 2015. "A Hybrid Model for River Water Temperature as a Function of Air Temperature and Discharge."*Environmental Research Letters* 10 (11): 114011. https://doi.org/10.1088/1748-9326/10/11/114011.

Washington Department of Fish & Wildlife. 2009. *Fish Passage Barrier and Surface Water Diversion Screening Assessment and Prioritization Manual*. Washington Department of Fish; Wildlife. Olympia, Washington. https://wdfw.wa.gov/sites/default/files/publications/00061/wdfw00061.pdf.

Wilson, Tim, and Ken Rabnett. 2007. *Fish Passage Assessment of Highway 16 and CN Rail in the Bulkley Watershed*. 124. https://data.skeenasalmon.info/dataset/fish-passage-assessment-highway-16-cn-rail-bulkley.

Woll, Christine, David Albert, and Diane Whited. 2017. *Salmon Ecological Systems*. The Nature Conservancy. https://www.conservationgateway.org/ConservationByGeography/NorthAmerica/UnitedStates/alaska/sw/cpa/Documents/TNC_A_Salmon_Ecological_Systems_Model_Nushagak_Kvichak.pdf.

Note that this reference was not actually cited in the body of the report but rather in the `index.Rmd` file `yaml` headerj under the `no_cite` key.

Session Info

Information about the computing environment is important for reproducibility. A summary of the computing environment is saved to `session_info.csv`, which can be viewed and downloaded from here.

```
## ─ Session info ───────────────────────────────────────────────────────────────
##  setting  value
##  version  R version 4.5.2 (2025-10-31)
##  os       macOS Tahoe 26.2
##  system   aarch64, darwin20
##  ui       X11
##  language (EN)
##  collate  en_US.UTF-8
##  ctype    en_US.UTF-8
##  tz       America/Vancouver
##  date     2026-08-06
##  pandoc   3.9.0.2 @ /opt/homebrew/bin/ (via rmarkdown)
##  quarto   1.4.553 @ /usr/local/bin/quarto
## 
## ─ Packages ───────────────────────────────────────────────────────────────────
##  package          * version    date (UTC) lib source
##  arrow            * 25.0.0     2026-07-16 [1] CRAN (R 4.5.2)
##  askpass            1.2.1      2024-10-04 [1] CRAN (R 4.5.0)
##  assertthat         0.2.1      2019-03-21 [1] CRAN (R 4.5.0)
##  base64enc          0.1-6      2026-02-02 [1] CRAN (R 4.5.2)
##  bit                4.6.0      2025-03-06 [1] CRAN (R 4.5.0)
##  bit64              4.8.0      2026-04-21 [1] CRAN (R 4.5.2)
##  blob               1.3.0      2026-01-14 [1] CRAN (R 4.5.2)
##  bookdown         * 0.46       2025-12-05 [2] CRAN (R 4.5.2)
##  boot               1.3-32     2025-08-29 [2] CRAN (R 4.5.2)
##  bslib              0.10.0     2026-01-26 [1] CRAN (R 4.5.2)
##  cachem             1.1.0      2024-05-16 [1] CRAN (R 4.5.0)
##  cd               * 0.3.0      2026-05-12 [1] Github
(NewGraphEnvironment/cd@a3e4733)
##  chk                0.10.0     2025-01-24 [1] CRAN (R 4.5.0)
##  class              7.3-23     2025-01-01 [2] CRAN (R 4.5.2)
##  classInt           0.4-11     2025-01-08 [1] CRAN (R 4.5.0)
##  cli                3.6.6      2026-04-09 [1] CRAN (R 4.5.2)
##  codetools          0.2-20     2024-03-31 [2] CRAN (R 4.5.2)
##  crayon             1.5.3      2024-06-20 [1] CRAN (R 4.5.0)
##  crosstalk          1.2.2      2025-08-26 [1] CRAN (R 4.5.0)
##  curl               7.1.0      2026-04-22 [1] CRAN (R 4.5.2)
##  data.table         1.18.2.1   2026-01-27 [1] CRAN (R 4.5.2)
```

```
##  DBI                 1.3.0      2026-02-25 [1] CRAN (R 4.5.2)
##  dbplyr              2.5.2      2026-02-13 [1] CRAN (R 4.5.2)
##  devtools            2.4.6      2025-10-03 [2] CRAN (R 4.5.0)
##  digest              0.6.39     2025-11-19 [1] CRAN (R 4.5.2)
##  doParallel          1.0.17     2022-02-07 [2] CRAN (R 4.5.0)
##  dplyr             * 1.2.1      2026-04-03 [1] CRAN (R 4.5.2)
##  DT                  0.34.0     2025-09-02 [2] CRAN (R 4.5.0)
##  e1071               1.7-17     2025-12-18 [1] CRAN (R 4.5.2)
##  elevatr             0.99.1     2025-09-10 [2] CRAN (R 4.5.0)
##  ellipsis            0.3.2      2021-04-29 [2] CRAN (R 4.5.0)
##  evaluate            1.0.5      2025-08-27 [1] CRAN (R 4.5.0)
##  farver              2.1.2      2024-05-13 [1] CRAN (R 4.5.0)
##  fasstr              0.5.3      2024-09-27 [2] CRAN (R 4.5.0)
##  fastmap             1.2.0      2024-05-15 [1] CRAN (R 4.5.0)
##  fishbc            * 0.2.1.9000 2026-01-15 [2] Github (lucy-
schick/fishbc@0607ffd)
##  forcats           * 1.0.1      2025-09-25 [2] CRAN (R 4.5.0)
##  foreach             1.5.2      2022-02-02 [2] CRAN (R 4.5.0)
##  fpr               * 1.2.0      2026-02-03 [2] Github
(NewGraphEnvironment/fpr@2817533)
##  fs                  2.1.0      2026-04-18 [1] CRAN (R 4.5.2)
##  generics            0.1.4      2025-05-09 [1] CRAN (R 4.5.0)
##  ggplot2           * 4.0.3      2026-04-22 [1] CRAN (R 4.5.2)
##  ggrepel             0.9.7      2026-02-25 [2] CRAN (R 4.5.2)
##  gh                  1.6.1      2026-07-20 [1] CRAN (R 4.5.2)
##  glue                1.8.1      2026-04-17 [1] CRAN (R 4.5.2)
##  gtable              0.3.6      2024-10-25 [1] CRAN (R 4.5.0)
##  here                1.0.2      2025-09-15 [2] CRAN (R 4.5.0)
##  hms                 1.1.4      2025-10-17 [1] CRAN (R 4.5.0)
##  htmltools           0.5.9      2025-12-04 [1] CRAN (R 4.5.2)
##  htmlwidgets         1.6.4      2023-12-06 [1] CRAN (R 4.5.0)
##  httr                1.4.8      2026-02-13 [1] CRAN (R 4.5.2)
##  iterators           1.0.14     2022-02-05 [2] CRAN (R 4.5.0)
##  janitor             2.2.1      2024-12-22 [1] CRAN (R 4.5.0)
##  jquerylib           0.1.4      2021-04-26 [1] CRAN (R 4.5.0)
##  jsonlite            2.0.0      2025-03-27 [1] CRAN (R 4.5.0)
##  kableExtra        * 1.4.0      2024-01-24 [2] CRAN (R 4.5.0)
##  Kendall             2.2.2      2025-12-22 [2] CRAN (R 4.5.2)
##  KernSmooth          2.23-26    2025-01-01 [2] CRAN (R 4.5.2)
##  knitr             * 1.51       2025-12-20 [1] CRAN (R 4.5.2)
##  labeling            0.4.3      2023-08-29 [1] CRAN (R 4.5.0)
##  lattice             0.22-9     2026-02-09 [2] CRAN (R 4.5.2)
##  leafem            * 0.2.5      2025-08-28 [2] CRAN (R 4.5.0)
##  leaflet           * 2.2.3      2025-09-04 [1] CRAN (R 4.5.0)
##  lifecycle           1.0.5      2026-01-08 [1] CRAN (R 4.5.2)
##  lubridate         * 1.9.5      2026-02-04 [1] CRAN (R 4.5.2)
##  magick              2.9.0      2025-09-08 [2] CRAN (R 4.5.0)
```

```
##  magrittr           2.0.5      2026-04-04 [1] CRAN (R 4.5.2)
##  Matrix             1.7-4      2025-08-28 [2] CRAN (R 4.5.2)
##  memoise            2.0.1      2021-11-26 [1] CRAN (R 4.5.0)
##  mgcv               1.9-3      2025-04-04 [2] CRAN (R 4.5.2)
##  ngr              * 0.0.1      2026-08-05 [1] Github
(NewGraphEnvironment/ngr@519c03b)
##  nlme               3.1-168    2025-03-31 [2] CRAN (R 4.5.2)
##  otel               0.2.0      2025-08-29 [1] CRAN (R 4.5.0)
##  pagedown         * 0.23       2025-08-20 [2] CRAN (R 4.5.0)
##  pak                0.11.1     2026-07-22 [1] CRAN (R 4.5.2)
##  pdftools         * 3.7.0      2026-01-30 [2] CRAN (R 4.5.2)
##  pillar             1.11.1     2025-09-17 [1] CRAN (R 4.5.0)
##  pkgbuild           1.4.8      2025-05-26 [2] CRAN (R 4.5.0)
##  pkgconfig          2.0.3      2019-09-22 [1] CRAN (R 4.5.0)
##  pkgload            1.5.0      2026-02-03 [2] CRAN (R 4.5.2)
##  plyr               1.8.9      2023-10-02 [1] CRAN (R 4.5.0)
##  png                0.1-9      2026-03-15 [1] CRAN (R 4.5.2)
##  prettyunits        1.2.0      2023-09-24 [1] CRAN (R 4.5.0)
##  processx           3.9.0      2026-04-22 [1] CRAN (R 4.5.2)
##  progress           1.2.3      2023-12-06 [2] CRAN (R 4.5.0)
##  progressr          0.18.0     2025-11-06 [2] CRAN (R 4.5.0)
##  proxy              0.4-29     2025-12-29 [1] CRAN (R 4.5.2)
##  purrr            * 1.2.2      2026-04-10 [1] CRAN (R 4.5.2)
##  qpdf               1.4.1      2025-07-02 [2] CRAN (R 4.5.0)
##  R6                 2.6.1      2025-02-15 [1] CRAN (R 4.5.0)
##  rappdirs           0.3.4      2026-01-17 [1] CRAN (R 4.5.2)
##  raster             3.6-32     2025-03-28 [1] CRAN (R 4.5.0)
##  rayshader          0.37.3     2024-02-21 [2] CRAN (R 4.5.0)
##  rbbt             * 0.0.0.9000 2026-01-15 [2] Github
(paleolimbot/rbbt@212680c)
##  RColorBrewer       1.1-3      2022-04-03 [1] CRAN (R 4.5.0)
##  Rcpp               1.1.1-1.1  2026-04-24 [1] CRAN (R 4.5.2)
##  RcppRoll           0.3.1      2024-07-07 [2] CRAN (R 4.5.0)
##  readr            * 2.2.0      2026-02-19 [1] CRAN (R 4.5.2)
##  readwritesqlite  * 0.2.0.9021 2026-01-15 [2] Github
(poissonconsulting/readwritesqlite@ccdb297)
##  remotes            2.5.0      2024-03-17 [2] CRAN (R 4.5.0)
##  rgl                1.3.31     2025-11-14 [2] CRAN (R 4.5.2)
##  rlang              1.3.0      2026-07-05 [1] CRAN (R 4.5.2)
##  rmarkdown        * 2.31       2026-03-26 [1] CRAN (R 4.5.2)
##  RPostgres        * 1.4.10     2026-02-16 [1] CRAN (R 4.5.2)
##  rprojroot          2.1.1      2025-08-26 [2] CRAN (R 4.5.0)
##  RSQLite            3.52.0     2026-05-10 [1] CRAN (R 4.5.2)
##  rstudioapi         0.18.0     2026-01-16 [2] CRAN (R 4.5.2)
##  rvest              1.0.5      2025-08-29 [1] CRAN (R 4.5.0)
##  s2                 1.1.9      2025-05-23 [1] CRAN (R 4.5.0)
##  S7                 0.2.2      2026-04-22 [1] CRAN (R 4.5.2)
```

```
##  sass               0.4.10     2025-04-11 [1] CRAN (R 4.5.0)
##  scales             1.4.0      2025-04-24 [1] CRAN (R 4.5.0)
##  sessioninfo        1.2.3      2025-02-05 [2] CRAN (R 4.5.0)
##  sf               * 1.1-0      2026-02-24 [1] CRAN (R 4.5.2)
##  snakecase          0.11.1     2023-08-27 [1] CRAN (R 4.5.0)
##  sp                 2.2-1      2026-02-13 [1] CRAN (R 4.5.2)
##  staticimports    * 0.0.0.9001 2026-01-15 [2] Github
(newgraphenvironment/staticimports@19cd73c)
##  stringi            1.8.7      2025-03-27 [1] CRAN (R 4.5.0)
##  stringr          * 1.6.0      2025-11-04 [1] CRAN (R 4.5.0)
##  svglite            2.2.2      2025-10-21 [2] CRAN (R 4.5.0)
##  systemfonts        1.3.1      2025-10-01 [2] CRAN (R 4.5.0)
##  terra            * 1.9-11     2026-03-26 [1] CRAN (R 4.5.2)
##  textshaping        1.0.4      2025-10-10 [2] CRAN (R 4.5.0)
##  tibble           * 3.3.1      2026-01-11 [1] CRAN (R 4.5.2)
##  tidyhydat          1.0.1      2026-06-11 [1] CRAN (R 4.5.2)
##  tidyr            * 1.3.2      2025-12-19 [1] CRAN (R 4.5.2)
##  tidyselect         1.2.1      2024-03-11 [1] CRAN (R 4.5.0)
##  tidyterra        * 1.1.0      2026-03-11 [2] CRAN (R 4.5.2)
##  tidyverse        * 2.0.0      2023-02-22 [2] CRAN (R 4.5.0)
##  tidyxl             1.0.10     2026-01-15 [1] Github
(nacnudus/tidyxl@7e2fbe7)
##  timechange         0.4.0      2026-01-29 [1] CRAN (R 4.5.2)
##  tzdb               0.5.0      2025-03-15 [1] CRAN (R 4.5.0)
##  units              1.0-1      2026-03-11 [1] CRAN (R 4.5.2)
##  usethis            3.2.1      2025-09-06 [2] CRAN (R 4.5.0)
##  vctrs              0.7.3      2026-04-11 [1] CRAN (R 4.5.2)
##  viridisLite        0.4.3      2026-02-04 [1] CRAN (R 4.5.2)
##  vroom              1.7.1      2026-03-31 [1] CRAN (R 4.5.2)
##  withr              3.0.2      2024-10-28 [1] CRAN (R 4.5.0)
##  wk                 0.9.5      2025-12-18 [1] CRAN (R 4.5.2)
##  xfun               0.57       2026-03-20 [1] CRAN (R 4.5.2)
##  xml2               1.5.2      2026-01-17 [1] CRAN (R 4.5.2)
##  yaml               2.3.12     2025-12-10 [1] CRAN (R 4.5.2)
##  zyp                0.11-1     2023-03-22 [2] CRAN (R 4.5.0)
##
##  [1] /Users/airvine/Library/R/arm64/4.5/library
##  [2] /Library/Frameworks/R.framework/Versions/4.5-
arm64/Resources/library
##  * -- Packages attached to the search path.
##
##
```

Attachment - Phase 1 Data and Photos

Data and photos for all Phase 1 fish passage assessments are provided online at https://www.newgraphenvironment.com/fish_passage_skeena_2025_reporting/appendix—phase-1-fish-passage-assessment-data-and-photos.html.

Attachment - Habitat Assessment and Fish Sampling Data

All field data collected is available here.

PSCIS reassessment data is available for download here https://github.com/NewGraphEnvironment/fish_passage_skeena_2025_reporting/blob/main/data/pscis_reassessments.xlsm. This is the workbook submitted to the Province, so it follows the provincial PSCIS submission format.

Attachment - Bayesian analysis to map stream discharge and temperature causal effects pathways

Two collaborative modelling studies with Poisson Consulting underpin the stream temperature work described in the Methods chapter.

The 2024 study mapped stream discharge and temperature causal effects pathways across the Nechako Watershed, settling on a four-parameter `air2stream` formulation with site-level random effects — Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b (Hill et al. 2024).

The 2025 study was carried out in this study area. It extended the approach to a larger spatial network and added Growing-Season Degree Days — accumulated thermal units during the season when stream temperature remains at or above 5 °C — as the productivity-relevant metric for age-0 salmonid growth — Spatial Stream Network Analysis of Skeena Watershed Stream Temperatures (Hill et al. 2025).

The water temperature observations feeding this work are organized in `water-temp-bc` (Irvine [2025] 2025b), which publishes Environment Canada hydrometric and water-quality records as a partitioned-parquet layer on Amazon S3.